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PROJECT AI-BASED GAP ANALYSIS OF ERP SYSTEM USAGE IN RETAIL ORGANISATIONS IN BAHRAIN

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Table of Acronyms

Table: Acronyms

ACRONYM	MEANING
AI	Artificial Intelligence
CRISP-DM	Cross-Industry Standard Process for Data Mining
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DSR	Design Science Research
DT	Decision Tree
EIS	Enterprise Information Systems
ERP	Enterprise Resource Planning
GCC	Gulf Cooperation Council
HR	Human Resources
ICT	Information and Communication Technology
INQ	Information Quality
IRB	Institutional Review Board
IS	Information Systems
ISK	ICT Skills
J48	Java 48 Decision Tree Algorithm
KPI	Key Performance Indicator
LR	Logistic Regression
MAE	Mean Absolute Error

ML	Machine Learning
NB	Naïve Bayes
PAL	Predictive Analysis Library
PEOU	Perceived Ease of Use
PUS	Perceived Usefulness
RF	Random Forest
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
ROI	Return on Investment
SAP	Systems, Applications, and Products
SCM	Supply Chain Management
SEQ	Service Quality
SP	Senior Project
SVM	Support Vector Machine
SYQ	System Quality
TAM	Technology Acceptance Model
TMS	Top Management Support
TOE	Technology, Organization, and Environment
TP	True Positive
UBA	User Behavior Analytics
UTAUT	Unified Theory of Acceptance and Use of Technology
WEKA	Waikato Environment for Knowledge Analysis

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Abstract

Enterprise Resource Planning (ERP) systems are widely implemented to improve operational efficiency and organizational integration. However, many retail organizations in Bahrain continue to experience ERP underutilization, where employees rely only on basic system functions while advanced features remain unused. This study investigates the gap between expected ERP usage and actual system utilization within the Bahraini retail sector.

The research adopts a combined Artificial Intelligence (AI) and Machine Learning (ML) approach to identify and classify ERP usage patterns. The study is theoretically grounded in the Technology–Organization–Environment (TOE) framework and the DeLone and McLean IS Success Model. Data was collected from 148 survey participants within Bahraini retail organizations, while an additional case study dataset containing 237 instances was used for validation and comparative analysis.

The experimental analysis was conducted using Waikato Environment for Knowledge Analysis (WEKA) and employed three classification algorithms: J48 Decision Tree, Random Forest, and Naïve Bayes. The findings revealed significant ERP underutilization, particularly in procurement and human resource functions, with many employees depending on external tools such as spreadsheets to complete operational tasks.

The results further indicate that insufficient training, limited monitoring practices, weak top management support, and low information quality are among the primary factors contributing to the ERP usage gap. Among the evaluated models, J48 provided the most interpretable results for the survey dataset, while Random Forest achieved the highest predictive accuracy in the case study dataset.

The study concludes that the ERP usage gap in the Bahraini retail sector is influenced more by organizational and human factors than by technological limitations alone. The findings demonstrate the practical value of machine learning techniques in identifying ERP usage patterns and supporting data-driven organizational decision-making.

Chapter 1: Introduction

Enterprise Resource Planning (ERP) systems are crucial for the integration of business processes like inventory management, sales processing, and reporting within retail enterprises. Many retail companies in Bahrain have invested heavily in ERP systems as part of the national digital transformation programs. Even so, ERP systems are poorly utilised in practice, as employees often circumvent the system or rely on external tools and manual workarounds. This leads to operational inefficiencies, reduced data accuracy, and lower return on investment (Solano & Cruz, 2024).

1.1 Overview of Enterprise Resource Planning (ERP) Systems

According to Gartner, ERP systems were originally designed to integrate and improve business processes and transactions within an organisation. Over time, ERP systems have expanded to include large integrated software packages that support cross-functional operations by providing a centralised platform for managing organisational routines and enabling information sharing across departments. By reducing duplicate data entry and improving access to consistent information, ERP systems can support faster reporting, better coordination, and more reliable information management (Coşkun, et al., 2022).

Despite these intended benefits, achieving sustained ERP value depends on effective post-implementation use. While many organisations anticipate that the return on their investment in an ERP system will continue after the organisation has completed implementation, several studies report that these expected benefits are not fully achieved in the post-implementation stage. Post-implementation issues cited are a lack of user knowledge regarding ERP systems and a lack of continuing learning by any/all users of ERP systems, which led to under-use of modules and poorer successful performance outcomes (Bravo, 2022). In addition, studies conducted in various levels of Middle East countries suggest many of the cultural and organisational barriers they experienced preventing effective use of ERP systems remain through the post-launch period, involving a lack of commitment to training of employees, resistance factors, as well as a weaker IT culture (Ali, Edghiem & Alkhalifah, 2022).

1.2 Background of the Study

Retail companies work in a very challenging and fast-paced environment. Success is dependent on the ability to make decisions accurately and quickly, as well as maintaining efficiency. ERP (Enterprise Resource Planning) is defined by Coursera (Team), 2026) as software used to manage and optimise a business's operations or business processes. ERP provides the retailer with a centralised system where multiple areas of the retailer's business can access information required to manage every aspect of their daily operations. For example, employees in store operations can access the same sales and inventory data as those in accounting, who may need to reconcile sales to inventory on hand. Thus, ERP is a method for cross-functional integration and coordination of an entire retailer's operations.

Many retailers turn to an ERP system to provide a highly effective method of managing their daily operations. An ERP system will provide a single digital platform within which multiple key business activities (like: inventory management, sales processing, procurement and reporting) are integrated, and on a real-time basis; therefore, when a customer makes a purchase at a retail store, it automatically updates the retailer's inventory records and related financial and managerial reporting at the time of sale.

Although organisations invest significant financial and technical resources in implementing ERP systems, challenges often arise after deployment. The real difficulty is not installing the system but ensuring that it is used effectively and consistently by employees. Research on ERP systems in the Middle East highlights issues such as resistance to change, limited user training, customisation complexity, and the absence of continuous monitoring mechanisms.

This situation creates what can be described as an ERP usage gap, a difference between how the system is designed to be used and how it is actually used in daily operations. When ERP modules are underutilised, organisations may experience inaccurate inventory records, delays in generating reports, and slower managerial decision-making.

In the Kingdom of Bahrain, digital transformation initiatives have encouraged retail organisations to adopt advanced enterprise technologies. However, systematic and automated evaluation of ERP usage remains limited.

At the same time, Artificial Intelligence (AI) and machine learning have become increasingly integrated into enterprise analytics to enhance data-driven decision-making and system optimization (Balázs, 2024).

By applying AI-based analytical techniques to structured survey data, organisations can move beyond descriptive feedback toward quantitative evaluation of ERP utilisation levels. Such approaches support proactive identification of underutilised modules and informed optimisation strategies within retail operations.

This study is therefore positioned at the intersection of ERP post-implementation evaluation and AI-support analytics. It proposes an intelligent approach to analysing quantitative ERP usage indicators collected through structured survey data in retail organisations in Bahrain to detect usage gaps and assess their impact on organizational performance.

1.3 Research Aim and Purpose

This project developed and applied an Artificial Intelligence (AI)-based analytical model to detect gaps between expected and actual ERP system usage in retail organisations in Bahrain, based on quantitative survey data collected from these organisations. Structured survey data were collected to examine ERP usage frequency, commonly used modules, and operational challenges faced by users. Machine learning techniques, implemented using Waikato Environment for Knowledge Analysis (WEKA), were applied to classify ERP usage patterns, identify indicators of underutilisation, and detect inconsistencies in system usage behaviour. The study utilised quantitative survey data as the primary data source, and the developed machine learning model successfully identified ERP usage gaps by classifying usage patterns and detecting inconsistencies in system utilisation behaviour. The results also provided insights into how these usage gaps are associated with organisational performance within retail organisations in Bahrain.

1.4 Problem Statement

ERP systems have become a staple in Bahrain's retail sector. Even with these systems in place, many businesses still struggle to use them effectively. These systems are mainly introduced to integrate and automate core business processes. In reality, the actual use of the system does not always match what was originally expected. Certain ERP modules may be overlooked or only

partly used, while different departments may rely on the system in inconsistent ways. This uneven usage reduces efficiency and prevents companies from fully benefiting from their investment (Bandara, et al., 2023).

A major concern is that organisations tend to focus heavily on implementing the ERP system but give limited attention to what happens afterwards. In many cases, there is no clear or continuous approach to evaluating how the system is being used. When regular monitoring is absent, process deviations, incomplete transactions, and uneven use across departments can continue for long periods without being detected.

At present, ERP usage is often reviewed through manual checks and occasional feedback from staff. This type of evaluation does not always capture how the system is actually used across different departments.

In many retail companies, the ERP system is expected to improve cooperation between various departments such as inventory, procurement, sales, and reporting. However, employees may still use manual spreadsheets, third-party software, or other modes of communication to complete their work. In some cases, for example, some departments may collect data from outside the ERP system due to difficulty using particular modules or a preference for more comfortable traditional means. Such behavior results in inconsistencies in business operations and a loss of trust in corporate data.

The magnitude of this issue extends beyond operational inconvenience. Previous studies have shown that ERP underutilisation can lead to reduced return on investment, process inefficiencies, duplication of work, and poor decision-making quality (Aalst, 2016). Although organisations invest heavily in ERP implementation, the expected benefits may not be fully achieved when employees do not consistently use the system as intended.

The issue acquires added importance within the context of the Bahraini retail industry, considering the increasing dependence on the processes of digital transformation and integration of business processes. Retail settings involve dynamism with constant business transactions, stock updates, dealing with customers, and coordination among operations. Inconsistency in using ERP applications may have an adverse effect on organizational performance and reduce the effectiveness of organisational workflows.

Even with the growing importance of ERP systems in Bahrain, there is limited research that deals with ERP adoption problems after implementation, especially studies adopting an analytical approach towards solving the problem using machine learning. Most previous studies have been largely concerned with the process of adoption and implementation, and not whether the systems were actually being used effectively after being put into operation. This research aims to fill the gap and analyze the behavior of ERP use as well as organizational and technical factors behind their underutilization in Bahrain's retail sector.

1.4.1 Problem Magnitude

According to research (Adebayo, 2023), Organizations typically utilize only 30-40% of their ERP system's capabilities within the first two years post-implementation. In the GCC context, studies report underutilization rates exceeding 60% for advanced modules such as analytics and HR. For a typical mid-sized retail organization in Bahrain, this translates to estimated annual losses of BD 15,000-25,000 in unrealized ROI.

This fiscal strain is further intensified by deep-seated operational friction. A significant portion of the ERP investment becomes functionally redundant when personnel revert to manual "workarounds" or isolated spreadsheets, a behavior observed in 36.5% of the underutilizing user base within the studied retail sector. Such fragmented usage patterns inevitably give rise to "data silos," which compromise the integrity of organizational intelligence. When executive decisions are informed by these incomplete or disconnected datasets, overall institutional performance falters (Nugroho, 2013).

1.5 Questions for Research

As described earlier, this research examines ERP underuse in retail businesses located in Bahrain through a structured, analytical lens, as a result of the previously identified problem. Prior research suggests that while ERP systems are highly adopted worldwide to integrate and streamline business processes, post-implementation challenges limit organisations' ability to realise the

anticipated value (Aalst, 2016). Many organisations tend to focus primarily on the implementation of the system and to neglect to continuously evaluate actual usage patterns. Consequently, inconsistencies in use per module and appropriate work procedures may not be discovered. Furthermore, research on ERP challenges in the Middle East indicates a need for improved monitoring mechanisms and structured evaluation methods to promote efficient use of the system (Shaiti, 2020).

Emerging methodologies and machine learning approaches to analytics offer new opportunities to analyse the usage of the systems systematically. Process-oriented data analysis and the use of pattern recognition techniques are effective methods for identifying behavioural deviations across enterprise systems (Chawla, 2019) ; (Aalst, 2016). However, few studies have applied these analytical models to quantitatively measure ERP usage within the retail industry in Bahrain, thus providing this study with the following research questions:

1. What measurable indicators can be used to identify ERP underutilization within retail organisations in Bahrain?

The purpose of this question is to find certain observable factors that demonstrate inconsistent use and incomplete adoption of complete modules that are only using a few of the 15 available modules or the use of ERP systems in a manner that differs from the prescribed company workflow as defined by the ERP system being implemented.

2. How can an AI Machine Learning model classify and identify ERP Usage Gaps?

The focus of this question will be to study the feasibility of using artificial intelligence techniques to systematically investigate ERP usage patterns, automate the identification of instances of inefficiency, and utilise the machine learning model's classification capabilities as a basis for determining the factors that contribute to the under-utilisation of ERP systems within the retail organisations in Bahrain.

3. Is there a correlation between ERP usage gaps and the performance of retail organisations in Bahrain?

This question aims to investigate whether there are correlations between levels of ERP use, such as how well the utilised modules of the ERP system have been utilised by organisations and their resulting levels of operational efficiency, that is organisation's total productivity, profitability, mall productivity, etc., and ultimately, the degree to which each organisation is successful.

In total, these 3 research questions provide the study direction by linking methods for evaluating ERP usage and analysing the relationship between productivity and ERP use, via AI technologies, within the Bahraini retail industry.

1.6 Research Objectives

To address the research questions, this study established a structured analytical framework to examine ERP underutilisation in retail organisations in Bahrain.

Prior to this project, there has been evidence produced by other studies (Aalst, 2016) to show that ERP systems are, in general, utilised as a way of integrating and optimising business processes, but that due to issues that arise after the system's implementation and lack of proper monitoring of how the systems are being used, most organisations do not experience the anticipated benefits of implementing an ERP system.

A significant amount of research conducted in the Middle East has revealed a necessity to create more formalised, consistent, systematic evaluation mechanisms to ensure that organisations are effectively utilising ERP systems to improve the performance of their businesses (Shaiti, 2020).

Recent advances in the areas of process-based analytics and machine learning technologies have also demonstrated that they have the ability to detect behavioural deviations from typical patterns of behaviour and can uncover underlying behaviours within enterprise systems (Chawla, 2019); (Aalst, 2016). However, there is limited empirical evidence from studies that have quantitatively assessed ERP system usage and implementation within the retail sector in Bahrain using these analytics-based methods.

Therefore, in direct alignment with the research questions, this project seeks to identify measurable ERP usage indicators, design and develop an AI/ML-based classification model to detect ERP usage gaps, and examine the relationship between ERP utilisation levels and organisational

performance. By translating the research questions into clear and measurable objectives, the study aims to provide both academic contribution and practical value to retail organisations seeking to improve ERP effectiveness. The project has some goals to help with the problems it is trying to solve.

1. To identify and define key ERP usage gap indicators within retail organisations in Bahrain.

The first goal is to find out what is going on with ERP systems in companies in Bahrain. This means looking at things that can be measured to see how well companies are using their ERP systems. This objective aims to identify measurable indicators of ERP underutilisation, including partial module usage, deviation from intended workflows, and inconsistent system adoption across departments, where they are only using part of the system, and where they are doing things in a way that's not what the system is meant for. By looking at these things, we can turn the idea of not using ERP systems into something that we can measure and understand.

2. To design, develop, and train an AI/ML-based analytical model for ERP gap detection.

This objective focused on applying a machine learning-based analytical model to analyse ERP usage data and detect patterns of underutilisation. The model was developed using survey data collected from retail organisations in Bahrain and was used to classify usage behaviour and identify areas where ERP systems are not effectively utilised. This approach replaces manual evaluation methods with a more systematic and data-driven analytical process.

3. To evaluate the model's accuracy and examine its implications for organisational performance.

The final objective evaluated the performance of the developed model and examined its implications for organisations. The analysis assessed how accurately the model identifies ERP usage gaps and explored the relationship between ERP utilisation levels and organisational performance. By doing so, the study demonstrated how data-driven and machine learning-based

approaches can support organisations in improving ERP usage and enhancing operational efficiency in the retail sector, as supported by (Shaiti, 2020) .

Overall, these objectives collectively contributed to developing a structured and context-specific solution tailored to retail organisations in Bahrain, utilising AI/ML techniques to address ERP underutilisation challenges.

1.7 Scope and Limitations

This section defines the scope and limitations of the study. It clarifies the boundaries within which the research was conducted and highlights the key constraints that may influence the interpretation of the findings.

1.7.1 Scope of the Research

This research is intended to examine the effect of applying artificial intelligence to perform the gap analysis within ERP systems used by retailers operating in the Kingdom of Bahrain. The retailing sector has been chosen for the study because it relies heavily on integrated ERP modules such as inventory management, sales processing, procurement, and reporting.

The aim of the study is to explore the employees' ERP usage patterns and to identify the potential usage gaps in environments where the system is already implemented. The main focus is on identifying the differences between the ideal use of the system functionality and its actual usage during daily operations, as well as examining how these usage gaps relate to operational performance indicators and using WEKA to develop predictive models that estimate performance outcomes based on ERP utilization patterns

Geographically, the research is confined to the retail organizations operating within the Kingdom of Bahrain. Therefore, the findings will reflect the performance expected advances in the context of our data sources only.

Methodologically, the research adopted a quantitative data approach supported by both primary survey data and a secondary dataset to generate measurable ERP usage indicators, which were

subsequently analysed using machine learning techniques in WEKA to develop the analytical model for gap detection.

1.7.2 Limitations of the Research

This study faced several limitations that may affect the research outcomes. The first limitation is the difficulty of gaining access to system-related data due to business confidentiality concerns. This reduces the size and diversity of the sample, potentially leading to the lack of some important examples, thus reducing the accuracy of the developed AI-model.

In addition to that, in order to develop the AI-based gap analyser, the study focuses only on certain major ERP modules rather than the entire system. This focus may limit the generalizability of the results to other modules or broader ERP contexts.

Finally, the study relies on employee-reported data, which could be influenced by perception biases or memory problems. While constructing data from surveys and case studies helps lower the severity of the problem, some level of subjectivity in the findings may remain.

1.8 Significance of the Study

This section explains the significance of the study from academic and practical perspectives. It highlights the contribution of the research to existing knowledge as well as its relevance to organisations and the broader regional context.

1.8.1 Academic-Significance:

This study contributes to the growing body of research on integrating Artificial Intelligence (AI) within Enterprise Resource Planning (ERP) systems, with specific emphasis on the post-implementation stage. While much of the ERP literature focuses on implementation and adoption, recent work increasingly stresses the need for continuous evaluation after go-live to ensure that ERP investments deliver sustained organizational benefits (Domagała, et al., 2021). By shifting attention from implementation success to utilization effectiveness, this research adds a usage-oriented perspective for assessing ERP outcomes. In addition, the study provides value in the Gulf

Cooperation Council (GCC) context, where empirical research applying machine learning to evaluate ERP utilization and post-implementation behaviour remains limited, particularly within the retail sector (Alkraijia, et al., 2020; Salih, et al., 2022). Therefore, this research contributes by providing empirical evidence from Bahrain and extending AI-driven ERP evaluation research within the GCC region. By proposing an AI-based approach to identify underutilization patterns in retail organizations in Bahrain, the study extends regional Information Systems research and clarifies how AI-based analytics can support long-term ERP effectiveness.

1.8.2 Practical-Significance:

This study provides retail organisations in Bahrain with a structured and data-driven approach to evaluate and improve ERP system usage. In contrast to conventional methodologies, where evaluation is made based on periodic feedback, the proposed machine learning model helps to detect underutilization of ERP features. The findings from this study can be used by businesses to enhance efficiency, report better, and make informed decisions. By identifying any inefficiency, organizations will be able to carry out training and streamline their operations.

1.8.3 Broader Implications:

This study also has broader implications for the Gulf Cooperation Council (GCC) region, as it demonstrates how artificial intelligence and machine learning can be applied to improve ERP utilisation and organisational performance. The approach presented in this research can support digital transformation initiatives and promote more effective use of enterprise systems across the retail sector.

1.9 Chapter Summary

This chapter begins with the Introduction, which explains the role of ERP systems in integrating retail operations in Bahrain and highlights the practical issue that many organisations do not fully utilise ERP capabilities after implementation. It then presents the Purpose of the project, describing the aim of developing an AI-based analytical approach to identify gaps between expected and actual ERP usage using quantitative indicators. The Problem Statement follows, clarifying the nature of ERP underutilization and the limitations of current evaluation practices, such as reliance

on manual checks and limited monitoring. The Background of the Study provides a wider context on why ERP is important in retail, what challenges commonly affect post-implementation ERP usage, and why AI and enterprise analytics can support better evaluation. The chapter then introduces the Research Questions, which specify what the study seeks to investigate regarding measurable usage indicators, AI-based classification of usage gaps, and the relationship between usage and performance outcomes. Next, the Research Objectives translate these questions into clear project goals related to defining indicators, building the AI/ML model, and evaluating performance implications. The Scope and Limitations section defines the study boundaries in terms of sector (retail), location (Bahrain), selected ERP modules, and the data collection approach, while also acknowledging constraints such as data accessibility and possible self-report bias. Finally, the Significance of the Study explains the academic value of focusing on post-implementation ERP utilisation and the practical value of providing organisations with a structured approach to improving ERP effectiveness and decision-making.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the academic foundations relevant to the study and positions the research within existing knowledge on Enterprise Resource Planning (ERP) systems and Artificial Intelligence (AI) in enterprise analytics. The literature review is structured into two main domains. The first domain examines ERP systems, focusing on post-implementation evaluation, system usage, and organisational performance. The second domain explores the role of Artificial Intelligence and machine learning techniques in analysing structured organisational data and supporting decision-making.

In addition, this chapter introduces the theoretical frameworks that guide the study. The Technology–Organisation–Environment (TOE) framework and the DeLone and McLean Information Systems Success Model are widely used models for understanding ERP usage and its influence on organisational performance. These frameworks provided a structured basis for defining ERP usage gaps and assessing system effectiveness within retail organisations in Bahrain. They supported the study by (1) structuring the key factors that influence ERP utilisation and (2) clarifying how ERP usage relates to organisational benefits, thereby strengthening the interpretation of the machine learning–based results.

2.2 Theoretical Foundation

This study is guided by two established Information Systems theories that support the investigation of ERP usage gaps in retail organisations in Bahrain. The Technology–Organisation–Environment (TOE) framework helps explain why ERP usage differs across organisations by examining influencing factors from three perspectives. The DeLone and McLean IS Success Model helps explain how ERP usage relates to outcomes by linking system usage to organisational benefits. Together, these frameworks provide a strong conceptual basis for collecting survey data, defining usage indicators, and interpreting the results of the machine learning model. This combined use is

supported in ERP research, where TOE is applied to explain continuance/usage behaviour, and IS Success logic is applied to connect usage with benefits (Jo & Bang, 2023).

2.2.1 Technology, Organisation, Environment (TOE) Framework

The (TOE) framework explains how the adoption, implementation, and continued effective use of technology are shaped by three contexts: technology, organisation, and environment. Although TOE is often applied to adoption decisions, it is also useful for understanding post-implementation utilisation, because continued and effective ERP usage depends on the system's characteristics, the organisation's readiness, and external pressures. Recent ERP studies have extended TOE to explain ERP continuance intention and post-adoption usage patterns (Jo & Bang, 2023).

TOE helps interpret differences in ERP usage patterns reported by retail organisations in Bahrain. The survey instrument included measurable indicators under each TOE dimension, and these indicators were used as explanatory variables for identifying usage gaps and underutilization. By structuring ERP usage indicators under the technological, organisational, and environmental dimensions, the TOE framework provides a structured basis for identifying where usage gaps occur and why they persist within retail organisations in Bahrain. In this study, these indicators are collected through structured survey items, for example, agreement scales, used to explain variations in reported ERP utilisation. This supports the interpretation of the model results by linking weak utilisation patterns to contextual drivers (technology/organisation/environment) rather than treating underutilization as an isolated outcome.

A) **Technological Context (T):** The technological context refers to the characteristics of the ERP system itself and the supporting technology infrastructure. These factors influence whether employees find the ERP system practical and efficient to use during daily operations. In the context of this study, the technological dimension includes aspects such as:

- The complexity of the interface: With a higher number of complicated user interfaces and/or processes, users may decide against using a function/module, ultimately resulting in lower utilisation.

- The consistency and reliability of the ERP system: With a lack of consistency and reliability within an ERP system in terms of slow response time, downtime, and errors, users may lose confidence in the system and therefore will not use the system consistently.
- Availability of valuable functionality within the ERP system: Reporting, dashboards, analytic tools, and automated functionality that meet a user's requirements will help promote dependence on ERP systems and promote consistency in usage.
- Integration of an ERP system with other systems (positive enabler): Tools such as point-of-sale systems, barcode scanners, and supplier systems enable users to perform their daily tasks more easily than if the ERP system did not have these types of integrations; users are therefore more likely to consistently use an ERP system with these types of integrations. This integration reduces duplicate work and improves workflow continuity.

B) **Organisational Context (O):** The organisational context refers to internal conditions within the organisation that affect how ERP systems are used. Even if the technology is strong, ERP usage may remain low if organisational support and readiness are limited. In this study, the organisational dimension includes:

- Training and user support: Inadequate training leads to low confidence in users and thus an inconsistent use of the ERP system. Adequate support and training build users' confidence and show them how to effectively utilize more modules.
- Management support and enforcement: When management provides support and guidelines for using ERP systems and continues to monitor for compliance, it tends to increase ERP system usage across departments. Management may also be able to provide clear governance rules, as well as through regular follow-up, which will reduce inconsistent use across departments.
- Staff skills and experience: The capabilities of employees determine whether or not they are able to effectively utilize advanced modules. Higher digital skills and ERP experience are associated with more efficient and consistent usage.
- Process standardization: When departments use different procedures, the workflows are more likely to deviate, leading to inconsistent ERP utilization.

C) **Environmental Context (E):** The environmental context refers to external pressures that influence ERP usage and the motivation to improve it. Retail organizations may be pushed to optimize ERP use due to competition, regulations, or supply chain requirements. In this study, the environmental dimension includes:

- Competitive pressure: Retailers operate under competition and must respond quickly and accurately to report results, which increases the importance of effective ERP use.
- Supplier and customer requirements: Suppliers and customers may have business or compliance requirements requiring retailers to provide sufficient documentation, traceability, or timely reporting.
- Regulatory requirements: Retailers may have compliance obligations requiring them to maintain accurate records and produce reports at the request of regulators using enterprise systems.
- Marketplace expectations: Retail customers expect retail businesses to provide quick service, offer prices that accurately reflect the cost of providing the service, and have goods available when needed, which depends on efficient internal operations that ERP systems can support.

2.2.2 DeLone and McLean IS Success Model

Through the DeLone and McLean Information Systems Success Model, the utilisation of a variety of metrics to determine if IS effective within an entity is possible. The metrics utilised to measure the success of IS include System Quality, Information Quality, Service Quality, usage, User Satisfaction, and Net benefits. Based upon the relationship between system and service quality, it is suggested that an improved level of performance will result in increased usage, user satisfaction, and ultimately improvements in the overall success of the organisation. Recent empirical studies continue to validate the model's logic by examining how quality dimensions influence Use/Usage and how usage contributes to decision-making and organisational outcomes (Lutfi, et al., 2022).

In this study, the DeLone and McLean model is used to support the relationship between ERP usage patterns and organisational performance outcomes in retail organisations in Bahrain. Since the focus of this project is ERP underutilization and its implications, the model is applied primarily through two key dimensions: Usage and Net Benefits.

A) **Usage (System Use):** The Usage dimension refers to how frequently and how effectively users engage with the ERP system in daily operations. In ERP environments, usage is not limited to logging in; it also reflects whether users rely on ERP modules consistently to complete transactions and whether system utilisation is balanced across departments.

In this study, ERP usage is operationalised through quantitative indicators that reflect the level of engagement with ERP modules and the consistency of system use in daily operations. These indicators are designed to capture variations in utilisation across departments and to support the identification of underutilization patterns within retail organisations. This aligns with DeLone and McLean because "Usage" is treated as a key success outcome that reflects whether ERP is actually embedded in routine work practices.

B) **Net Benefits (Organisational Benefits):** The Net Benefits dimension refers to the positive outcomes the organization gains from ERP use. In ERP contexts, net benefits are typically reflected in operational improvements such as efficiency, accuracy, visibility, and faster decision-making.

In this study, net benefits are operationalised through quantitative indicators that reflect the organisational outcomes associated with ERP usage. In retail contexts, these outcomes commonly relate to improvements in operational efficiency, accuracy of information, visibility of operations, and decision-making support. This supports the evaluation aim of the study because underutilization is expected to reduce the organisational benefits that ERP systems are designed to provide.

The DeLone and McLean model supports the logic of this project by connecting Usage to Net Benefits. Within this project, the machine learning model used the defined usage indicators to classify ERP utilisation levels. The study then examines how these utilisation levels relate to reported organisational outcomes, consistent with the relationship between Usage and Net Benefits in the model.

2.2.3 Framework Justification and Selection

This study adopts both the TOE and the DeLone and McLean Model because they play complementary roles. TOE provides an explanatory lens to understand factors that influence ERP utilisation (why usage differs), while DeLone and McLean provide an evaluative lens that links utilisation to organisational outcomes (what benefits result from usage). Using both frameworks strengthens the study by enabling a more complete interpretation of ERP usage gaps: contextual drivers are examined using TOE, and performance implications are assessed using the Usage → Net Benefits logic of DeLone and McLean. This combined approach is consistent with prior ERP research that integrates contextual and success/effectiveness perspectives to understand continued ERP use (Jo & Bang, 2023).

2.2.4 Comparison of Theoretical Frameworks for ERP Research

While the Technological-Organizational-Environmental (TOE) framework provides a comprehensive lens for institutional adoption, ERP research often benefits from a multi-perspective approach. Table 1 compares the primary frameworks used to investigate system usage and performance gaps, highlighting their specific applicability to the GCC retail context.

Table 1 : Comparative Analysis of Theoretical Frameworks in ERP Research

Dimension	TOE Framework	UTAUT2	Institutional Theory	DeLone & McLean
Primary Focus	Contextual factors (Tech, Org, Env)	Individual user acceptance & behavior	Social/Political pressures (Isomorphism)	System, Info, & Service Quality
Level of Analysis	Organizational/Firm level	Individual/Employee level	Institutional/Industry level	Product/Outpu level
Usage Gap Explanation	Attributes gaps to lack of infrastructure or environmental pressure.	Attributes gaps to low "effort expectancy" or "habit."	Attributes gaps to "mimetic" behavior (copying others without internal fit).	Attributes gaps to poor data quality or interface friction.

GCC Context Applicability	High: Captures specific local regulatory/market pressures.	Medium: Focuses on personal motivation rather than firm-wide policy.	High: Explains adoption based on cultural prestige or peer trends.	High: Relevant for assessing localized Arabic interface quality.

2.3 ERP Systems in Retail

ERP systems have become foundational technologies within retail organisations, supporting the integration of operational processes and enabling data-driven decision-making. In line with the objectives of this chapter, this section reviewed the academic literature on ERP systems in retail, with particular emphasis on post-implementation usage, system effectiveness, and their relationship to organisational performance.

ERP systems are designed to integrate core business functions into a unified platform that centralises data and standardises workflows. According to (M.A. Abd Elmonem, 2017), ERP systems collect, integrate, and distribute data across functional units such as inventory, finance, procurement, and sales, thereby reducing information silos and enhancing organisational coordination. In retail environments, where operational processes are highly interdependent, such integration is essential for maintaining inventory accuracy, financial consistency, and timely reporting.

2.3.1 Core ERP modules

Retail ERP systems typically consist of several core modules, including:

- **Inventory-Management**: The inventory management module enables real-time stock monitoring, automated replenishment, and demand tracking. Effective use of this module supports improved stock accuracy and reduces both overstocking and stockouts. However,

underutilization, such as relying on manual spreadsheets instead of system-based tracking, may reduce these benefits.

- **Supply Chain Management (SCM):** The module supports procurement coordination, supplier management, and distribution processes. ERP-enabled supply chain visibility enhances operational responsiveness and reduces delays across retail networks.

- **Financial and Reporting Modules:** Reporting and financial modules provide analytical capabilities that transform transactional data into actionable insights. (Maruf, 2025) highlights that ERP-integrated systems enhance forecasting accuracy, financial visibility, and operational optimisation in retail environments. When fully utilised, these modules improve managerial decision-making and performance monitoring.

However, literature indicates that organisations often use only basic ERP functionalities, leaving advanced analytical and reporting capabilities underexploited. This underutilization directly relates to the "system use" dimension of the DeLone and McLean model and raises concerns regarding unrealised net benefits.

2.3.2 Benefits of the ERP System

The literature consistently identifies three primary benefits of ERP systems in retail organisations: system integration, operational efficiency, and improved data accuracy and decision quality. However, the realisation of these benefits depends largely on the extent of post-implementation system use.

1. System Integration

One of the most significant benefits of ERP systems in retail environments is system integration. ERP platforms consolidate multiple business functions, including inventory management, sales processing, procurement, finance and reporting, into a unified information system. The integration will help avoid data silos and ensure that all departments work with synchronized and standardized information. According to Abd Elmonem (M.A. Abd Elmonem, 2017), ERP systems enhance data sharing across enterprise functions and improve communication between departments by

centralising information flows. In retail organisations, where inventory data, sales transactions, and supplier coordination must be aligned in real time, integration reduces duplication of effort and minimizes inconsistencies in reporting.

Furthermore, (Maruf, 2025) emphasises that integrated ERP systems strengthen cross-functional visibility, allowing financial, operational, and supply chain units to operate within a single coordinated framework. This unified structure enhances transparency and supports more consistent managerial oversight.

In light of the Technology-Organization-Environment (TOE) framework, integration represents a technological characteristic that influences effective utilization. When ERP systems are perceived as compatible, reliable, and capable of connecting multiple business processes, employees are more likely to use the system consistently. Therefore, integration not only improves operational alignment but also strengthens the technological foundation required for sustainable ERP usage (H.O. Awa, 2017).

2. Operational Efficiency

ERP systems significantly improve operational efficiency by automating routine business processes and standardising workflows across retail operations. Automation reduces manual intervention in tasks such as order processing, stock updates, financial reconciliation, and report generation.

Abd Elmonem, (M.A. Abd Elmonem, 2017) reports that ERP implementation contributes to reduced lead times, improved productivity, and lower operational costs through enhanced coordination and streamlined workflows. By centralising transactional processing and eliminating redundant manual entries, ERP systems minimize human error and accelerate business operations.

More recent empirical evidence by (J.P. Kallunki, 2017) indicates that ERP systems positively influence organisational performance by improving process control, internal reporting accuracy, and operational transparency. Their findings suggest that firms utilizing ERP systems effectively experience measurable improvements in efficiency due to enhanced information integration and standardized procedures.

In retail environments, operational efficiency is particularly critical because high transaction volumes require rapid data processing and accurate stock management. When ERP modules are fully utilized, retailers can automate replenishment cycles, synchronize warehouse updates, and reduce delays in financial reporting. However, when employees rely on manual spreadsheets or external tools instead of ERP functionalities, these efficiency gains may not be fully realized.

From the DeLone and McLean IS Success Model perspective, operational efficiency reflects the realization of "net benefits," which depend on the degree of consistent and meaningful system use (Müller, 2020).

3. Data Accuracy and Decision Quality

Another critical benefit of an ERP system in retail organizations is improved data accuracy and enhanced decision quality. ERP systems provide a centralized database that ensures a single source of truth for financial, inventory, and sales information. This reduces discrepancies between departments and enhances reliability in managerial reporting.

(Maruf, 2025) highlights that ERP-integrated systems improve forecasting accuracy, financial consolidation, and operational monitoring in retail businesses. Accurate and timely data enables managers to make evidence-based decisions regarding pricing strategies, inventory replenishment, and supplier performance.

Supporting this, (J.P. Kallunki, 2017) demonstrates that ERP adoption enhances internal information quality and strengthens management control mechanisms, ultimately contributing to improved decision-making performance. Their study confirms that organizations with higher ERP utilization levels achieve stronger financial and operational outcomes.

Within the DeLone and McLean IS Success Model, improved data accuracy contributes to both "information quality" and "net benefits." Reliable information enhances user satisfaction and supports better organizational performance outcomes. In retail contexts, accurate inventory records

reduce stock discrepancies, improve customer service levels, and strengthen supply chain coordination.

However, these benefits depend heavily on the degree of ERP utilization. If reporting modules or advanced analytics features remain underused, the potential improvements in decision quality may not materialize. Therefore, measuring ERP usage levels becomes essential for evaluating whether expected organizational benefits are fully achieved.

2.4 ERP Implementation Challenges in the Middle East and Bahrain

The implementation of ERP in the Middle East, such as Bahrain, faces a range of organizational and cultural challenges, as well as the well-known technical and managerial obstacles described in the global literature of these types of implementations. Also, although ERP systems are implemented in order to bring the various functions of a business together to become more efficient, many organisations in the Middle East do not see the expected results after implementation.

In Bahrain specifically, (Kamhawi, 2008) found that even when organisations adopt ERP systems to improve their integration and to make them more competitive, they often face obstacles to their successful implementation, including: the readiness of the organisation to implement ERP; the skills of the employees; and the support from their management. Kaamhawi's study indicates that organisations have strong motivations for the adoption of ERP, but, as a result of human and structural limitations, they do not see the same level of performance after the implementation of ERP as they had anticipated.

(Coşkun, et al., 2022) Conducted a systematic review of ERP failure studies and confirmed that most ERP projects fail to meet expectations, largely because organisations underestimate the complexity of organisational change associated with ERP implementation. (S.A. Menon, 2019) identifies critical ERP challenges spanning technological, organisational, and human resource dimensions, with human resource-related challenges being the most significant. Supporting these findings, recent studies by (Light, 1999), (Corbett, 2007) and (Nelson, 2001) argue that ERP success depends on effective system installation, comprehensive training, management commitment, and alignment between business processes and ERP functionalities.

2.4.1 Common Barriers in the Middle East

Even though Enterprise Resource Planning (ERP) systems are designed to improve operational efficiency and integration across an organisation, in practice, these systems experience obstacles such as significant time delays and project failure. As indicated by multiple studies, these challenges are exacerbated when there is not sufficient organizational readiness and cultural fit.

According to (K. Al-Assaf, 2024) The various phases of the ERP implementation process rely heavily on change management and user involvement. He goes on to say that resistance during the early stages of the project may result in negative outcomes for the overall success of the ERP system. Even (Coşkun, et al., 2022) used a systematic approach to map all the literature available on ERP failure and concluded that the most common reasons for the failure of ERP systems are user-related and include resistance to change and inadequate training.

According to (Corbett, 2007); (Nelson, 2001), critical success factors in overcoming ERP implementation barriers include strong top management support, comprehensive user training, effective communication, and alignment between business processes and ERP systems. (M.I. Mahraz, 2020) emphasise that organisational readiness and continuous support are essential to achieving ERP success. If these factors are not present, an organisation can have a technically functioning ERP system, but the organization to not realise the benefits that were anticipated when the ERP system was implemented.

From a socio-cultural perspective, (Moss, 2018) states that the dynamics of social and cultural factors greatly influence the decision to adopt ERP systems. Organizational culture, employee communication patterns, and attitudes of employees towards both authority and technology may either facilitate the successful implementation of ERP systems or impede that success. This becomes even more pertinent in environments such as those found in the Middle East, where organisational structures and established work processes may not be compatible with established ERP standardised processes.

1. Resistance to Change

Change resistance stands out as a key obstacle in ERP rollouts. Shifting to an ERP usually means rearranging how workflows, who does what, and how reports are handled - these shake things up for staff. People might resist because they worry about losing their jobs, how updates mess with daily habits, heavier tasks, or feeling unsure about handling newer tools (K. Al-Assaf, 2024); (Coşkun, et al., 2022).

In Middle Eastern companies, top-down control often fuels pushback because workers rarely shape decisions. If people stay on the sidelines while systems are built, scepticism grows. Without buy-in early on, rollout tends to meet resistance or half-hearted adoption. According to (Salih, et al. ، 2022), missing user approval leads to deteriorating ERPs across local markets.

2. Lack of Training

One big issue? Not enough training. Those holes ERP uses and speed right away. Complex software needs clear guidance. Without it, workers get lost in tasks. Mistakes pile up when tools are unclear. Frustration grows fast under those conditions (Coşkun, et al.2022 ،).

One thing (K. Al-Assaf, 2024) pointed out is how good training needs to include both tool knowledge and workflow awareness so workers can adjust well to changes. Several firms across the Middle East only run basic onboarding when systems launch, then drop off. Yet (Salih, et al., 2022) noted back that steady learning plus help after setup matters a lot for building staff trust and lasting ERP performance. When workers lack clear guidance, they often fall back on old habits like paper tracking or separate software - this weakens what the ERP system can do.

3. Cultural Fit

What fits culturally depends on how well a company's everyday way of working matches up with the fixed procedures built into ERP tools. Not every ERP follows regional habits or leadership methods, since many copy standard global approaches instead (Moss, 2018).

Across much of the Middle East and Bahrain, workplace norms tend to value close personal connections, respect for rank, and adaptability in choices. If ERP tools demand stiff processes or standardised steps, friction can follow. (Corbett, 2007) argue that organisational culture and

leadership commitment significantly influence communication quality, user engagement, and overall ERP success.

When culture isn't considered, companies often go too far in adapting tools - adding confusion and expense. Other times, they push rigid formats, which could lower motivation and willingness to use new systems (Coşkun, et al., 2022).

These challenges are often interconnected, where resistance to change, insufficient training, and cultural misalignment collectively influence ERP usage outcomes. Addressing these barriers requires not only technical solutions but also organisational strategies that focus on user engagement, continuous learning, and alignment between system functionality and business practices.

2.4.2 The concept of the ERP Usage Gap (Expected vs. Actual usage)

The term "ERP Usage Gap" describes how a business's perception of an established Enterprise Resource Planning (ERP) system differs from the reality of that ERP being used in practice. Worldwide companies spend large sums of money installing an ERP system, and underutilization can prevent companies from achieving the full potential of the ERP system, resulting in minimal or negative performance. There are several factors that need to be evaluated to create an understanding of the ERP Usage Gap, such as issues related to the technology being implemented, user/purchaser acceptance, and how well the ERP system has been integrated into the organisation (S.A. Menon, 2019).

A significant difficulty with implementing ERP systems is achieving and maintaining overall end-user acceptance. Research indicates that although ERP systems are designed to assist with managing various functions such as finance, human resources, supply chain and customer relationship and to give organizations consolidated information among each area of the business, numerous organizations experience ongoing challenges with users accepting and utilizing the ERP system after being deployed because they have not accepted it in its full capacity, according to research. The Technology Acceptance Model suggests that perceived

usefulness and ease of use have a substantial influence on the user's intention to use and the utilisation of ERP systems. Additionally, the user's attitude and overall satisfaction with the system have a direct relationship with their adoption of ERP systems and the continued utilisation of those systems (Tanis, 2000); (Salam, 2004). Furthermore, empirical studies using UTAUT in ERP systems have established that performance expectancy, effort expectancy, and facilitating conditions have a significant relationship with user acceptance and user behaviour related to using ERP systems (M.R. Mohamad Zain, 2023). In addition, the literature supports that a lack of automation, coupled with manual configuration efforts made by users in ERP systems, decreases workflow efficiency and real-time decision making, creating "usage gaps" where the end user's optimal process is not uniformly used (Z.T. Butarbutar, 2023).

Expected usage, user behaviour that is assumed and expected for the period planned and realistic in the planning and realisation phases. See Expected system behaviour for more information about what is assumed in this respect in the case of the ERP system. The assumptions of expected system use are based on a general strategy for the company and an assumed function of the ERP system. According to (Olsen, 2021) the post-implementation value-in-use from an ERP system implementation depends on whether companies can use and create value from the dynamic capabilities created by utilising the ERP system; capabilities such as internal integration and adaptability. According to (M.M. Queiroz, 2020), companies will utilise ERP-based digital integration across functions to enhance supply chain planning efficiency and to promote fact-based decision-making. In other words, it is normative and strategic, representing the management’s perceptions of how the system should be used in the company.

Actual usage, on the other hand, reflects how end-users actually behave when interacting with the system after it has gone live. Actual ERP usage often differs greatly from expectations of management based on empirical studies. Employees will often turn to parallel systems, such as spreadsheets, do not use all modules in the ERP system, or have only been trained to use basic transaction features, but do not have any knowledge of how to use advanced analytical tools and automation features. As explained by (M.I. Mahraz, 2020), there are numerous reasons for under-usage of an ERP system, including lack of training, resistance to change, and

misalignment of the business processes with how the ERP system has been configured. (Z.T. Butarbutar, 2023) also indicated that usability issues, limited perception of usefulness, and inadequate support after go-live can also greatly affect ongoing usage of an ERP system. Therefore, the findings suggest that actual usage is based more on user perceptions, the culture of the organisation, and the day-to-day realities of how it operates than it is on the system design.

By measuring how much of an organisation's ERP system is being used against the desired usage levels, the ERP Usage Gap can be identified, which is the difference between the two. The established theory is that while the planned (expected) usage of an ERP system is designed to support an organisation's strategic, standardised, and goal-oriented objectives, the actual (usage) of the ERP system will be predominantly influenced by the context in which it is employed, human behaviour, and organisational characteristics. Diminished return on investment and limited performance improvements will be the result of employees not following optimised workflows on their own or not making use of all integrated modules. For ERP systems to be successful long-term, as stated by (S.V. Grabski, 2019), there needs to be a continual alignment of system capabilities and user behaviours. Therefore, reducing the size of the ERP Usage Gap requires an organisation to employ training, change management, user-experience improvements, and or management reinforcement for as long as required in order to bring actual usage to the level of intended usage.

2.5 Artificial Intelligence in Enterprise Systems

The term 'Artificial Intelligence' originated in 1956 at the Dartmouth Conference, organized by John McCarthy. Although the early AI was rule-based and limited, it sparked decades of research. As the time passes and the computational power increases, this limited rule-based technology has shifted toward systems that can learn from massive data and has huge impact in all aspects of life.

Today, Artificial Intelligence has a growing usage in Enterprise Information Systems (EIS) which changes the way business software works. AI-enabled systems can analyse data patterns and support real time decision making processes within enterprise environment, detect unusual activity and study users' behaviors by performing techniques like process mining, behavioral analytics, machine learning, and anomaly detection to improve the system efficiency and optimize the usage (Zdravković & Panetto, 2022).

2.5.1 Introduction to Process Mining

In the business environment, maintaining strong integration of business processes and ensuring data consistency across the enterprise' incorporation are critical factor for achieving competitive advantage (Hlatshwayo, 2023). As the organization integrates its information systems, organisations are exposed to large volumes of structured process data in a format of dated logs that can give an idea of how the business processes are actually executing.

In the triangle of business processes management, artificial intelligence and data mining, process mining comes as an intersection area between them. It is an analytical approach that uses event log data generated by the enterprise information systems to explore, analyze and improve the business processes (Hlatshwayo, 2023).

According to Hlatsshwayo, process mining goes through several stages as follows:

1. **Process Discovery:** To automatically construct business process models from event logs.
2. **Conformance Checks:** To compare the event logs with some predefined process models to detect deviations.
3. **Process Enhancement:** To improve the existing process models
4. **Predictive Business Process Management:** To use advanced analytics to predict future logs and make decisions accordingly.

When it comes to ERP systems, process mining plays a critical role in assessing the system utilization and identifying the usage gaps. By analyzing the data logs, organizations can easily find

whether their employees are following the supported ERP functionalities or relying on manual work. Also, conformance checks can show where users are not following the intended system processes, while process enhancement can highlight the problems similar to the repeated manual entries. Finally, Predictive process mining can also help us in detecting patterns of improper use, thus, suggesting some kind of lack in training. Consequently, process mining serves as a foundational element for ERP usage gap analysis.

2.5.2 User Behavior Analytics in Enterprise Software

User Behavior Analytics (UBA) is a part of the Behavioral Analytics field that focuses on understanding, analyzing, and predicting user behavior using machine learning techniques. UBA is used in many areas, such as spotting suspicious activities on a network, recommending products to customers, or analyzing customer behavior (Martín, et al., 2021). In enterprise resource planning (ERP) systems, it can be used to analyze users' interactions and identify special patterns in their behaviors, indicating usage gaps.

2.5.3 Anomaly Detection in Enterprise Systems

While most analytics focus on normal operations, anomaly detection identifies events that are different from expected process behavior. It is important in business because anomalies can be a sign of inefficiencies, insufficiently trained employees, or even potential (Nolle, et al., 2019). Importantly, these anomalies can serve as indicators of usage gaps in enterprise systems.

When it comes to ERP systems, an anomaly is usually used to describe a "manual workaround". If the software is too complex or the user is not properly trained, it is very common that the user may bypass the standard system procedures and cause deviations and show up in the event logs (Nolle, et al., 2019).

To demonstrate how AI can actually detect such anomalies, the following sections discuss two case studies from enterprise systems. These examples show how anomaly detection in ERP systems can find manual workarounds, unusual process steps, and parts of the system that are not being used properly.

2.5.3.1 Case Study: SAP

SAP HANA Predictive Analysis Library (PAL) could be a great example to explain how things can be done practically. SAP defines anomalies as data objects that do not comply with the general behavior of the system and often refers to them as "discordant observations" or "exceptions" (SAP, n.d.). In an enterprise context, these exceptions are translated into actionable information related to the system health and process integrity.

In SAP, anomaly detection runs directly inside the SAP HANA database through the Predictive Analysis Library (PAL) so that the data does not need to be moved to another system for analysis. This makes the process is fast and secure, it also allows large amounts of enterprise data to be analyzed efficiently without any duplications (SAP, n.d.).

The SAP HANA framework provides a two-step machine learning approach to identify these gaps:

- **Clustering:** The system groups normal business data into clusters using the K-Means algorithm, based on patterns from past activity.
- **Distance Identification:** The system then identifies data points that are located far from any cluster center and flags them as anomalies.

Technically, SAP's anomaly detection can adjust parameters like **outlier percentage and distance metric** to detect unusual behavior which allows the system to identify employees who bypass standard procedures and identify the possible usage gaps.

However, since SAP is using the K-Mean clustering algorithm, it faces some limitations related to the number of clusters, the value of K is challenging to select, the selection of initial class center and the detection of outliers (Chong, 2021).

2.5.3.2 Case Study: Oracle

Another example of anomaly detection is the Oracle Data Mining toolset. Unlike the SAP toolset, which relies on clustering algorithms, Oracle's approach is based on One-Class Classification, a method well suited to enterprise environments with large amounts of normal data and little to no data describing bad behavior or usage gaps (Oracle, n.d.).

The Oracle toolset uses the One-Class Support Vector Machine (SVM) algorithm for anomaly detection. This algorithm builds a profile of normal enterprise behavior using historical data only and follows a positive security model, where deviations from normal behavior are treated as anomalies.

At the enterprise level, this approach provides a different way to identify usage gaps. First, the Oracle toolset allows organizations to measure how abnormal behavior is by ranking users or entities based on how much their behavior differs from the normal profile. This helps enterprises focus their investigations and take corrective actions where they are most needed (Oracle, n.d.) Second, because the model is trained only on normal data, it is effective at detecting new and previously unseen ways of bypassing ERP system procedures.

However, the Oracle toolset also has its own limitations related to One-Class Classification. Since the model does not compare normal data with clearly defined bad behavior, the definition of normal behavior may become too broad, which can reduce detection accuracy compared to methods that use both normal and abnormal data (Oracle, n.d.).

2.6 Machine Learning Techniques

Machine learning has become a widely adopted approach and is attracting a lot of attention. It is a rapidly growing field of computing algorithms that aims to enable systems to learn patterns from data and their environment, instead of being programmed for every task without external intervention (El-Naqa & Murphy, 2015). This section introduces the topics of supervised learning, unsupervised learning and anomaly detection while providing some case studies related to them.

2.6.1 Supervised Learning (Classification)

Machine learning works with datasets that consist of labelled examples, which can be in the form of binary, categorical, or continuous data (Dridi, 2022). Supervised learning is a training approach in artificial intelligence where the model uses prior knowledge along with labelled input data to learn patterns and predict future outcomes (R. Saravanan & Pothula Sujatha, 2018).

Saravanan and Sujatha (2018) suggest that supervised machine learning techniques can be grouped under two main categories, probabilistic classifiers and linear classifiers, based on the way they predict.

Probabilistic classifiers predict class labels by estimating the probability that an input data belongs to each class. Probabilistic classifiers also can be classified into several types, one of the simplest types is Naïve Bayes. It assumes that all features are independent and uses Bayes' Theorem to calculate the Probability of data belonging to a certain class. Another famous type is Bayesian Networks which identify the relationships between input data and making decisions accordingly. However, Maximum Entropy classifiers work in a different way. They assign weights to features to estimate the most likely. These models are sometimes called generative classifiers because they model how data is generated for each class (R. Saravanan & Pothula Sujatha, 2018).

On the other hand, linear classifiers separate data into different classes using a straight line in the feature space. It sets a clear limit in how well they can handle small, intentional changes in the input, called adversarial perturbations. The accuracy of the classification depends on how far apart the classes are: the closer the classes, the more likely the classifier will make mistakes. This means that carefully designed examples can trick even a well-trained linear classifier (Fawzi, et al., 2018).

Linear classifiers include several common methods for dividing data into different classes. Support Vector Machines (SVMs) draw a boundary that separates two classes in the best possible way and can use kernel functions to handle more complex patterns. While Logistic Regression (LR) estimates the chance that a data point belongs to a particular class, Decision Trees (DTs) divide the data based on feature values. In addition, Random Forests (RFs) improve accuracy by combining the results of many decision trees (R. Saravanan & Pothula Sujatha, 2018).

2.6.2 Unsupervised Learning (Clustering)

In contrast to supervised learning, unsupervised learning is not about predicting a specific thing, it is more about exploring data to find hidden patterns. It does not need labeled examples. Instead, it tries to understand the data by finding unusual structures like groups or outliers (Wong, 2021). Because there are no absolute right or wrong labels, it usually determines the results by how much they help to understand the data or how they improve other tools being used.

Clustering is one of the most widely used unsupervised machine learning techniques. It involves mining data to identify meaningful groups and to gain a better understanding of the typical behavioral patterns exhibited by these groups within a computational psychometrics assessment study (Wong, 2021).

According to Wong (2021), clustering techniques are usually categorized in three main ways:

- **One Dimension vs. Many Dimensions:** monothetic techniques look at only one feature, but most modern versions (polythetic) look at all the data dimensions at once.
- **Top-Down vs. Bottom-Up:** divisive methods start with one big group and break it into smaller ones. On the other hand, agglomerative start with individual points and join them together into larger groups.
- **Hard vs. Soft Boundaries:** In hard clustering, a data point belongs to only one group. However, in soft clustering, a point can be a part of multiple groups at the same time.

As demonstrated in the SAP case study, K-Means is a widely used divisive clustering method that applies hard boundaries. The algorithm requires the number of clusters to be defined in advance and iteratively assigns data points to the nearest cluster center, updating these centers until stable groupings are formed. While this approach is computationally efficient and easy to interpret, it does not automatically determine the optimal number of clusters, requiring prior knowledge or experimentation by the analyst (Wong, 2021).

Hierarchical clustering comes as an alternative clustering technique that offers different strengths. It follows a bottom-up approach by merging the closest data points into larger groups and representing the results as a dendrogram, a diagram that is similar to a tree that shows exactly how these groups are related. Using it, analysts can identify meaningful subgroups within the data. DBSCAN, in contrast, is a method that identifies clusters based on areas of high data concentration. Data points that do not belong to any density region are labelled as noise or outliers (Wong, 2021).

In ERP systems, clustering techniques are particularly useful for sorting thousands of individual actions into behavioral personas. By looking at these groups, organizations can identify which users follow the standard system processes and who does not. This insight helps identify the usage gaps.

2.7 Chapter Summary

This chapter established the theoretical and academic foundations of the study by linking traditional business management perspectives with modern Artificial Intelligence approaches. It examined the problem of ERP underutilization and introduced the concept of the "usage gap" as a central issue addressed in this research. There are two primary theoretical approaches that have been introduced. First, TOE approach has been used to elaborate on different influences that may affect the implementation of ERP through various perspectives, including technological, organizational, and environmental ones. Another theoretical approach introduced is the model suggested by DeLone & McLean that demonstrates a connection between the effective usage of a system and organization's gains that are associated with such usage.

The key concept introduced in this chapter was the ERP usage gap which was defined as the difference between the intended system use by management and the actual usage by employees in daily operations. In the retail context, full ERP utilization is associated with improved inventory management and faster decision-making. However, issues such as limited training, resistance to change, and cultural misalignment in the workplace often lead users to rely on "shadow IT" tools such as spreadsheets. This chapter also explored how Artificial Intelligence and machine learning techniques can be used as diagnostic tools to analyze ERP usage behavior. Process mining was discussed as a method for reconstructing real workflows from system logs, allowing organizations

to identify deviations from intended processes. Moreover, anomaly detection was introduced as a way to highlight unusual or non-standard system behavior that may indicate workarounds or underutilization.

Finally, machine learning approaches such as classification and clustering were reviewed. Classification techniques were used to predict whether users fall into high or low usage categories, while clustering methods were described as a way to group users into behavioral segments, helping organizations better understand usage patterns across departments. This chapter aims to deliver the concept of usage gaps as behavioral errors rather than technical ones. Traditional management oversight is often insufficient to catch these gaps; therefore, the study justifies the use of machine learning to provide a data-driven, objective way to identify underutilization and trigger targeted interventions like specialized training.

Chapter 3: Research Methodology

3.1 Introduction

This chapter presents the research methodology used to achieve the objectives of this study. The study aimed to develop an AI model to identify ERP usage gaps in the retail sector in Bahrain and evaluate how these gaps influence users' continuance intention. The methodology of this study includes the research philosophy, research approach, data collection procedures, and the analytical framework used for designing and evaluating the artificial intelligence model.

Accordingly, the overall objective of this study was to identify the characteristics of ERP underutilization, develop a machine learning-based detection model, and examine the factors influencing ERP usage behaviour and continuance intention. This research was conducted using a quantitative research approach supported by the principles of Design Science Research (DSR), as the study focused on developing and evaluating a technological artefact represented by an AI-based predictive model capable of detecting ERP usage gaps.

In the first stage of the analysis, a secondary case study dataset derived from a published ERP adoption study (Jo & Bang, 2023) was used to demonstrate the implementation of the CRISP-DM methodology and develop the initial machine learning models using WEKA. The dataset included ERP users' responses related to demographic characteristics, system quality, information quality, usability, management support, and continuance intention. Using this dataset allowed the study to test and validate the machine learning modelling process before applying the same methodology to the primary survey data collected from retail organizations in Bahrain.

In addition, the research followed a methodology aligned with the Cross-Industry Standard Process for Data Mining (CRISP-DM).CRISP-DM is a process framework specifically designed for data mining and analytics((Shearer, 2000); (Wirth & Hipp, 2000)).

CRISP-DM organizes the data mining process life cycle into six phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment (Mariscal, Marbán & Fernández, 2010). The following sections describe how each phase of CRISP-DM is applied in this research to develop and validate the AI model for detecting ERP usage gaps and predicting continuance intention.

Similarly, the research philosophy adopted in this study is positivism. Positivism is a research philosophy which suggests that knowledge can be obtained through the observation of real-world phenomena and empirical analysis of measurable data. This philosophy is appropriate for this study because it focuses on quantifiable ERP usage and perception variables, allowing the research to identify patterns that explain ERP continuance intention and usage gaps.

In addition, the study adopts the principles of Design Science Research (DSR) in order to propose a solution to the research problem in the form of a technological artefact. Design Science focuses on designing and developing technological solutions that address real-world problems (Hevner, et al., 2004). In this study, the technological artefact is an AI-based predictive model that identifies ERP usage gaps and predicts users' continuance intention. The model is designed, trained, and evaluated using several machine learning algorithms implemented in WEKA software based on the case study dataset.

The methodological framework of this study, therefore, integrates:

- CRISP-DM for structuring the data mining and machine learning workflow
- Design Science Research for developing the AI-based predictive artefact
- Quantitative data analysis for evaluating ERP usage patterns and continuance intention

In addition, the methodology incorporated several data preprocessing and feature engineering procedures before machine learning modelling. Survey responses were cleaned, transformed into numerical values, and prepared using Microsoft Excel and WEKA. New analytical variables such as GAP_SCORE and MODULE_LEVEL were created to support ERP usage gap classification and improve model interpretability. Reliability analysis using Cronbach's Alpha was also conducted to evaluate the internal consistency of the questionnaire constructs before the modelling stage. This integrated methodological framework enabled the study to analyse ERP usage survey

data and supported the development of a predictive model capable of identifying ERP usage gaps within retail organisations in Bahrain.

The chapter further presents the machine learning environment used in the study, project planning activities, risk management procedures, data collection strategy, validity and reliability assessment, ethical considerations, and descriptive analysis of the collected responses before discussing the modelling and evaluation stages.

3.2 Research Hypothesis

Research hypotheses provide a structured foundation for examining the factors associated with ERP system underutilization within retail organizations in Bahrain. Based on the Technology-Organization-Environment (TOE) framework and the DeLone and McLean Information Systems Success Model, the study proposes several hypotheses to investigate the relationship between organizational, behavioural, and technical factors influencing ERP usage gaps. These hypotheses also support the machine learning modelling process by identifying the variables expected to contribute most significantly to ERP underutilization and continuance intention.

- H1: Information quality significantly contributes to reducing ERP system usage gaps.
- H2: Perceived usefulness (PUS) of the ERP system positively influences ERP utilization and reduces underutilization among retail employees.
- H3: Higher levels of top management support (TMS) increase ERP system adoption and reduce reliance on manual or external tools.
- H4: Active organizational monitoring of ERP activities reduces the likelihood of ERP usage gaps.
- H5: Specialized ERP training positively influences employees' ability to utilize ERP modules effectively.
- H6: Organizational factors demonstrate higher predictive significance in identifying ERP usage gaps compared to technical factors when analysed using machine learning techniques.

- H7: ERP usage gaps negatively influence users' continuance intention toward ERP systems.

The proposed hypotheses were developed in alignment with the research questions presented in Section 1.5. Each hypothesis contributes to examining specific aspects of ERP usage gaps, organizational behaviour, and machine learning-based analysis within the Bahraini retail sector. Table 2 provides the relationship between the research question and the hypothesis used in the study.

Table 2: Mapping Between Research Questions and Research Hypotheses

Research Question	Related Hypothesis	Purpose
RQ1: What measurable indicators can be used to identify ERP underutilization within retail organisations in Bahrain?	H1, H3, H4, H5	To identify measurable organisational and technical indicators associated with ERP underutilization.
RQ2: How can an AI Machine Learning model classify and identify ERP Usage Gaps?	H1, H2, H3, H4, H5, H6	To examine the predictive capability of AI/ML models and determine the most influential ERP gap factors.
RQ3: Is there a correlation between ERP usage gaps and the performance of retail organisations in Bahrain?	H2, H6, H7	To evaluate how ERP usage gaps influence continuance intention and organisational effectiveness.

The mapping shown in Table 2 demonstrates how the hypotheses support the overall analytical structure of the study. It also ensures that the proposed machine learning models and theoretical constructs are directly connected to the main research questions and objectives of the research.

3.3 Research Approach

This study adopted a quantitative research approach, as it enabled the identification of patterns, relationships, and trends in ERP system usage based on structured numerical data. The use of quantitative data also supported the application of machine learning techniques, which helped identify patterns that may not be easily observed through traditional analysis methods.

In addition, the study followed the principles of Design Science Research (DSR), which focused on solving practical problems using analytical and technological approaches. In this study, DSR was applied through the use of machine learning techniques to analyse ERP usage data and identify factors related to underutilisation. The combination of quantitative research and DSR was appropriate for this study, as it supported a structured analysis of ERP usage issues while also providing practical insights for organisations.

3.4 Process Model (CRISP-DM)

To ensure a clear and organized approach to data mining and analysis, this study uses the Cross-Industry Standard Process for Data Mining (CRISP-DM). According to (Fernando Martínez-Plumed, 2021 (Originally appeared online/under review around 2019)) CRISP-DM is the most recognized framework for structuring data science projects. It offers a dependable, repeating sequence of six phases that keeps the analytical process closely connected to research goals.

The six phases of the CRISP-DM framework are defined as follows:

1. **Business Understanding**: This initial phase is the foundation of the project and focuses on understanding the research objectives from a domain perspective. It involves defining the specific problem, the underutilization of ERP systems in the Bahraini retail sector, and translating these organizational goals into a technical data mining problem definition. In this stage, the researcher needs to define the stakeholders and set the success criteria. Contemporary research recognizes that this stage is important because it helps guarantee that value is generated from the project and not just results. (Stanton, 2017).

2. **Data Understanding**: The second phase involves the collection of initial data and actions taken in order to gain familiarity with the data properties. This study, it included reviewing both the secondary case study dataset and the primary survey data collected from retail organizations. This phase requires an exploration of the data to identify quality issues, such as missing values or inconsistencies, and to discover preliminary insights that may suggest hidden patterns of ERP usage. By understanding the "shape" of the data early on, researchers can form more accurate hypotheses regarding how specific variables might influence user

continuance intention (Witten, et al., 2016).

3. **Data Preparation**: It is usually the most demanding process, as data preparation entails all activities necessary for the creation of the final data set from raw data. It includes intense data cleaning, missing data treatment, and selecting relevant attributes for the model. (Stephan Huber, 2019)highlight that modern data preparation must ensure that data is transformed and normalized correctly so that machine learning algorithms, such as those used in WEKA, can process the information without bias or technical error.

4. **Modelling**: At this stage, several machine learning methods are chosen and used on pre-processed data. For this research, the stage involves running the data through analytical classifiers to identify factors contributing to weak ERP usage. This is an iterative process where researchers calibrate parameters to achieve the highest predictive accuracy. (Witten, et al., 2016)note that selecting the appropriate algorithm is essential for understanding the complex relationships between variables like system quality and continuance intention.

5. **Evaluation**: The subsequent stage involves evaluation of the developed models to ensure that the goals of the research are achieved. The purpose is not only to analyze the model but to determine whether it answers the research question. (Fernando Martínez-Plumed, 2021 (Originally appeared online/under review around 2019))emphasize that this phase serves as a critical quality gate, determining if the findings regarding ERP underutilization are robust enough for real-world application or if the process needs to return to the business understanding stage.

6. **Deployment**: The final phase involves the communication of the knowledge gained from the analysis. The insights must be made useful to stakeholders through documentation and actionable recommendations. In the context of this study, deployment focuses on providing strategies for retail organisations in Bahrain to improve ERP effectiveness. (Stanton, 2017)state that successful deployment ensures that analytical results are integrated into organizational decision-making to solve the practical problems identified at the start.

3.5 Machine Learning Environment (WEKA)

This study employed WEKA as the primary machine learning environment for data pre-processing, model training, and performance evaluation. WEKA is a machine learning and data mining platform developed by the University of Waikato and is widely used in academic and research environments for data analysis and predictive modelling (Asha Kiranmai & Jaya Laxmi, 2018).

The use of WEKA was suitable for this study because the research focused on analysing ERP survey data and identifying ERP usage gaps using classification techniques. The software provides a range of built-in machine learning algorithms and preprocessing tools that support the different stages of the CRISP-DM methodology adopted in this research. During the analysis process, WEKA was used to prepare and transform the collected survey data before modelling. This included handling nominal attributes, relabelling class values, filtering variables, and preparing the dataset for machine learning classification. The software also supported iterative experimentation during the modelling stage, allowing different classification models to be trained and evaluated efficiently (Hemlata, 2018).

In addition, the WEKA software had several visualization and evaluation capabilities that made it easier to interpret classification outcomes. Machine learning algorithms such as J48 Decision Trees and Random Forest were implemented through WEKA to analyse ERP usage behaviour and identify the factors associated with ERP underutilization within retail organizations in Bahrain.

Overall, WEKA provided an effective analytical environment that supported the preprocessing, modelling, and evaluation stages of this study and contributed to the development of the AI-based predictive model proposed in this research.

3.6 Project Plan

To apply the CRISP-DM properly, the project followed a structured timeline. It is divided into four primary operational stages, starting from collecting raw facts and pieces of information to extracting insights.

The following table outlines the estimated duration and key activities for each phase of the research:

Table 3: Project plan phases

Phase	***Duration***	***Key Activities***
Phase 1: Data Collection	Week 1-2	Secondary data acquisition Survey distribution and monitoring
Phase 2: Data Preparation	Week 3	Data Cleaning
Phase 3: Modelling	Week 4-7	Algorithm selection iterative model training and parameter tuning
Phase 3: Evaluation	Week 7-8	Performance analysis Validation and documentation

3.7 Risk Management

To help ensure the research is completed successfully, several potential risks were identified. This section discusses the anticipated risks and how they will be managed throughout the project.

The table below highlights the main risks related to this study, along with the measures that will be taken to manage them:

Table 4: Risk management

Potential Risk	Risk Level	Strategy
Limited/Insufficient Data	High	A secondary dataset from an earlier ERP study was included to support and validate the analysis. This reduced the dependence on primary data alone.
Low Response Rate	Medium	The survey was distributed through professional networks and retail industry organizations in Bahrain. Survey design was kept concise to encourage participation.
Low Model Accuracy	Medium	The data was thoroughly prepared, and relevant attributes were carefully selected. An iterative approach was used in WEKA: experimenting with multiple algorithms and adjusting parameters to improve the accuracy and reliability of the results.
Data Quality Issues	Low	During the data preparation stage, the data was carefully cleaned and normalized to handle any missing values or inconsistencies before modelling.

Note:

- **High Impact:** Risks categorized as high impact are those that could fundamentally compromise the study's ability to answer the primary research questions. Such risks would necessitate a significant change in research design or could lead to the cessation of the project if not mitigated.

- **Medium Impact:** These risks may hinder the quality or depth of the analysis but will not prevent the completion of the study. A medium impact risk might lead to delays in the project timeline or require adjustments to the technical parameters of the modelling phase.
- **Low Impact:** Risks with a low impact are considered minor operational challenges. These can typically be managed through standard data processing procedures or administrative checks without affecting the overall scope or validity of the research findings.

3.8 Data Collection and Sampling Strategy

This section details the techniques employed in collecting the primary survey data used in the research, covering the population targeted, the sampling method used, survey distribution, and the response rate. Given the nature of the research, which examines the application of ERP in the retail industry in Bahrain, it is clear that the data collection process was aimed at retail employees using ERP.

This section highlights the determination of the sample size needed for sufficient data that could be used for predictive analysis through machine learning techniques.

3.8.1 Target Population

To ensure the statistical adequacy of the study, the target population was estimated using current labor market data in Bahrain. According to the Employment Skills Platform (ESP Bahrain), the total private sector workforce is approximately 583,958 employees. Based on estimates published by Tamkeen, around 23% of this workforce operates within the retail sector, representing approximately 134,310 employees. In addition, data from the Labour Market Regulatory Authority (LMRA) indicates that large organizations with 100 or more employees account for approximately 40% of the workforce. Accordingly, the relevant target population for this study was estimated at approximately 53,724 employees working in large retail organizations in Bahrain.

Using a 95% confidence level and a 5% margin of error, the minimum required sample size was calculated to be 129 respondents. The study successfully collected and validated 148 responses,

exceeding the minimum requirement and providing a statistically adequate basis for machine learning analysis and predictive modelling.

3.8.2 Sampling Strategy

A convenience sampling strategy was adopted in this study due to accessibility constraints and the absence of a centralized database of ERP users within Bahraini retail organizations. The survey was distributed electronically to a wide range of retail companies in Bahrain, including supermarkets, fashion retailers, pharmacies, and electronics stores. It is estimated that about 125 companies or business organizations received emails and invitations for their participation in the research through the distribution of the questionnaire to the employees utilizing ERP systems in their business operations on a daily basis. In order to increase the validity of the research and encourage organizational participation, the entire process was conducted using a letter of authorization provided by the University of Bahrain. The survey invitation also explained the academic purpose of the study and assured participants that all responses would remain anonymous and confidential.

3.8.3 Sample Size Adequacy and Power Consideration

A statistical power analysis was conducted to evaluate whether the collected sample size was sufficient for the machine learning and statistical analysis performed in this study. According to general recommendations in behavioural and information systems research, a minimum sample exceeding 100 responses is considered acceptable for classification-based predictive modelling and exploratory organisational analysis.

The final dataset consisted of 148 valid survey responses collected from employees working in the Bahraini retail sector. This sample size was considered sufficient to support the supervised machine learning techniques applied in this research, including J48, Random Forest, and Naïve Bayes classifiers.

Furthermore, the use of stratified 10-fold cross-validation enhanced the reliability of the experimental evaluation by ensuring that all observations were repeatedly used for both training and testing purposes. This approach reduced the risk of sampling bias and improved the generalizability of the obtained results.

Therefore, the sample size used in this study was considered adequate to achieve statistically meaningful classification performance and support the research objectives.

3.8.4 Survey Distribution and Response Rate

A total of approximately 125 invitation emails were distributed to retail organizations operating in Bahrain. However, some emails were undeliverable due to inactive addresses, while others did not result in participation. The survey ultimately received 149 responses.

During the data preparation stage, one response was excluded due to incompleteness or invalidity, resulting in a final dataset of 148 valid responses used for the analysis.

The obtained sample size was considered adequate for the purposes of machine learning classification and predictive analysis. Furthermore, statistical power analysis supported the adequacy of the sample size for the selected analytical methods and model evaluation procedures.

3.9 Data Validity and Reliability

In this study, the survey instrument's validity and reliability of the data were carefully considered to ensure the accuracy and consistency of the collected data. The questionnaire questions were drawn based on the constructs usually used in ERP and information systems literature, such as perceived usefulness, perceived ease of use, system quality, information quality, support from top management, ICT skills, and continuance intention. This helped support the content validity of the instrument by ensuring that the questions measured concepts relevant to ERP usage and underutilization.

In addition, the questionnaire was reviewed and refined to improve clarity, consistency, and relevance to the Bahraini retail context before distribution. The bilingual format of English and Arabic also helped reduce misunderstandings and improve response accuracy among participants. To support construct reliability, multiple questionnaire items were used to measure each major

construct. For example, perceived usefulness, ease of use, system quality, information quality, top management support, ICT skills, and continuance intention were each measured using multiple Likert-scale questions. The responses for these items were aggregated using average calculations in Microsoft Excel before being transformed into categorical variables for machine learning analysis.

Reliability analysis was conducted using Cronbach's Alpha to evaluate the internal consistency of the questionnaire constructs. The obtained values exceeded the acceptable threshold of 0.70, indicating satisfactory reliability and consistency of the measurement instrument.

Table 5: Reliability Analysis Result

CONSTRUCT	ITEMS	CRONBACH'S ALPHA
PERCEIVED USEFULNESS	PUS1 + PUS2	0.764
EASE OF USE	PEOU1 + PEOU2	0.743
SYSTEM QUALITY	SYQ1 + SYQ2	0.716
INFORMATION QUALITY	INQ1 + INQ2	0.778
TOP MANAGEMENT SUPPORT	TMS1 + TMS2	0.830
ICT SKILLS	ISK1 + ISK2	0.819
CONTINUANCE INTENTION	COI1 + COI2 + COI3	0.860

The reliability analysis results demonstrated acceptable internal consistency across all questionnaire constructs, as all Cronbach's Alpha values exceeded the recommended threshold of 0.70. This indicates that the measurement items used in the study were sufficiently reliable for

analysing ERP usage behaviour and continuance intention within the Bahraini retail sector. Several data preparation and preprocessing procedures were also performed to improve data reliability before modelling. The data obtained from the surveys were cleaned, coded numerically, and validated to remove incomplete or inconsistencies responses, such as missing data. One invalid response was excluded during the cleaning process, resulting in 148 valid responses used for the final analysis.

Furthermore, feature engineering techniques were applied to improve analytical consistency. New variables such as GAP_SCORE and MODULE_LEVEL were created based on ERP usage behavior indicators. In addition, categorical values were standardized and transformed into nominal classes within WEKA to ensure compatibility with the selected machine learning algorithms.

The preprocessing process also included merging related categories, converting textual responses into binary variables, and normalizing construct classifications using threshold-based transformations. These steps improved the consistency and interpretability of the machine learning models developed in this study. These preprocessing and transformation procedures helped improve data consistency, reduce classification ambiguity, and enhance the overall quality of the machine learning analysis performed in this study.

3.10 Ethical Considerations

This study focused on ERP usage and underutilization in retail organizations in Bahrain, and all collected responses were used for academic purposes only. Before conducting the survey, the purpose of the research was explained to the participating organizations by means of an invitation email. Taking part in the survey was optional, and participants had the right to decide if they wished to fill out the questionnaire or not. It was not necessary to obtain any personal information from the subject, for example, his/her name, phone number, CPR number, or any other information about the company that could be considered confidential. This ensured that the participants remained anonymous in the data collection process.

In addition, the survey invitation included an official letter from the University of Bahrain approved by the project supervisor. This letter was included to add credibility to the research process and encourage organizations participate in the study. Following the collection of the data, the database was checked and cleaned, after which analysis began. The number of invalid responses identified after the dataset was cleaned was one, leaving 148 valid responses. The collected responses were only used by the research team during the analysis process using Microsoft Excel and WEKA software.

In general, the study complied with academic ethics in relation to issues of confidentiality, voluntariness, and the proper management of data collected.

3.11 Data validation (Responses Overview):

The survey collected data from a diverse group of professionals within the retail sector in Bahrain to ensure a comprehensive understanding of ERP system usage and underutilization, and to identify the usage gap in ERP system utilisation in Bahrain. The following descriptive analysis presents the demographic and professional characteristics of the survey respondents and provides an overview of ERP usage patterns within retail organizations in Bahrain. Data was collected and analyzed using machine learning techniques implemented in Weka.

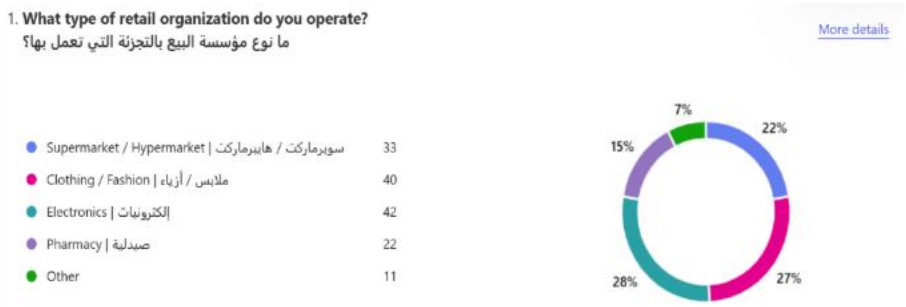

Figure 1: Type of Retail Organization

This chart illustrates the diversity of the retail sectors represented in the study, showing a relatively even distribution across major industries in Bahrain. The Electronics sector constitutes the largest portion of respondents at 28% (42 individuals), followed closely by Clothing / Fashion at 27% (40 individuals). Supermarket / Hypermarket organizations account for 22% (33 individuals), while the Pharmacy sector represents 15% (22 individuals). Only 7% (11 individuals) fall into the "Other" category, ensuring the findings are grounded in specific, high-traffic retail environments.

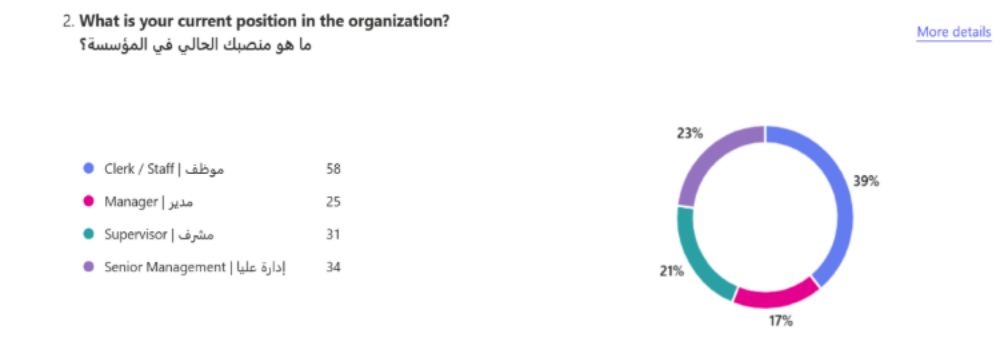

Figure 2: Current Position in the Organization

This figure categorizes the respondents by their professional roles to ensure the data reflects both operational and strategic viewpoints. The majority of participants are Clerk / Staff members, making up 39% (58 individuals) of the sample, which provides insight into daily system interactions. Senior Management represents 23% (34 individuals), offering a high-level view of

system ROI and implementation. Supervisors and Managers account for 21% (31 individuals) and 17% (25 individuals) respectively, completing a comprehensive organizational hierarchy.

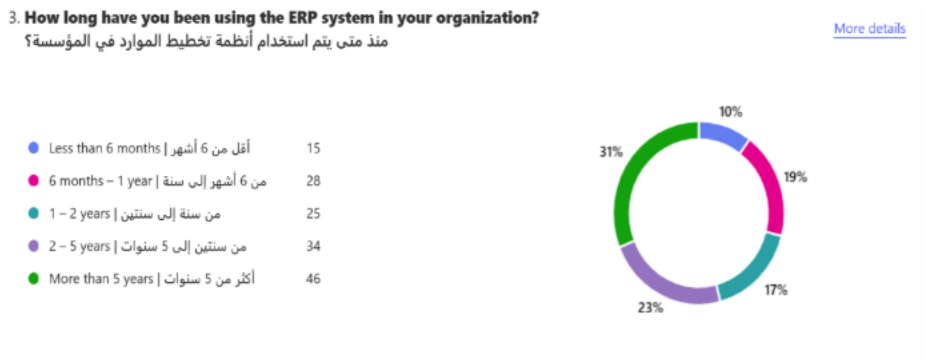

Figure 3: Duration of ERP System Usage

This chart validates the reliability of the data by showing the experience levels of the users. A significant majority of the respondents are seasoned users; 31% (46 individuals) have used their ERP system for more than 5 years, and 23% (34 individuals) have between 2 to 5 years of experience. Users with 1 to 2 years of experience make up 17% (25 individuals), while 19% (28 individuals) have been using the system for 6 months to 1 year. Only 10% (15 individuals) are new users with less than 6 months of experience, ensuring that the feedback on underutilization is based on established usage patterns.

Figure 4: Dominant ERP Systems in Bahraini Retail

This figure identifies the specific ERP platforms currently implemented within the surveyed organizations. By capturing the variety of systems in use ranging from international solutions like Microsoft Dynamics, SAP, or Oracle to localized or niche retail platforms this data allows for the analysis of whether certain software architectures are more susceptible to underutilization. Identifying the "Which ERP system" variable is the first step in determining if usage gaps are system-specific or a broader industry trend in Bahrain.

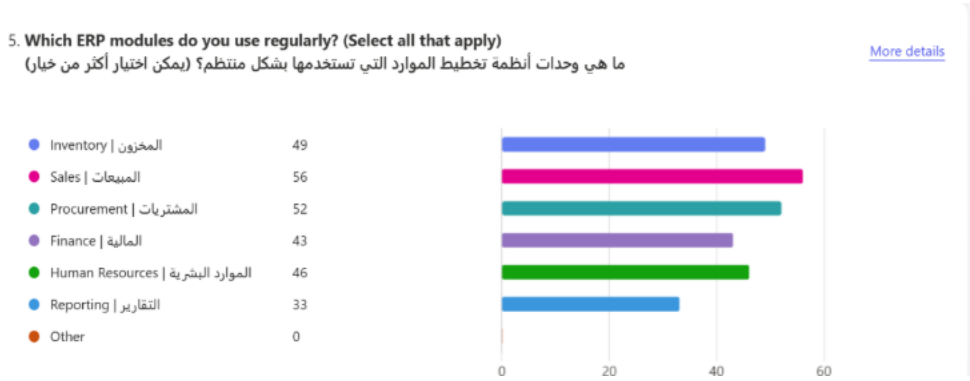

Figure 5: Local Resource Planning (ERP) Modules Used

This figure illustrates the diverse use of ERP modules within the Bahraini retail sector, clearly showing that basic operational modules are the most frequently used. The Sales module recorded 56 entries, followed by the Purchases module with 52 entries, and then the Inventory module with 49 entries. This reflects the operational nature of the retail sector, which relies primarily on the daily flow of goods. In contrast, the Human Resources (HR) and Finance modules show relatively similar usage (46 and 43 entries, respectively), while the Reporting module, at the very end, only uses 33 entries for communication purposes. The high usage of the Reporting module compared to the operational modules suggests a significant underutilization of advanced ERP features, where employees are limited to procedural functions without the tools that support active operations.

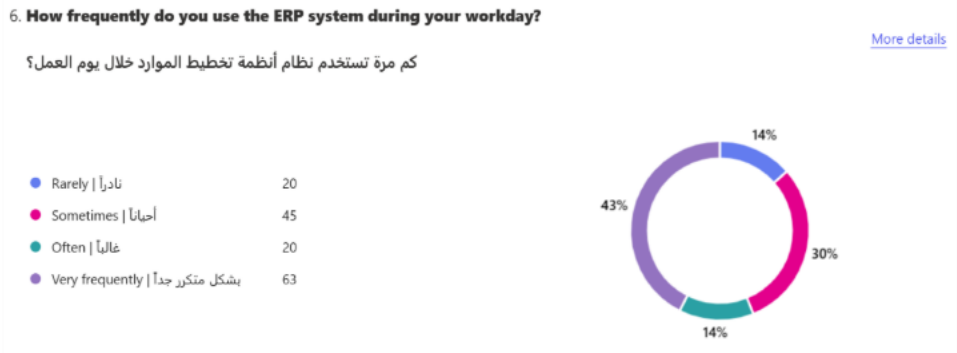

Figure 6: Frequency of ERP System Usage During the Workday

This figure illustrates the intensity of ERP integration into the daily routines of employees within Bahrain's retail sector. The data shows a significant lean toward heavy usage, with 63 respondents (43%) stating they use the system "Very frequently" and 20 respondents using it "Often." This combined majority suggests that the ERP is a core component of the operational workflow. However, a notable segment of the workforce still falls into the "Sometimes" (45 respondents) and "Rarely" (20 respondents) categories. This discrepancy is a key indicator of the "Usage Gap," suggesting that while the system is present, a substantial portion of the staff only interacts with it sporadically, potentially relying on manual processes or external tools for a large part of their shift.

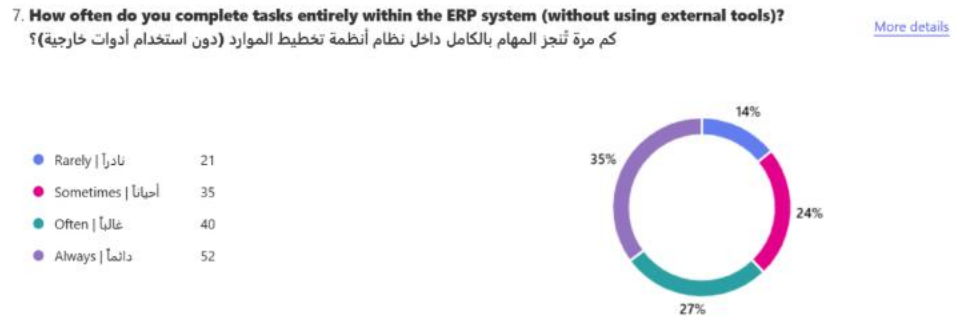

Figure 7: Frequency of Task Completion Entirely Within the ERP System

This figure examines the extent of "system reliance" versus the use of external workarounds, such as manual spreadsheets or paper-based methods. The results show that 52 respondents (36%) "Always" complete their tasks within the system, and 40 respondents do so "Often." While these

figures represent a strong core of integrated users, a significant 38% of the sample (combining 21 "Rarely" and 35 "Sometimes") indicates a frequent departure from the ERP environment to complete their duties. This reliance on external tools is a definitive symptom of underutilization, suggesting that either the ERP system lacks specific necessary functionalities or that employees find the system too cumbersome for certain tasks, leading to fragmented data and decreased organizational efficiency.

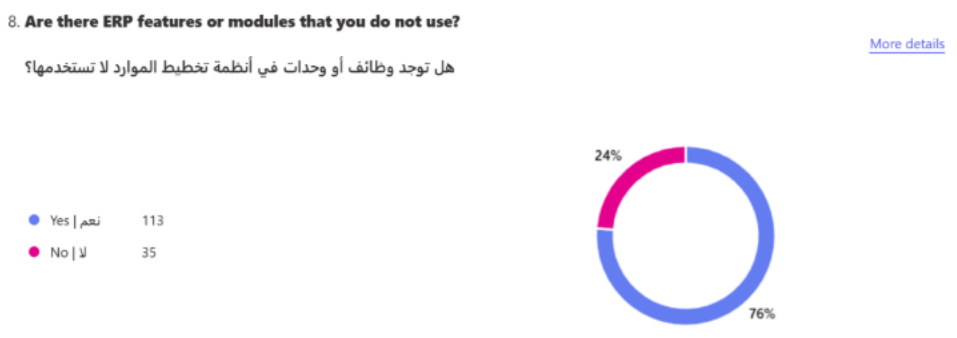

Figure 8: Existence of Unused ERP Features or Modules

This figure highlights a critical finding regarding system underutilization within retail organizations in Bahrain. A significant majority of respondents, 113 individuals (approximately 76%), affirmed that there are specific ERP features or modules that they do not use. In contrast, only 35 respondents (24%) reported utilizing the system in its entirety. This overwhelming "Yes" response provides empirical evidence of a widespread "Usage Gap." It suggests that while organizations invest heavily in comprehensive ERP suites, a vast portion of the software's capabilities remains dormant, representing a missed opportunity for maximizing operational efficiency and return on investment.

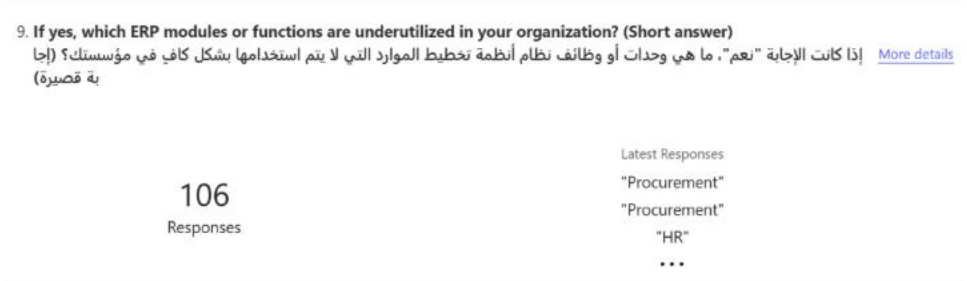

Figure 9: Specific ERP Modules and Functions Identified as Underutilized

This qualitative analysis provides a deeper look into the specific areas where the "Usage Gap" is most prevalent. According to the 106 short-answer responses **,** modules such as Procurement and Human Resources (HR) and were frequently cited as being underutilized. and some answers was Reporting units. suggests that while core retail functions like Sales and Inventory are heavily used, back-office or support modules are often neglected. The repetitive mention of these specific functions indicates that employees may perceive these modules as secondary to their primary roles or that the modules themselves lack the localized features necessary for the Bahraini retail context. This finding is crucial for organizations looking to bridge the usage gap, as it points directly to the departments that require better integration or more targeted training.

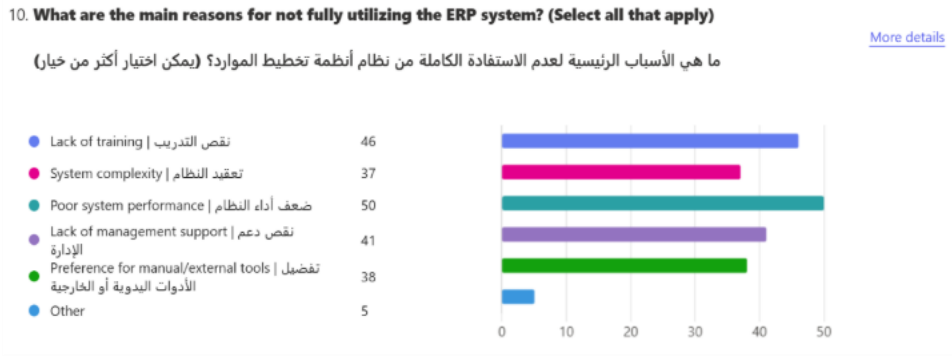

Figure 10: Main Reasons for ERP Underutilization

This figure identifies the primary barriers preventing retail employees in Bahrain from fully leveraging their ERP systems. The data reveals that "Poor system performance" is the most significant obstacle, cited by 50 respondents, followed closely by a "Lack of training" (46 respondents) and "Lack of management support" (41 respondents). Additionally, 38 respondents

admitted a preference for manual or external tools, and 37 found the system too complex. These results suggest that underutilization is not caused by a single issue but is a combination of technical shortcomings and organizational failures. The high frequency of "Lack of training" and "System performance" issues highlights a critical need for organizations to invest in both technical infrastructure and human capital to close the usage gap.

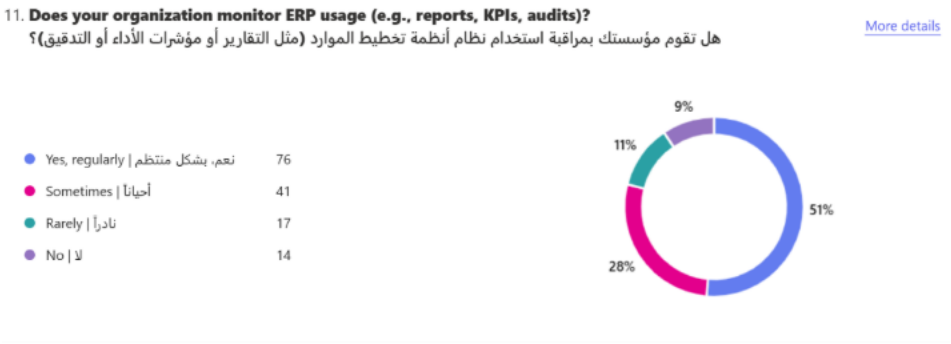

Figure 11: Organizational Monitoring of ERP Usage

This figure examines the level of oversight retail organizations in Bahrain maintain over their ERP systems, which is a key factor in preventing underutilization. The results show that 76 respondents (52%) work in environments where ERP usage is monitored "Regularly," and 41 respondents report "Sometimes." This indicates that a majority of businesses recognize the importance of tracking KPIs and audits to ensure system ROI. However, a significant group of 31 respondents combined reported that monitoring is "Rarely" or "No" (never) performed. This lack of oversight is a primary driver of the usage gap; without active monitoring, underutilization often goes unnoticed by management, allowing inefficient manual workarounds to persist without intervention.

Figure 12: Perceived Usefulness (PUS)

This figure evaluates the extent to which retail employees believe the ERP system enhances their job performance and efficiency. Based on the 5-point Likert scale responses for the items "Using ERP helps me complete tasks more quickly" and "Using ERP improves my productivity," the data reveals a generally positive perception. A majority of respondents "Agree" or "Strongly Agree" that the system provides tangible benefits to their daily output. However, the presence of "Neutral" and "Disagree" responses highlights a segment of the workforce that does not perceive the system as a value-add. This lack of perceived usefulness is a critical factor in the "Usage Gap," as employees who do not see the system as a tool for improvement are less likely to explore its advanced features, leading to basic-level utilization only.

Figure 13: Perceived Ease of Use (PEOU)

This figure assesses the user-friendliness of the ERP interfaces within the Bahraini retail sector, specifically looking at the items "The ERP system is easy to use" and "Learning to use ERP is easy for me." The data reflects a mixed sentiment: while a significant portion of users find the systems accessible, there is a prominent "Neutral" and "Disagree" cluster. This indicates that a subset of employees struggles with the technical complexity of the software. A low score in Ease of Use is

a primary driver of the usage gap, as employees who find the system difficult to navigate will often default to the simplest possible functions or avoid the system entirely, leaving more advanced, productivity-enhancing features untouched.

Figure 14: System Quality (SYQ)

This figure evaluates the technical performance of ERP systems within Bahraini retail organizations, specifically analyzing the metric "ERP functions meet my work requirements." Based on the data, there is a clear divide in user satisfaction: while approximately 50.6% of respondents express positive feedback (29.7% Agree and 20.9% Strongly Agree), a significant portion of the workforce experiences a functional mismatch. Specifically, 25.6% of users actively disagree (10.1% Strongly Disagree and 15.5% Disagree) that the system meets their professional needs, while 23.6% remain neutral. This suggests that for nearly half of the surveyed employees, the ERP system lacks the specific features or flexibility required to perform their daily tasks efficiently. This functional gap is a primary driver of underutilization, as users who find the system's capabilities insufficient are highly likely to seek out external tools or manual workarounds to bridge the gap between the software’s output and their actual work requirements.

Figure 15: Information Quality (INQ)

Based on the survey data regarding whether the "ERP provides useful and relevant information," this figure highlights a significant "information gap" within Bahraini retail organizations. While a combined total of 48.7% of respondents view the system's output favorably (31.1% Agree and 17.6% Strongly Agree), there is a substantial level of uncertainty, with 33.1% of users remaining Neutral. This large neutral block suggests that a significant portion of the workforce does not find the ERP's data particularly impactful or tailored to their specific roles. Furthermore, 18.3% of respondents expressed dissatisfaction (9.5% Strongly Disagree and 8.8% Disagree). In the context of retail, where the relevance of data directly influences inventory and sales decisions, these findings indicate that if the information is perceived as "noise" rather than "insight," employees will naturally underutilize the system, preferring to rely on their own external records or intuition.

Figure 16: Top Management Support (TMS)

This figure evaluates the level of organizational commitment toward ERP systems by measuring how strongly leaders encourage their staff to use the technology. The data shows that while a majority of 51.3% of respondents feel encouraged by their leadership (32.4% Agree and 18.9% Strongly Agree), a significant portion of the workforce lacks this motivation. Specifically, 24.4% of employees expressed a negative sentiment (12.2% Strongly Disagree and 12.2% Disagree), indicating they do not feel pressured or supported by management to utilize the ERP. Another 24.3% remain Neutral, suggesting that management in many Bahraini retail firms may be passive regarding system adoption. This lack of active "top-down" encouragement is a major driver of the usage gap; when leadership does not explicitly champion the system, employees are more likely to view ERP tasks as optional and revert to familiar, manual workflows.

Figure 17: ICT Skills

This figure assesses the self-perceived technical competency of employees, specifically their belief in having "sufficient skills to use ERP effectively." The data reveals that while a majority of 58.7% of respondents feel confident in their abilities (37.8% Agree and 20.9% Strongly Agree), a significant portion of the workforce identifies a personal "skill gap." Specifically, 21.6% of respondents admitted to lacking the necessary skills (12.8% Disagree and 8.8% Strongly Disagree), while 19.6% remained Neutral, indicating they may only feel competent in basic functions. This combined 41.2% of users who are either unsure or clearly under-skilled highlights a major human-capital barrier. Without the technical proficiency to navigate complex software, a large segment of the Bahraini retail workforce is forced into underutilization, likely using only the most basic features while avoiding more advanced modules that could improve operational efficiency.

Figure 18: Behavioral Intention

This figure provides a critical look at the growth potential of ERP utilization within the Bahraini retail sector. The data reveals that while 46.6% of respondents have a positive intention to expand their use of the system (25% Agree and 21.6% Strongly Agree), the most significant finding is the

high level of stagnation, with 36.5% of users remaining Neutral. This large neutral block indicates a "usage plateau," where over a third of the workforce is content with their current level of interaction and feels no drive to explore more advanced features. Furthermore, 16.9% expressed a lack of plan to increase usage (8.1% Strongly Disagree and 8.8% Disagree). These results suggest that unless organizations provide active incentives or advanced training, the "Usage Gap" will persist, as more than half of the employees (the combined neutral and negative groups) do not currently see a personal or professional reason to integrate the ERP more deeply into their daily workflows.

3.12 Chapter Summary

This chapter presented the research methodology adopted in this study to investigate ERP usage gaps in retail organizations in Bahrain. The study employed a quantitative research approach supported by Design Science Research principles and implemented the CRISP-DM framework to structure the data mining and machine learning process.

The chapter also discussed the machine learning environment used in the research, specifically WEKA, alongside the project planning and risk management procedures followed throughout the study. In addition, the chapter explained the data collection process, sampling strategy, validity and reliability measures, ethical considerations, and data preprocessing techniques applied before the modelling stage.

Furthermore, descriptive analysis of the survey responses was presented to provide an overview of ERP usage behaviour, organizational characteristics, and underutilization patterns within the Bahraini retail sector. Overall, the methodology provided a structured foundation for developing and evaluating the AI-based predictive models used in this research.

Chapter 4: Data Analysis by AI Model Development

4.1 Introduction

This chapter presents the data analysis and the application of artificial intelligence (AI) techniques to identify ERP usage gaps in retail organizations in Bahrain. The analysis is conducted using machine learning algorithms implemented through WEKA, following the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework introduced in the previous chapter, as shown in Figure 19.

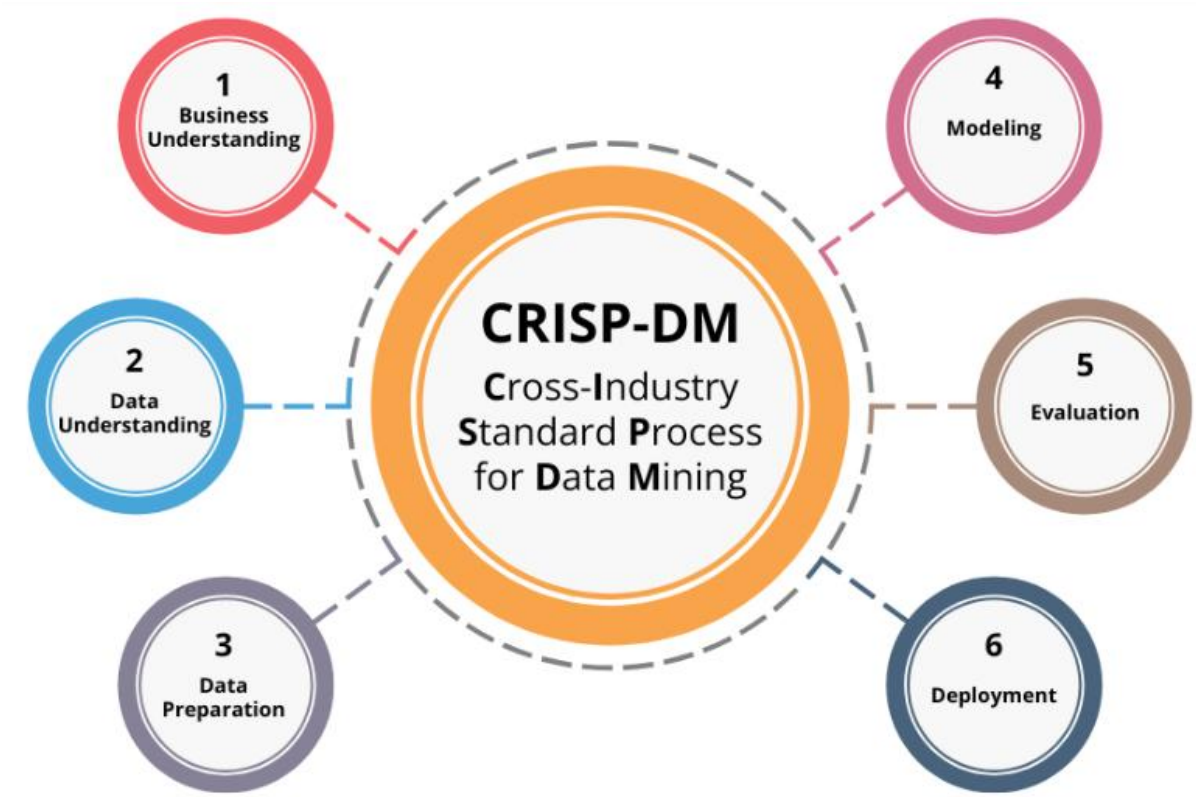

Figure 19: CRISP-DM Framework

The main objective of this chapter is to analyse ERP system usage behaviour, identify patterns of

underutilisation, and examine the factors that contribute to weak or ineffective use of ERP systems. In addition, the analysis considers how these usage gaps may influence users' continuance intention and, consequently, organisational performance. To ensure a structured and reliable analysis, this study applies the CRISP-DM methodology in practice across two different datasets.

The first dataset is a secondary case study dataset obtained from a previously published ERP study. This dataset is used in the initial stage to test the analysis process and the selected machine learning models before applying them to the main research context.

The second dataset is the primary survey dataset collected from retail organisations in Bahrain. The survey dataset is considered the main part of this study, as it is collected from retail organisations in Bahrain and reflects how ERP systems are actually used in this context. This allows the study to directly analyse ERP usage gaps based on real user responses.

The same analysis steps are applied to both datasets in order to maintain a consistent approach when using the selected machine learning models. This also helps in comparing the results between the two stages of the study.

4.2 Case Study Analysis

This section presents the application of the CRISP-DM methodology on the case study dataset. It serves as a preliminary validation step before applying the same methodology to the primary dataset collected from retail organisations in Bahrain.

4.2.1 Phase 1: Business Understanding

The initial phase of the CRISP-DM lifecycle is critical as it defines the project's objectives from a business perspective. According to (Jo & Bang, 2023), Users are more likely to adopt and continue using systems that are both useful and easy to use. ERP systems are considered a core type of Information Systems (IS). The successful use of such a system will always ensure the users' intention to keep on using it. Thus comes the IS success model, according to which, the success of an IS depends on two critical factors: system quality and information quality.

4.2.4.1 ERP Usage Gap

In the competitive retail landscape of Bahrain, organisations invest heavily in Enterprise Resource Planning (ERP) systems to streamline supply chains and customer data. However, a significant "Usage Gap" exists between the system's technical potential and the actual utilisation by ERP users.

In a study by (Jo & Bang, 2023), which integrates the Technology-Organisation-Environment (TOE) framework with the Technology Acceptance Model (TAM) and the IS Success Model, they identified three usage problems gaps for ERP systems:

1. **Technological Gaps:** These involve deficiencies in System Quality (SYQ) and Information Quality (INQ), which are foundational to user trust. A critical component within this level is ICT Skill (ISK). As defined by (Jo & Bang, 2023), ICT skill represents the expertise individuals have in utilising IT components and software. The research highlights that users with lower ICT mastery often feel dissatisfied, creating a "usage gap" where they are less inclined to "delve deeper into the intricacies of ERP systems." Conversely, higher proficiency allows users to "acclimatise to ERP frameworks with more ease," directly increasing their intention to persistently engage with the system.
2. **Organisational Gaps:** This level focuses on internal corporate support. A primary gap is the lack of Top Management Support (TMS), which (Jo & Bang, 2023) Identify the most predominant organisational characteristic for ERP success. Without executive endorsement, the motivation to bridge usage gaps remains low.
3. **Environmental Gaps:** This refers to the external and support-related factors surrounding the system. Specifically, poor Service Quality (SEQ) from technical staff reduces user satisfaction. (Jo & Bang, 2023) argue that high service quality engenders a positive user experience and paves the way for continued system utilisation.

- **Gap Impact on Operation**

This gap manifests as decreased ROI, the persistence of data silos, and a decline in organisational performance. When continuance intention (COI) is low, users often revert to manual "workarounds," rendering the ERP investment redundant and failing to achieve the desired strategic alignment, as described by (Jo & Bang, 2023).

4.2.4.2 Data Mining Goal: Classifying Utilisation Levels

The technical objective of this AI project is to transition from descriptive statistics to Predictive Classification. Using the dataset of 237 respondents, this aligns with the research objective of identifying ERP usage gaps in the Bahraini retail sector using AI-based classification techniques. The project aims to:

- **Classification Task**: Develop and compare robust AI models utilising algorithms such as Decision Trees (J48), Random Forest, and Naive Bayes to classify users into "High" vs. "Low" utilisation categories. This classification is primarily based on Continuance Intention (COI) metrics. To ensure a statistically sound classification, the target variable was discretised into two categories: "Low" (scores ≤ 4) and "High" (scores > 4), reflecting the neutral midpoint of the Likert scale and providing a clear threshold for the "usage gap."

The three models are utilised as follows:

1. Decision Trees (J48): This algorithm identifies the most significant variables (such as PUS1 or TMS3) to create a flowchart-like structure. It is highly valued for its "white-box" transparency, allowing retail managers to see the exact decision path that leads a user to the usage gap.
2. Random Forest: As an "ensemble" method, it generates multiple decision trees and merges their results. This provides superior predictive accuracy and ensures that the feature importance identified is stable across the entire Bahraini retail dataset, reducing the risk of errors from outlier responses.
3. Naive Bayes: This probabilistic model calculates the likelihood of a user falling into the "Low" utilisation category based on their survey inputs. It is particularly effective for baseline comparisons and works efficiently even with smaller datasets, providing a statistical "risk profile" for different ERP modules.

- **Predictive Insight**: The goal is to perform a Feature Importance Analysis to identify whether internal factors (like ICT Skills) or external/organisational factors (like Service Quality and Top Management Support) are the primary predictors of the usage gap. Initial tree results from the J48 algorithm indicate that Perceived Usefulness (PUS1) and Top Management

Support (TMS3) act as the primary root nodes, confirming that these variables are the most critical determinants of whether an employee in the Bahraini retail context intends to continue using the ERP system (Jo & Bang, 2023).

4.2.4.3 Success Criteria

To ensure that our project provides tangible value, the following success criteria have been established:

- Business Success: The delivery of a Gap Matrix that provides actionable recommendations for Bahraini retail managers. Success is defined by the ability to propose specific interventions, such as Module-specific training for General Managers or System infrastructure upgrades for Finance modules.

- Technical Success: The AI model, developed in platforms like Weka, must achieve a statistically significant performance level, such as an accuracy rate of >70% or a high F1-Score, in predicting user adherence patterns.

- Organisational Impact: Successfully establishing a correlation between ERP quality metrics and the user's intent to continue usage. By bridging these gaps, the organisation can ensure that the ERP system acts as a catalyst for, rather than a barrier to, organisational performance.

4.2.2 Phase 2: Data Understanding

In the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, the second phase, Data Understanding, is the stage where the research transitions from business goals to the actual data. Its primary purpose is to familiarise the researcher with the data, identify data quality problems, and discover first insights into the "Usage Gap." By performing initial descriptive analysis and verifying the data's volume and variety, this phase ensures that the dataset is capable of supporting the predictive goals established in the Business Understanding phase ((DSPA), 2022).

To achieve this, the project focuses on the following data-specific tasks:

1. Collect initial data: Acquire the necessary dataset and load it into the Weka analysis tool.
2. Describe data: Examine and document surface properties like the 237 records and field identities.
3. Explore data: Dig deeper through visualisation to identify relationships among variables.
4. Verify data quality: Check for any "dirty" data or missing values to ensure accuracy.

Specifically, the structured description below details the main attributes in the case study dataset, which are used to execute these tasks:

1. **ID:** Unique identifier for each respondent. This attribute is used for record identification only and does not carry analytical meaning, so it was removed in WEKA.

2. **Gender:** Respondent's gender coded numerically (such as 1 = Male, 2 = Female). This attribute is used to explore gender-related differences in ERP usage or perceptions. During the data quality verification, an inconsistency was identified where 3 records were incorrectly coded (as 3 or 4) and subsequently corrected to 1 (Male) to ensure the dataset aligns with the demographics. (Jo & Bang2023 ،), This correction ensures the final count of 211 Males and 26 Females, matching the validated table 1 of the original research.

3. **Usage Period:** Duration for which the respondent has been using the current system or tool. It is coded into ordered categories, for example:

   - 1 = Less than 6 months
   - 2 = 6 months – 1 year
   - 3 = 1 year – 2 years
   - 4 = 2 years – 5 years
   - 5 = More than 5 years

   This attribute helps analyse whether longer exposure to the system is associated with higher usage or continuance intention, and based on this phase, we made sure that all recorded from 1 to 5.

4. **Age:** Respondent's age is grouped into ranges, for example:

   - 1 = 10s
   - 2 = 20s

- 3 = 30s
- 4 = 40s
- 5 = 50s

Age categories can be used to examine whether user perceptions and ERP usage differ across age groups, and based on this phase, and it's the first demographic attribute on DB (Age, Position, Module), we made sure that all were recorded from 1 to 5.

5. **Position:** Respondent's job position or role in the organisation, such as:

- 1 = Clerk
- 2 = Assistant Manager
- 3 = Manager
- 4 = Deputy Manager
- 5 = General Manager
- 6 = Executive Director

This allows analysis of how role responsibility may affect ERP usage patterns, and based on this phase, we made sure that all recorded from 1 to 6.

6. **Module:** Main module or functional area of the system that the respondent primarily uses, for example:

- 1 = Operation Management
- 2 = Design
- 3 = Sales/Business Management
- 4 = Materials/Procurement
- 5 = Quality Management
- 6 = Management/Human Resources
- 7 = Finance/Accounting
- 8 = Other

This attribute links user experience to specific functional modules, which is important when thinking about module underutilization in ERP systems, and based on this phase, we made sure that all recorded from 1 to 8.

7. **SYQ1, SYQ2, SYQ3 – System Quality items**

These attributes represent Likert-scale questions related to the System Quality construct, capturing perceptions such as system reliability, response time, and technical performance. Higher values typically indicate a more positive evaluation of system quality. During the explore data task, Weka's visualisation panel was used to inspect the score distributions, revealing whether retail employees generally perceived the system as stable or if there were significant clusters of low scores (1–3). Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

8. INQ1, INQ2, INQ3 – Information Quality items

These items measure Information Quality, including accuracy, relevance, completeness, and timeliness of the information produced by the system. After being collected and described as 7-point Likert scale numeric variables, the explore data task utilised Weka's visualisation panel to inspect score distributions; this revealed whether retail employees perceived information quality as high or if clusters of low scores (1–3) suggested a data reliability gap. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to classification modelling.

9. PUS1, PUS2, PUS3 – Perceived Usefulness items

These attributes reflect the extent to which users perceive the system as useful for improving their job performance and effectiveness in daily tasks. After being collected and described as numeric variables on a 7-point Likert scale, the exploratory data task utilised Weka's visualisation panel to inspect score distributions; this revealed whether retail employees generally perceived the system as beneficial or if significant clusters of low scores (1–3) indicated a lack of perceived utility. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

10. SEQ1, SEQ2, SEQ3 – Service Quality items

These questions relate to Service Quality, such as responsiveness and support provided by IT or ERP support teams when users face problems or need assistance. After being collected and described as numeric variables on a 7-point Likert scale, the exploratory data task utilised Weka's visualisation panel to inspect score distributions; this revealed whether retail employees generally perceived the support teams as helpful or if significant clusters of low scores (1–3) highlighted a lack of adequate assistance. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

11. PEOU1, PEOU2, PEOU3 – Perceived Ease of Use items

These items capture how easy users find it to learn, navigate, and operate the system. They are conceptually aligned with the perceived ease of use construct in technology adoption models. After being collected and described as numeric variables on a 7-point Likert scale, the exploratory data task utilised Weka's visualisation panel to inspect score distributions; this revealed whether retail employees generally perceived the system as user-friendly or if significant clusters of low scores (1–3) indicated a steep learning curve or complex interface. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

12. IFL1, IFL2, IFL3 – ICT Infrastructure items

These measures include perceptions of the ICT infrastructure, such as hardware adequacy, system availability, and network reliability that support ERP usage. After being collected and described as numeric variables on a 7-point Likert scale, the explore data task utilized Weka's visualization panel to inspect score distributions; this revealed whether retail employees perceived the ICT infrastructure as adequate or if low scores (1–3) indicated system limitations. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

13. ISK1, ISK2, ISK3 – ICT Skills items

These items relate to users' ICT skills and competence, indicating how comfortable and capable they feel when working with systems and digital tools. After being collected and described as numeric variables on a 7-point Likert scale, the explore data task utilized Weka's visualization panel to inspect score distributions; this revealed whether retail employees generally felt capable or if significant clusters of low scores (1–3) indicated a deficiency in digital proficiency. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

14. TMS1, TMS2, TMS3 – Top Management Support items

These attributes measure perceived Top Management Support, such as visible backing from senior management, encouragement to use the system, and provision of resources for training and system improvement. measuring visible backing from senior management and the provision of resources for training, after being collected and described as numeric variables on a 7-point Likert scale, the explore data task utilized Weka's visualization panel to inspect score distributions; this revealed whether retail employees generally perceived management as encouraging or if significant clusters of low scores (1–3) suggested a lack of institutional support. Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

15. COI1, COI2, COI3 – Continuance Intention items

These items capture the user's Continuance Intention, that is, their willingness and motivation to continue using the system in the future. This construct is a strong candidate for the target variable in the classification model. For example, high vs. low continuance. Capturing the user's willingness and motivation to continue using the system, after being collected and described as numeric variables on a 7-point Likert scale, the explore data task utilized Weka's visualization panel to inspect score distributions; this revealed whether retail employees generally intended to persist with ERP usage or if significant clusters of low scores (1–3) identified a critical "usage gap." Finally, the data was verified by auditing the Min/Max range in the summary panel to ensure

all 237 records fell strictly within the valid 1–7 range, confirming the absence of outliers or "dirty" data before proceeding to the classification modelling.

During the data exploration stage, Weka's visualisation tools were used to inspect score distributions across all variables. In addition, Min/Max values were verified to ensure that all attributes fall within the valid 1–7 Likert scale range, confirming data quality and consistency.

Appendix A. Lists of Measurement Items

Continuance Intention is derived from Bhattacherjee [63].
COI1: I intend to continue my use of ERP in the future.
COI2: I intend to increase my use of ERP in the future.
COI3: I will keep using the ERP as regularly as I do now.
ICT skill is derived from Lin and Lin [115].
ISK1: I am generally aware of the functions of ERP.
ISK2: I am well trained in ERP.
ISK3: I am supported by specialized or knowledgeable personnel for ERP.
ICT Infrastructure is derived from Premkumar and Ramamurthy [60].
IFR1: Our organization has a good ICT infrastructure for implementing ERP.
IFR2: Our organization has the available IT resources (i.e., computer, internet, etc.) to implement ERP.
IFR3I: There is an alignment between IT and business strategy in implementing ERP in our organization.
Perceived usefulness is derived from Davis and Bagozzi [66].
PUS1: Using ERP helps me accomplish things more quickly.
PUS2: Using ERP increases my productivity.
PUS3: Using ERP increases the quality of output for the same amount of effort.
Perceived ease of use is derived from Davis and Bagozzi [66].
PEOU1: Learning how to use ERP is easy for me.
PEOU2: It is easy for me to become skillful at using ERP.
PEOU3: Overall, using ERP is easy for me.
System quality is derived from Tam and Oliveira [29].
SYQ1: ERP is well structured.
SYQ2: ERP offers appropriate functionality.
SYQ3: ERP allows me to easily use it whenever I want.
Information quality is derived from Urbach and Müller [86].
INQ1: The information provided by ERP is necessary.
INQ2: The information provided by ERP is adequate.
INQ3: The information provided by ERP is reliable.
Top management support is derived from Swink [93], Hsu and Liu [116,117].
TMS1: In recent years, top management has committed to making an ERP project successful.
TMS2: In recent years, top management has defined the aims of the ERP project explicitly.
TMS3: In recent years, top management has been involved throughout the ERP project process.
Service quality is derived from Urbach and Müller [86].
SEQ1: The responsible service personnel has sufficient knowledge to answer my questions concerning the ERP.
SEQ2: The responsible service personnel provides personal attention when I experience problems with the ERP.
SEQ3: The responsible service personnel provides services related to the ERP at the promised time.

Figure 20: Dataset Appendix

The measurement instrument adopts a multi-item scale to ensure internal consistency, with 27 variables adapted from established literature, including works by Bhattacharjee, Davis, and Urbach, to evaluate ERP continuance intention (Jo & Bang2023 ،). In Weka, these items were collected and described as numeric attributes on a 7-point Likert scale. During the explore data

task, Weka's visualisation panel was used to inspect score distributions, identifying whether employees perceived the system positively or if clusters of low scores (1–3) signalled significant barriers. Finally, data was verified by auditing the Min/Max range in the summary panel, ensuring all 237 records fell strictly within the valid 1–7 range and confirming the absence of outliers before proceeding to classification modelling.

Please answer the following questions.

Q1. Do you agree to participate in this study?

Q2. The results of this study may be published academically. Do you agree with this?

Q3. What is your gender?

Q4. How many years have you worked at the company?

Q5. How long have you been using ERP in your organization?

Q6. How old are you?

Q7. Which department or team are you currently a part of?

Q8. What is your job title?

Q9. What are the main functions (ERP modules) that you handle with your current ERP system? (Please select only one option)

Q10. Please provide any recommendations for improving your current ERP system.

Below is a questionnaire regarding your current ERP system. Please carefully read each item and select the appropriate box that reflects your response. Each item is rated on a seven-point scale, ranging from strongly disagree (1) to strongly agree (7).

No.	Question	1	2	3	4	5	6	7
1	I intend to continue my use of ERP in the future.							
2	I intend to increase my use of ERP in the future.							
3	I will keep using the ERP as regularly as I do now.							
4	I am generally aware of the functions of ERP.							
5	I am well trained in ERP.							
6	I am supported by specialized or knowledgeable personnel for ERP.							
7	Our organization has a good ICT infrastructure for implementing ERP.							
8	Our organization has the available IT resources (i.e., computer, internet, etc.) to implement ERP.							
9	There is an alignment between IT and business strategy in implementing ERP in our organization.							
10	I intend to continue my use of ERP in the future.							
11	I intend to increase my use of ERP in the future.							
12	I will keep using the ERP as regularly as I do now.							
13	I am generally aware of the functions of ERP.							
14	I am well trained in ERP.							
15	Using ERP helps me accomplish things more quickly.							
16	Using ERP increases my productivity.							
17	Using ERP increases the quality of output for the same amount of effort.							
18	Learning how to use ERP is easy for me							
19	It is easy for me to become skillful at using ERP.							
20	Overall, using ERP is easy for me.							
21	ERP is well structured.							
22	ERP offers appropriate functionality.							
23	ERP allows me to easily use it whenever I want.							
24	Using ERP helps me accomplish things more quickly.							
25	Using ERP increases my productivity.							
26	The information provided by ERP is necessary.							
27	The information provided by ERP is adequate.							
28	The information provided by ERP is reliable.							
29	In recent years, top management has committed to making an ERP project successful.							
30	In recent years, top management has defined the aims of the ERP project explicitly.							
31	In recent years, top management has been involved throughout the ERP project process.							
32	The responsible service personnel has sufficient knowledge to answer my questions concerning the ERP.							
33	The responsible service personnel provides personal attention when I experience problems with the ERP.							
34	The responsible service personnel provides services related to the ERP at the promised time.							

Thank you for your cooperation.

Figure 21: Survey Questions

The survey instruments for the constructs listed above (SYQ, INQ, PUS, SEQ, PEOU, IFL, ISK, TMS, and COI) were adapted from the validated scales used by Jo and Bang (2023). Each item is measured using a 7-point Likert scale, ranging from 1 (Strongly Disagree) to 7 (Strongly Agree). This granular scale allows the AI model to capture subtle variances in user sentiment, which is essential for identifying the "usage gap" in the Bahraini retail context.

The questioning logic is designed to capture the Technology-Organisation-Environment (TOE) dimensions; for instance, the ICT Skills (ISK) items specifically ask users to rate their own technical competence, while the Top Management Support (TMS) items evaluate the perceived commitment of corporate leadership. During the data understanding phase in Weka, these 27 construct items are treated as numeric predictors. The distribution of these scores, specifically the clustering of responses around the neutral midpoint of 4.0, serves as the mathematical basis for the Discretisation filter applied in Phase 3, allowing the project to classify respondents into distinct "High" and "Low" utilisation groups for the final J48 Decision Tree analysis.

4.2.3 Phase 3: Data Preparation

In the data preparation phase, we cleaned and prepared the case study dataset for machine learning analysis in WEKA with the aim of predicting ERP users' continuance intention. The first step was to remove non-informative attributes, namely the ID field, since it was only a unique identifier and had no real impact on predicting ERP users' continuance intention. All other attributes remained. These attributes represented demographic, contextual, and ERP-related perception variables that could influence users' intention to continue using the ERP system. This step is essential to ensure that the dataset is suitable for building accurate and reliable machine learning models.

Next, the ReplaceMissingValues filter in WEKA was used to address missing values. Missing numeric values were filled using the mean of the attribute, while missing nominal values were replaced using the most frequent category (mode). By using these methods, we maintained consistency in our dataset without unnecessarily decreasing the number of usable records. Once this step was completed, the variables were reviewed again to ensure that their coding and value ranges were suitable for analysis related to ERP continuance intention prediction.

For feature selection, the study retained all attributes that were theoretically relevant to ERP usage and continuance intention. These included demographic variables (Gender, Usage Period, Age, Position, Module) as well as ERP-related construct items (SYQ, INQ, PUS, SEQ, PEOU, IFL, ISK, TMS). These variables were preserved because they may influence how users experience and continue using the ERP system and therefore affect their continuance intention. However, the individual COI1, COI2, and COI3 items were removed from the predictor set after being used to construct the target variable in order to prevent target leakage during the prediction of ERP continuance intention.

To create a class variable, the three items representing continuance intention (COI1, COI2, and COI3) were grouped using the AddExpression filter in WEKA to create an average value, referred to as COI_mean. This variable represents the overall continuance intention score of ERP users in this case study dataset. The composite score was then discretised using the WEKA Discretise filter to convert the numeric value into two categories: Low and High continuance intention, allowing it to be used as a binary class variable for classification in the prediction models.

Finally, the prepared dataset was used for supervised learning in WEKA to build models capable of predicting ERP users' continuance intention. To evaluate different approaches, three classification algorithms were tested: J48 decision tree, Random Forest, and Naive Bayes. These models were used to examine how accurately ERP continuance intention could be predicted based on the demographic and ERP perception variables included in the dataset. Model performance was assessed using 10-fold cross-validation, in which the dataset was split into ten parts, and each part was used once for testing while the remaining parts were used for training. Each subset was used once as a test set while the remaining subsets were used for training, providing a reliable estimate of model performance while making efficient use of the available sample.

4.2.4 Phase 4: Modelling

During this phase, the dataset was used to build predictive models aimed to identify ERP usage gaps by selecting certain machine learning algorithms and setting them up within the WEKA working environment to convert the survey data into a practical analytical tool.

Given the wide range of available machine learning algorithms, the focus was primarily placed on three algorithms. These were selected based on their effectiveness and were subsequently tested and evaluated in Phase 5:

4.2.4.1 J48 (Decision Tree)

J48 is an implementation of the C4.5 algorithm, which balances computational efficiency with clear, rule-based results (Kapoor & Rani, 2015). C4.5 algorithms are decision tree machine learning algorithms that build classification models by repeatedly dividing a dataset along the attribute with the greatest information gain. This helps the model identify the most informative attribute. The output of the J48 algorithm is a flowchart-like decision tree where each internal node represents a decision based on a particular attribute, and each leaf node is a final class. Simply, the algorithm breaks down complicated relationships in the data into a set of straightforward if-then statements, making the model both interpretable and useful for analysis and decision-making.

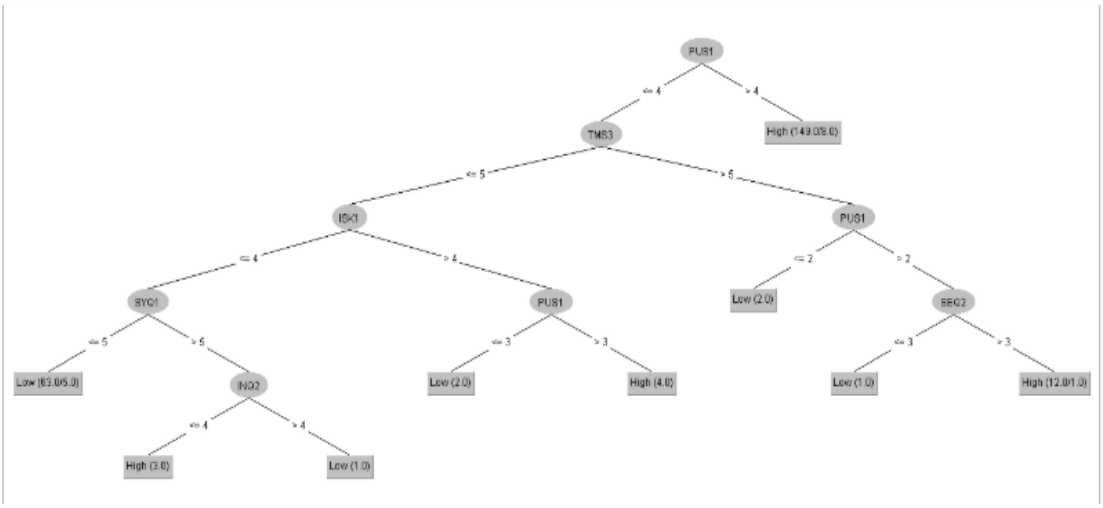

Figure 22: J48 decision tree generated by Weka (case study)

Figure 22 shows the decision tree generated by the J48 algorithm using 237 survey responses and 30 attributes to classify ERP usage as “Low” or “High.” The tree displayed the main decision rules, where the machine learning model identified Perceived Usefulness (PUS) as a primary factor. Where each node represents an attribute and each branch shows its outcome. This concise structure keeps only the most important variables and makes the results easier to interpret.

To optimize the model's reliability, the J48 algorithm was configured with a confidence factor of 0.15 to mitigate the impact of subjective survey noise through effective post-pruning. This process resulted in a parsimonious 9-leaf model, as shown in the text-based classification rules detailed in Figure 23.

```
J48 pruned tree
------------------

PUS1 <= 4
|   TMS3 <= 5
|   |   ISK1 <= 4
|   |   |   SYQ1 <= 5: Low (63.0/5.0)
|   |   |   SYQ1 > 5
|   |   |   |   INQ2 <= 4: High (3.0)
|   |   |   |   INQ2 > 4: Low (1.0)
|   |   ISK1 > 4
|   |   |   PUS1 <= 3: Low (2.0)
|   |   |   PUS1 > 3: High (4.0)
|   TMS3 > 5
|   |   PUS1 <= 2: Low (2.0)
|   |   PUS1 > 2
|   |   |   SEQ2 <= 3: Low (1.0)
|   |   |   SEQ2 > 3: High (12.0/1.0)
PUS1 > 4: High (149.0/8.0)

Number of Leaves  :     9

Size of the tree :      17


Time taken to build model: 0 seconds
```

Figure 23: pruned tree classification rules and tree dimensions (case study)

To ensure that the model’s findings are generalizable, the 10-fold cross-validation method was performed. The analysis yielded an accuracy rate of 87.34%, indicating that the selected attributes are very informative. Also, the Kappa statistic was calculated at 0.7031. This score reflects a strong consistency between the predicted and actual outcomes.

Table 6: Confusion matrix of the J48 algorithm (case study)

	PREDICTED: LOW (A)	**PREDICTED: HIGH (B)**

ACTUAL: LOW (A)	58	15
ACTUAL: HIGH (B)	15	149

Table 6 demonstrates that the model achieved a high level of correct classification across both categories, which reflects the high predictive capability of the model. However, there is a relatively low error rate, including a Mean Absolute Error of 0.1819. The minor misclassifications observed more within the "Low" category may indicate the lack of some important examples.

4.2.4.2 Random Forest

A Random Forest is a machine learning method. It aggregates multiple decision trees that are independently trained on a random portion of the total dataset to generate a prediction by majority voting. This approach reduces the overfitting and improves the ability to generalize to novel data. This method improves the model's stability and accuracy, which makes it suitable for analyzing data that may contain noise or irregular patterns (Salman, et al., 2024).

In this study, the Random Forest algorithm was implemented in the WEKA environment using 100 trees (iterations), with default parameter settings. Similarly to the J48 algorithm, the model was evaluated using stratified 10-fold cross-validation to ensure reliable and generalizable results using the same dataset.

The Random Forest model achieved high classification performance, with 211 instances correctly classified, resulting in an overall accuracy of 89.03%. Only 26 instances were misclassified, corresponding to an error rate of 10.97%. The detailed classification results are presented in Table 7, which illustrates the confusion matrix of the Random Forest model.

Table 7: Confusion matrix of the Random Forest algorithm (case study)

	PREDICTED: LOW (A)	**PREDICTED: HIGH (B)**
ACTUAL: LOW (A)	58	15

ACTUAL: HIGH (B)	11	153

The weighted average F-measure was 0.889 and the Kappa statistic of 0.7387, confirming balanced predictive effectiveness across classes. Additionally, the model produced a ROC Area of 0.938. The Mean Absolute Error of 0.177 and a Root Mean Squared Error of 0.2906, indicating low prediction error.

4.2.4.1 Naïve Bayes

In order to get a statistical baseline, this probabilistic classifier was included. It works on the assumption that input variables are assumed to be independent and calculates the probability of belonging to a certain class based on Bayes' Theorem (Yang, 2018). Naïve Bayes is useful in this study because it checks whether the selected ERP factors, such as System Quality and Top Management Support, have enough information to predict the outcome without using a complex model like a decision tree.

In this study, the same dataset and 10-fold cross-validation method were used to follow the same experimental design applied to the previous algorithms. It also achieved strong classification performance, with 205 instances correctly classified with an overall accuracy of 86.50%. However, a total of 32 instances were misclassified, corresponding to an error rate of 13.50%. The detailed classification results are presented in Table 8, which illustrates the confusion matrix of the Naive Bayes model.

Table 8: Confusion matrix of the Naïve Bayes algorithm (case study)

	PREDICTED: LOW (A)	**PREDICTED: HIGH (B)**
ACTUAL: LOW (A)	66	7

ACTUAL: HIGH (B)	25	139

4.2.5 Phase 5: Evaluation

In the evaluation phase, the developed machine learning models were assessed to determine how well they achieved the objective of this study, which was to predict ERP users' continuance intention based on the collected survey variables. The main purpose of this phase was to examine the ability of the models to classify ERP users into Low and High continuance intention categories using the prepared ERP case study dataset. To ensure a reliable assessment, the models were evaluated in WEKA using 10-fold cross-validation, which allowed all records in the dataset to be used for both training and testing in different iterations.

Three classification algorithms were evaluated in order to identify the most suitable model for predicting ERP continuance intention: J48 decision tree, Random Forest, and Naive Bayes. Their performance was compared using several evaluation measures, including accuracy, precision, recall, F1-score, Kappa statistic, and ROC area. These metrics were used to measure how accurately the models could classify ERP users based on their responses to the ERP perception constructs included in the dataset.

The results showed that all three models achieved good classification performance, which indicates that the ERP-related variables used in the study, such as perceived usefulness, system quality, information quality, system knowledge, and top management support, contain meaningful information for predicting ERP continuance intention.

Among the tested models, Random Forest achieved the best overall performance, with the highest accuracy and strongest ROC area. This suggests that the Random Forest model was the most effective at identifying patterns in the ERP user data and predicting whether users had high or low intention to continue using the ERP system.

The J48 decision tree also produced strong results and was particularly valuable for this study because it generated clear decision rules that helped explain the factors influencing ERP

continuance intention. The structure of the decision tree showed that Perceived Usefulness (PUS1) played a central role in determining whether users demonstrated high or low continuance intentions. Other ERP-related factors, such as top management support, system knowledge, system quality, and information quality, also appeared in the tree and contributed to the classification process.

The Naive Bayes model also performed well, although its accuracy was slightly lower than the other two models. Despite this difference, the results still demonstrate that the selected ERP perception variables can be used effectively to classify ERP users' continuance intention.

Table 9: Performance comparison of classification models (case study)

Model	Accuracy	Precision	Recall	F1-score	Kappa	ROC
J48	87.34%	0.873	0.873	0.873	0.703	0.888
Random Forest	89.03%	0.889	0.890	0.889	0.739	0.938
Naive Bayes	86.50%	0.882	0.865	0.868	0.703	0.929

The results indicate that all three models achieved relatively high classification performance when predicting ERP continuance intention using the survey variables collected in the case study. The highest values of accuracy, Kappa statistic, and ROC area were achieved by the Random Forest model (89.03%, 0.739, and 0.938, respectively). This suggests that the Random Forest model extracted the most relevant information from the ERP data and, therefore, classified the users' intention to continue using the ERP system with greater accuracy.

The J48 decision tree also produced strong performance with an accuracy of 87.34%. Although its predictive performance was slightly lower than Random Forest, it provided a clear and interpretable tree structure that helps explain the factors influencing ERP continuance intention. From the analysis of the tree structure, Perceived Usefulness (PUS1) was identified as a dominant factor which can strongly influence the intention to use the ERP system and determine whether the intention to use the ERP system will be at a high or low level. In addition, other factors related to

the ERP system, such as top management support, system knowledge, system quality, and information quality, also showed influence on the intention to use the ERP system.

The accuracy of the Naive Bayes model was 86.50%, which is slightly lower than the other two models. Although it is lower than the other two models, the predictive performance is still very high. This proves that the ERP perception variables used in this study have real meaning and can also be used as a strong basis for the classification of users' continuance intention.

The confusion matrices were also analysed to understand the classification errors made by the models. The results showed that the models were slightly more accurate in identifying users with high continuance intention compared to users with low continuance intention, although the overall classification performance remained balanced and acceptable.

Overall, the evaluation results confirm that machine learning techniques can effectively analyse ERP user perceptions and identify patterns related to continuance intention. The findings also support the study objective of understanding how different ERP-related factors influence users' willingness to continue using the system. These findings validate the effectiveness of AI-based classification in analysing ERP usage behaviour.

4.2.6 Phase 6: Deployment

In the deployment phase, the results of the developed machine learning models are considered in terms of how they could be used in practice to support ERP management and decision-making. The identified ERP usage gaps directly impact organisational performance through reduced efficiency, poor decision-making, and underutilisation of system capabilities. Since this phase uses an academic case study dataset to demonstrate the machine learning modelling process using survey data in WEKA, the models were not implemented directly in a live ERP environment. Instead, this phase focuses on how the findings and predictive models developed in this project could be applied by organisations to improve ERP utilisation and support user continuance intention.

The results from the evaluation phase showed that the Random Forest model achieved the highest predictive performance, while the J48 decision tree provided clear and interpretable decision rules explaining the factors that influence ERP continuance intention. These models can therefore be used as analytical tools to help organisations understand ERP user behaviour and detect patterns related to the usage gap identified in the business understanding phase.

In a real organisational environment, companies could collect similar survey data or system usage indicators from ERP users and apply the developed models to classify users into High or Low continuance intention groups. Such a classification can be beneficial to ERP administrators in identifying employees or departments that may be using the ERP system at a minimal level, thus giving them an early warning sign to take appropriate actions and improve both system use and user satisfaction.

The outcomes of the decision tree analysis can also provide ERP managers with valuable insights into ERP continuance intentions. In this study, variables such as perceived usefulness, system quality, information quality, system knowledge, and top management support were all statistically significant predictors of ERP continuance intention and should therefore be the focus of organisations when attempting to improve these elements through training programs, enhanced technical support, upgraded systems, or greater management support for using ERP systems.

In addition, the predictive models created as part of this project can serve as a guide for creating a decision-support or monitoring tool that evaluates ERP user feedback continually and can indicate potential declines in continuance intention. The use of this tracking tool could enable organisations to respond proactively when ERP engagement begins to decrease, allowing them to anticipate and address problems related to declining system usage.

It should be noted that due to the limitations of the data collected in this study, particularly the limited sample size of 237 respondents for the case study, the deployment described in this section has not been completely operationalised and therefore remains conceptual. This research could be expanded in future studies by applying the developed models to larger datasets, integrating them with actual ERP systems, and developing dashboards that regularly monitor user perceptions and ERP usage trends.

To sum up, this project demonstrates that the results can provide practical value for organisations beyond the academic context. By using predictive models to understand ERP user behaviour, organisations can better address the usage gap and improve employees' continuance intention to use ERP systems. This phase demonstrates how the developed models can be extended into real-world ERP environments, supporting data-driven decision-making in retail organisations.

4.3 Transition to Survey Data

Following the structured application of the CRISP-DM methodology, the analysis was extended to the primary survey dataset collected from retail organisations in Bahrain.

This transition marked the shift toward the main contribution of the study, where real-world ERP usage data was analysed to identify usage gaps among employees. The survey dataset provided direct insights into user behaviour, system interaction, and organisational factors affecting ERP utilisation.

To ensure consistency in the analysis process, the same CRISP-DM framework was applied to this dataset. This enabled a systematic examination of ERP usage patterns through the phases of business understanding, data understanding, data preparation, modelling, and evaluation.

In this stage, Artificial Intelligence techniques, particularly Machine Learning classification models, were utilised. Tree-based models such as J48 were implemented to analyse the data and identify patterns associated with ERP usage gaps, as well as the key factors influencing system utilization.

The use of two different datasets allowed the study to examine ERP behaviour from two complementary perspectives. The case study dataset focused on continuance intention and represented expected user behaviour toward ERP systems, while the Bahraini survey dataset reflected actual organisational practices and operational conditions. This combination strengthened the analytical framework by enabling the comparison between expected ERP usage and actual usage behaviour in real organisational environments. It also supported the identification of ERP usage gaps from both behavioural and operational perspectives.

4.4 Survey Data Analysis

This section presents the analysis of the primary survey dataset collected from retail organisations in Bahrain. The analysis was conducted using Artificial Intelligence techniques, specifically Machine Learning classification models, following the CRISP-DM methodology.

The dataset consisted of 148 valid responses after data cleaning. The purpose of this analysis was to identify ERP usage gaps and examine the key factors influencing user behaviour. The CRISP-DM phases were applied sequentially, including Business Understanding, Data Understanding, Data Preparation, Modelling, and Evaluation.

4.4.1 Phase1: Business Understanding

In this phase, the main problem addressed in this study was the existence of an ERP usage gap within retail organisations in Bahrain. Despite significant investment in ERP systems, it was observed that many employees did not fully utilise the system's features and capabilities. This underutilisation reduced the overall effectiveness of the system and limited the expected organisational benefits.

The ERP usage gap was defined as the difference between the expected level of ERP system utilisation and the actual usage behaviour of employees. Based on the survey data, this problem was influenced by several factors, including lack of training, system complexity, poor system performance, limited top management support, and reliance on external or manual tools.

To address this problem, the main objective of this phase was to analyse ERP usage behaviour and identify the key factors contributing to the usage gap. This was achieved by applying Artificial Intelligence techniques, specifically Machine Learning classification models, to detect patterns in user behaviour.

A classification task was defined, where ERP users were categorized into two groups:

- Gap: Users with low ERP utilisation
- No_Gap: Users with effective ERP utilisation

The basis of this classification was behavioral attributes that were found through the survey data and these include the number of times that the system is being used, level of task completion within the ERP system, and the presence of unused modules.

Furthermore, Tree-Based Machine Learning models such as the J48 model was implemented in order to generate interpretable results. Through this model, variables which had a great influence on the usage of ERP could be identified.

4.4.2 Phase 2: Data Understanding

In this phase, the dataset was examined in order to understand its structure, variables, and overall quality after preprocessing. The survey dataset consisted of 148 responses collected from employees working in different retail organisations in Bahrain.

The dataset included demographic variables, ERP system characteristics, behavioural usage indicators, and perception-based variables. These attributes were initially collected in categorical and Likert-scale formats. During the preprocessing stage in Excel and WEKA, they were transformed into numerical and nominal values to make them suitable for analysis.

The survey instrument consisted of five main sections, each addressing a different aspect of ERP usage, including organisational information, usage behaviour, usage gaps, system evaluation, and continuance intention. These sections were later reflected in the structure of the dataset used in this phase.

4.4.3.1 Section 1: Organisational and Respondent Attributes

Variables used included several characteristics related to organisational features and demographics of the respondents. These variables describe the organisational context and background of participants.

- **ORG_TYPE:** Organisational type (supermarket, clothes, electronics, pharmacy, or any other), encoded into numeric values.

- **POSITION:** Employee role, including staff, supervisor, manager and senior management.
- **USG_PERIOD:** Duration of ERP system usage by the respondent, from less than 6 months to more than 5 years.
- **ERP_SYS:** refers to the type of ERP system used, such as SAP, Oracle, Microsoft Dynamics, Odoo, and other systems.

These attributes provided context about the organisational environment and user background, as they were collected in Section 1 of the survey. They helped in understanding whether ERP usage behaviour differs based on organisation type, employee role, experience level, or system type. This context is important when interpreting the results of the machine learning models, as variations in usage patterns may be influenced by these background characteristics rather than system-related issues alone.

Additionally, these attributes supported segmentation of respondents, allowing the analysis to consider how ERP usage varies across different organisational and professional profiles.

4.4.3.2 Section 2: ERP Usage Behaviour Attributes

Several attributes were derived to capture actual ERP usage behaviour:

- **FREQ_USE and TASK_COMP:** These variables originally contained multiple categories (such as rarely, sometimes, often, always). They were converted into binary values where: Low usage = 1, High usage = 0
- **IS_THERE_UNUSED:** Shows if there are any unused ERP modules: 0 = No, 1 = Yes.

These variables were collected from Section 2 of the survey and later used to construct the GAP_SCORE and GAP variable. This transformation allowed the study to move from subjective survey responses into measurable behavioural indicators. By simplifying usage into low vs high categories, it became easier for the machine learning models to detect patterns related to underutilisation and identify users who are not fully engaging with the ERP system.

This step was essential in converting qualitative responses into structured inputs suitable for AI-based classification models.

Target Variable (ERP Usage Gap Classification)

The main target variable in this study was:

- **GAP:** Represents whether an ERP usage gap exists: Gap = Low utilisation, No_Gap = Effective utilisation

This variable was created based on a calculated GAP_SCORE, which combined the frequency of use, task completion within ERP, and the presence of unused modules. Users with higher GAP scores were classified as having a usage gap. This variable is central to the study, as it represents the main problem being analysed. By converting multiple behavioural indicators into a single classification variable, the machine learning models were able to predict which users are experiencing a usage gap and which are effectively utilising the ERP system.

It also served as the class attribute in the machine learning models, enabling the training of predictive classification algorithms.

ERP Module Utilisation Attributes

To measure ERP module utilisation:

- **Module:** Individual module usage (Inventory, Sales, Procurement, Finance, HR, Reporting) was converted into binary indicators.
- **MODULE_COUNT:** A new variable was created to calculate how many modules each user actively uses.

Based on this, MODULE_LEVEL was derived: Low (≤2 modules), Medium (3–4 modules) and High (>4 modules)

This helped in identifying the depth of ERP system usage across users. Rather than only measuring whether ERP is used or not, this approach provided a more detailed view of how extensively the system is utilised. Users with low module usage may rely on limited system functionality, which indicates a potential usage gap, while users with higher module engagement reflect more effective system adoption.

This provided an additional layer of behavioural insight beyond basic usage frequency.

4.4.3.3 Section 3: ERP Monitoring and Gap Factors

Section 3 of the survey captured both organisational monitoring practices and key factors contributing to ERP usage gaps.

The MONITOR variable originally had four categories: No, Rarely, Sometimes, and Yes (regularly). During preprocessing in WEKA, these values were merged into Low monitoring No, Rarely, Sometimes and High monitoring Regularly.

This variable reflects how organisations track ERP usage. The transformation simplified the analysis and allowed the model to better distinguish between weak and strong monitoring practices. This step was important because frequent monitoring of ERP usage is often associated with better system utilisation, and it enabled the model to capture whether organisations actively track ERP usage as a factor influencing user behaviour.

This variable represents an organisational control mechanism that may influence user accountability and system adoption. The dataset also included key factors contributing to ERP usage gaps:

- TRAINING: lack of training
- COMPLEXITY: system complexity
- PERFORMANCE: system performance issues
- MGT_SUPPORT: lack of management support
- EXTERNAL_TOOLS: use of manual/external tools

These variables were collected from Section 3 of the survey and transformed into binary format: Low = 0, High = 1. These attributes were used as predictors in the machine learning models. They represent the main reasons behind ERP underutilisation, as reported directly by users. For instance, a higher score in the TRAINING factor shows that there is a lack of training, while a higher score in the COMPLEXITY factor suggests that the users find the system hard to use, and using external aids implies that the users are using the ERP system less.

Incorporating these variables helped the model not only identify the presence of a usage gap but also provide insight into why this gap existed. This would prove helpful in formulating strategies since organizations could work toward improvement in any one of the areas mentioned, including training, ease of system use, better performance, or managerial support.

These variables were critical in linking user behaviour with underlying organisational and technical challenges.

4.4.3.4 Sections 4 and 5: Perception-Based Constructs

In the second modelling approach, Likert-scale variables were grouped and averaged to form composite constructs:

- EASE_OF_USE: PEOU1,PEOU2
- USEFULNESS: PUS1,PUS2
- SYSTEM_QUALITY: SYQ1,SYQ2
- INFORMATION_QUALITY: INQ1,INQ2
- TOP_MGT: TMS1,TMS2
- SKILLS: ISK1,ISK2
- INTENTION: COI1,COI2,COI3

These variables were grouped to represent a single construct. These variables were obtained from Sections 4 and 5 of the survey, where respondents indicated their level of agreement through a five-point Likert scale. The calculation for each construct involves getting the average of survey

questions related to it and then classifying them into two categories: Low ≤ 2.5 and High > 2.5, via the MathExpression filter in WEKA.

This transformation reduced data complexity while preserving the meaning of the original responses. Instead of analysing multiple individual questions, each construct represented a broader concept, allowing the model to identify which high-level perception factors have the strongest impact on ERP usage behaviour.

This also improved model efficiency by reducing dimensionality while maintaining interpretability.

4.4.3.5 Data Quality Verification

In this regard, the dataset was verified to ensure that all data was accurate and fit for analysis in subsequent steps. Verification involved checking all variables and confirming that each variable had valid values without any missing values. It is important to note that verification of data ensures its accuracy and makes it ready for machine learning analysis.

In this way, there is an assurance that any conclusions made from the analysis will be accurate and valid.

4.4.3 Phase 3: Data Preparation

Basically, data prep is that third step in CRISP-DM where all the actual heavy lifting happens. You're taking this huge pile of messy, raw data and scrubbing it down until it’s finally ready to be used for something. It’s a massive deal because any model you build, whether it's a Decision Tree or a Random Forest, is only going to be as good as the information you actually give it.

Truth sits plainly here: precision matters most when details align without clutter. When formatting finds its proper shape, consistency follows close behind. Mess distracts; clarity invites progress - so removing glitches hands the system cleaner material to work through. Patterns emerge easier once distractions dissolve into order. Function thrives where structure supports effort quietly. Models perform best inside tidy environments built ahead of time.

In this study, Data Preparation acted as the bridge between the raw responses collected from Bahraini retail employees and the analytical environment of WEKA. It involved four key technical processes:

4.4.3.1 Data Cleaning Procedures

Based on the technical transformation from our DB in EXAL and WEAKA steps we taken during the Data Cleaning Procedures:

i. Filtering and Case Removal

The first step was to ensure the integrity of the rows.

- **Action:** The first record in the raw dataset (ID 1) was identified as a "dummy" or "test" entry as it contained null values across all survey dimensions.
- **Result:** First row was deleted due to many missing values to test the survey running, leaving 148 valid responses for analysis.

ii. Attribute Renaming and Mapping

Raw headers from the survey were long, bilingual questions, Therefore we renamed all to :

1. Demographic & Organizational Profile

These attributes establish the background of the respondents and their organizations.

- **ORG_TYPE**: Renamed from "What type of retail organization do you operate?"
- **POSITION**: Renamed from "What is your current position in the organization?"
- **USG_PERIOD**: Renamed from "How long have you been using the ERP system?"
- **ERP_SYS**: Renamed from "Which ERP system is currently used?"

2. Module Usage & Behavioral Patterns

These attributes were extracted from multi-select questions to identify specific areas of utilization.

- **Inventory, Sales, Finance, Human_Resource, Reporting, Procurement**: These are the individual binary columns created from the "Which ERP modules do you use regularly?" multi-select question.
- **MODULE_COUNT**: A new attribute representing the total number of selected modules.
- **MODULE_LEVEL**: A categorical attribute defining usage as High, Medium, or Low.
- **FREQ_USE**: Renamed from "How frequently do you use the ERP system?"
- **TASK_COMP**: Renamed from "How often do you complete tasks entirely in ERP?"
- **IS_THERE_UNUSED**: Renamed from "Are there ERP features you do not use?"

3. Perception & Quality Constructs (Likert Scale Items)

Individual survey questions from the perception section were renamed to standardized codes for model processing.

- **PUS1, PUS2**: Renamed from the Perceived Usefulness items (Task speed and Productivity).
- **USEFULNESS**: The calculated average of PUS1 and PUS2.
- **PEOU1, PEOU2**: Renamed from the Perceived Ease of Use items (Navigability and Learning).
- **EASE_OF_USE**: The calculated average of PEOU1 and PEOU2.
- **SYQ1, SYQ2**: Renamed from the System Quality items (Reliability and Requirements).
- **SYSTEM_QUALITY**: The calculated average of SYQ1 and SYQ2.
- **INQ1, INQ2**: Renamed from the Information Quality items (Accuracy and Relevance).
- **INFORMATION_QUALITY**: The calculated average of INQ1 and INQ2.
- **TMS1, TMS2**: Renamed from the Top Management Support items (Support and Encouragement).
- **TOP_MGT**: The calculated average of TMS1 and TMS2.
- **ISK1, ISK2**: Renamed from the User Skills items (Skills and Training).
- **SKILLS**: The calculated average of ISK1 and ISK2.

4. Usage Barriers & Gap Indicators

These attributes identify why the "Usage Gap" exists and how the organization monitors the system.

- **TRAINING, COMPLEXITY, PERFORMANCE, MGT_SUPPORT**: Individual binary columns created from the "Reasons for not fully utilizing" question.
- **EXTERNAL_TOOLS**: Renamed from the question regarding the use of tools like Excel.
- **MONITOR**: Renamed from "Does organization monitor ERP usage?"
- **GAP_SCORE**: A numeric metric calculated to identify the depth of the usage gap.
- **GAP**: A binary indicator (Gap/No Gap) used for high-level classification.

5. Target Class Variables

The final variables used to classify the user's future behavior.

- **COI1, COI2, COI3**: Renamed from the three items regarding Continuance Intention (Future use, increased use, and regular reliance).
- **INTENTION**: The final classification target, representing the averaged and discretized category (High/Low) for the machine learning models.

iii. Bilingual Text Removal and Coding

Since the primary survey was distributed to retail employees in Bahrain, the raw dataset contained responses in both English and Arabic, such as "Clerk / Staff | موظف" or "Odoo | أودو". To prepare this data for machine learning analysis in WEKA, a rigorous cleaning and coding process was applied as follows:

Bilingual Text Stripping

Starting fresh, the raw data had every Arabic character stripped out. Special symbols disappeared too, including separators like the pipe "|". A clean slate emerged - English only, nothing else. That way, machine learning models would not mistake mixed-language entries as distinct categories. Consistency came through uniformity. For instance, any variation of "Supermarket" or its Arabic equivalent was standardized to a single "Supermarket" label.

Numeric Coding for Processing

Following the cleaning of bilingual text, all categorical responses were converted into numeric codes to enable statistical analysis and formula-based calculations within Excel. The following is a comprehensive list of all text values converted into numeric codes:

> "A few adjustments were made to **FREQ_USE** and **TASK_COMP** to match the logic of our **GAP_SCORE** formula (where higher numbers indicate a potential gap), and the **MONITOR** values were aligned with our WEKA merging process."

1. Organizational and Demographic Codes

- **ORG_TYPE (Organization Type):**
  - Supermarket / Hypermarket = 1
  - Clothing / Fashion = 2
  - Electronics = 3
  - Pharmacy = 4
  - Other = 5
- **POSITION (Current Position):**
  - Clerk / Staff = 1
  - Supervisor = 2
  - Manager = 3
  - Senior Management = 4
- **USG_PERIOD (Usage Period):**
  - Less than 6 months = 1
  - 6 months – 1 year = 2
  - 1 – 2 years = 3
  - 2 – 5 years = 4
  - More than 5 years = 5
- **ERP_SYS (ERP System):**
  - SAP = 1
  - Oracle = 2
  - Microsoft Dynamics = 3

- Odoo = 4
- Other = 5

2. Behavioral and Gap Score Codes

- **FREQ_USE (Frequency of Use):**
  - Daily / Very frequently = 0
  - Often = 0
  - Sometimes = 1
  - Rarely = 1
- **TASK_COMP (Task Completion in ERP):**
  - Always = 0
  - Often = 0
  - Sometimes = 1
  - Rarely = 1
- **IS_THERE_UNUSED (Unused Features Presence):**
  - No = 0
  - Yes = 1

3. Monitoring and Management

- **MONITOR (Monitoring Levels):**
  - No = 0
  - Rarely = 1
  - Sometimes = 2
  - Yes, regularly = 3 *(Note: In WEKA, values 0, 1, and 2 were merged into "Low", and value 3 was named "High")*

4. Final Class Label

- **GAP (Target Classifier):**
  - No_Gap (Gap Score < 2) = 0
  - Gap (Gap Score ≥ 2) = 1

Nominal Conversion and Merging

WEKA Transformation and Labeling: Once imported into WEKA, these numeric results were treated as Nominal values to support classification algorithms. To handle the MONITOR field, values 0, meaning No, along with 1 for Rarely and 2 called Sometimes got grouped under one label: Low - achieved through a merge operation blending several inputs at once. Value 3 stood apart, tagged instead as High without combining further elements. Later on, coded outcomes changed into clear terms via renaming steps applied across named categories; here, code 1 turned into Gap while 0 became No_Gap, keeping interpretations grounded in everyday store-level understanding.

iv. Binary Decomposition of Multi-Select Units

Inside the raw survey records appeared questions allowing several answers per person. From the first CSV download, every picked choice sat inside one cell as plain text. Separate entries showed up divided by semicolons within that field. An example looked like "Inventory; Sales; Finance" all packed together. This format bundled multiple picks without splitting them across columns. Because machine learning algorithms cannot process grouped text strings, the following "explosion" procedure was applied:

- **Logic Extraction:** Using the Excel formula =IF(ISNUMBER(SEARCH("reasons/module", $N2)), 1, 0), the single "Module" column was decomposed into six distinct binary attributes: **Inventory, Sales, Finance, Human_Resource, Reporting,** and **Procurement.**
- **Binary Assignment: * Value 1 (Used):** Assigned if the specific module name was detected within the respondent's text string.
    - **Value 0 (Not Used):** Assigned if the module name was absent from the respondent's selection.
- **Dimensionality Reduction:** Once the individual binary columns were successfully created, the original raw text columns (**MODULE, UNUSED_MODULE,** and **UNUSED_REASON**) were deleted. This streamlined the dataset and removed redundant information.

- **Analytical Impact:** This procedure transformed qualitative list-data into a quantitative format. It allowed the WEKA models (such as J48 and Random Forest) to evaluate the statistical weight of each specific ERP module individually, identifying exactly which functional areas contribute most significantly to the "Usage Gap."

v. Feature Engineering and Construct Normalization

To move beyond raw data and uncover deeper patterns regarding the "Usage Gap," a series of sophisticated feature engineering steps were performed in Excel. This process involved creating composite scores and normalizing qualitative perceptions into quantitative metrics.

- **Behavioral Feature Engineering:** New analytical columns were derived to quantify the extent of ERP underutilization:
  - **MODULE_COUNT**: Calculated the breadth of system usage by summing the six binary module columns: =[@INVENTORY]+[@SALES]+[@PROCUREMENT]+[@FINANCE]+[@[HUMAN_RESOURCES]]+[@REPORTING].
  - **MODULE_LEVEL**: Classified users into three intensity tiers based on their count: =IF(MODULE_COUNT<=2,"Low",IF(MODULE_COUNT<=4,"Medium","High")).
  - **GAP_SCORE**: A diagnostic sum of behavioral red flags: =[@[FREQ_USE]]+[@[TASK_COMP]]+[@[IS_THERE_UNUSED]].
  - **GAP (Target Class)**: Defined the presence of a gap using a threshold logic: =IF([@[GAP_SCORE]]>=2,1,0).
- **Construct Consolidation (Second Model Preparation):** To analyze psychological and organizational factors, individual Likert items were aggregated into unified constructs by calculating their mathematical means:
  - **EASE_OF_USE**: =AVERAGE(PEOU1, PEOU2)
  - **USEFULNESS**: =AVERAGE(PUS1, PUS2)
  - **SYSTEM_QUALITY**: =AVERAGE(SYQ1, SYQ2)
  - **INFORMATION_QUALITY**: =AVERAGE(INQ1, INQ2)
  - **TOP_MGT**: =AVERAGE(TMS1, TMS2)
  - **SKILLS**: =AVERAGE(ISK1, ISK2)

- **INTENTION**: =AVERAGE(COI1, COI2, COI3)

vi. Data Type Audit and WEKA Preparation

The dataset was prepared for use with J48, Random Forest, and Bayes in WEKA:

- **Attribute Selection and Dimensionality Reduction:** Removing extra columns helped cut down noise while boosting how well the model works. Columns like ID or GAP_SCORE, which only supported temporary math steps, had little value later. Even separate modules and frequency entries got dropped along the way. Fewer inputs meant cleaner data flowing into analysis. This step simplified things without losing key details. Less clutter often leads to sharper results in modeling tasks

  The final model focused exclusively on high-level predictors: ORG_TYPE, POSITION, USG_PERIOD, ERP_SYS, MONITOR, GAP (Target), and the specific usage barriers.

- **WEKA Merging and Discretization:**
  - **Value Merging**: The MergeManyValues filter was applied to the MONITOR attribute to simplify the labels into a binary Low (merged from No, Rarely, Sometimes) and High (Regularly) classification.
  - **Threshold Binarization**: For the averaged scale constructs, a WEKA MathExpression filter, `ifelse(A<=2.5,0,1),` was utilized to discretize continuous averages into binary categories (0 for Low, 1 for High) based on a 2.5 midpoint.
- **Human-Readable Labeling:** The RenameNominalValues filter was applied across all attributes to ensure the decision tree output remained interpretable:
  - **Target (GAP)**: 0 → No_Gap, 1 → Gap.
  - **Barriers, such as TRAINING, COMPLEXITY**: 0 → Low, 1 → High.
  - **Averaged Scales**: 0 → Low, 1 → High.
  - **Demographics**: Mapped numeric IDs back to their categories, such as ORG_TYPE 1 → Supermarket, POSITION 4 → Senior_Mgt).

4.4.3.2 Handling Missing Values

The management of missing data is a vital step in the CRISP-DM Data Preparation phase. Incomplete datasets can lead to biased results or errors when training machine learning algorithms like J48 and Random Forest.

i What is the Handling Missing Values?

Values missing show up when certain variable entries are blank within a dataset, especially common in numerical studies. When people skip questions, mistakes happen in file transfers, or someone enters information incorrectly - gaps appear. These omissions matter because they weaken how strong and correct the analysis becomes. Without addressing them properly, outcomes risk being misleading or less reliable.

ii Listwise Deletion of Invalid Records

Upon reviewing the initial database (DB) of 149 responses, a data auditing process was conducted. One specific record (the first response) was identified as containing inconsistent and incomplete data that did not meet the research criteria.

- One entry got removed during listwise deletion using Excel. This case was dropped entirely from the data set. Entire rows tied to missing values disappeared. The cleanup left the remaining records intact. Missingness triggered full-row exclusion. A single observation existed analysis at that point
- Eventually, 148 usable answers made up the final set - this shift came about because only clear, trustworthy inputs were kept. Quality mattered more than quantity when shaping the dataset used for training machine learning systems.

iii Proactive Validation Strategy (Mandatory Response)

For the remaining 148 records, a proactive prevention strategy was used to ensure zero missing values:

- **Mandatory Field Constraints:** The digital survey instrument was configured with a "Required" constraint for every question. This technical setting ensured that no participant could submit the form without answering all Likert-scale items, demographic questions, and behavioral usage fields.
- **Database Audit:** Following the data export to Excel, a manual audit confirmed that all 148 rows and their respective columns were 100% populated.

iv Data Integrity and Final Status

Because the survey design enforced total completion, the dataset achieved a 100% completion rate. This eliminated the need for reactive statistical techniques, such as Imputation, such as replacing missing values with the mean or mode using WEKA's ReplaceMissingValues filter. By maintaining the original responses exactly as provided by the retail employees, the study ensures maximum integrity in identifying the patterns behind the ERP usage gap.

v Feature Selection

In the CRISP-DM framework, **feature selection** is the process of identifying and retaining the most relevant attributes for use in model construction. This step is critical because including redundant or irrelevant data can lead to "overfitting," where the model learns noise instead of actual patterns, thereby reducing its predictive accuracy on new data.

vi Data Dimensionality Reduction

To streamline the analysis, several types of attributes were removed from the primary dataset:

- **Intermediate Calculation Columns:** Once the target class (GAP) and engineered features (**MODULE_COUNT**, **MODULE_LEVEL**) were successfully created, the raw columns used for those calculations (such as FREQ_USE, TASK_COMP, and IS_THERE_UNUSED) were removed to prevent "data leakage" and redundancy.

Qualitative Text Fields: Original raw text columns, such as REASON, MODULES, and UNUSED_REASON, were deleted, as they had already been transformed into binary numerical formats suitable for WEKA.

vii Attribute Relevance for Classification

The selection process focused on retaining high-impact predictors that influence ERP underutilization. For the final modeling phase, the following features were selected based on their theoretical and practical relevance:

- **Organizational Context:** ORG_TYPE, POSITION, and USG_PERIOD.
- **System Factors:** ERP_SYS and MONITOR.

Usage Complexity: MODULE_LEVEL and the binary usage barriers, such as TRAINING, COMPLEXITY, and MGT_SUPPORT.

viii Final Model Readiness

By narrowing the dataset to 148 valid responses and the most significant attributes, the feature selection process ensured that the J48 Decision Tree would focus only on the variables that statistically contribute to the "Usage Gap." This resulted in a cleaner, more interpretable tree structure that clearly illustrates the factors driving ERP underutilization in Bahrain's retail sector.

4.4.3.3 Creation of the Target Variable

In classification-based machine learning, the target variable (or class label) is the primary attribute that the model is trained to predict. For this research, two distinct target variables were engineered to analyze different dimensions of ERP underutilization in the Bahraini retail sector.

i The Usage Gap Target (Model 1)

The first model aimed to identify whether a significant gap exists between system potential and actual employee usage. This target was engineered using a composite logic derived from three behavioral indicators:

- **Engineering Logic:** A intermediate metric, GAP_SCORE, was calculated by summing the scores of FREQ_USE, TASK_COMP, and IS_THERE_UNUSED.
- **Binarization:** To create a clear classification, the continuous score was converted into a binary target using the Excel formula: =IF(GAP_SCORE>=2, 1, 0).

- **Class Labels:** In WEKA, these were labelled as "Gap" (1), representing underutilization, and "No_Gap" (0), representing effective system integration.

ii Continuance Intention Target (Model 2)

The second model focused on the psychological aspect, whether users intend to keep using the ERP system. This required consolidating multiple Likert-scale items into a single, actionable class.

- **Engineering Logic:** The target variable INTENTION was created by calculating the mathematical average of three specific survey items: COI1, COI2, and COI3, such as =AVERAGE(COI1:COI3).
- **Discretization:** Since decision trees perform best with categorical data, the continuous average was discretized using a threshold of 2.5, the neutral midpoint.
- **WEKA Implementation:** The MathExpression filter ifelse `(A<=2.5, 0, 1)` was applied, where 0 was labeled as "Low Intention", and 1 was labeled as "High Intention".

iii Strategic Importance

By engineering these target variables, the raw survey responses were transformed into structured outcomes. This allows the J48 and Random Forest algorithms to pinpoint exactly which factors (such as lack of training or system complexity) are the strongest predictors of a "Usage Gap" or a "Low Intention" to use the system in the future in Bahrain.

4.4.4 Phase 4: Modelling

In this phase, three supervised learning algorithms were selected to classify and predict the ERP "Usage Gap" within the Bahraini retail sector. These models were chosen for their ability to handle the categorical nature of the survey data and provide interpretable insights into employee behavior.

4.4.4.1 J48 (Decision Tree)

The J48 algorithm is the WEKA implementation of the C4.5 decision tree, which classifies data by creating a flowchart-like structure. It was utilized to provide a clear visual map of the "decision

path" leading to system underutilization. By splitting the data based on information gain, it identifies which specific factors,

such as lack of training or organizational position, act as the strongest predictors of a usage gap, making the results easy to interpret for practical management decisions.

Figure 24: J48 decision tree of the survey generated by Weka (Survey)

Figure 24 illustrates the visual output of the J48 Decision Tree, which serves as a practical roadmap for understanding the specific factors that drive the ERP usage gap in Bahrain's retail sector. At the heart of the model, the "MONITOR" attribute emerges as the most critical predictor, appearing as the root node and suggesting that whether an organization actively tracks system usage is the primary dividing line between effective integration and underutilization. From this starting point, the tree branches into secondary influences such as the employee's "POSITION" and their "USG_PERIOD" (usage duration), demonstrating how a person's specific role and their level of experience further refine the likelihood of a performance gap. By tracing these logical paths to the final "Gap" or "No_Gap" classifications, the figure provides a clear, data-driven narrative that helps management pinpoint exactly where interventions, such as enhanced monitoring or targeted training, are most needed to bridge the usage divide.

```
J48 pruned tree
------------------

MONITOR        = Low
|   ORG_TYPE = Supermarket: Gap (20.0/6.0)
|   ORG_TYPE = Clothing
|   |   TRAINING = Low: Gap (11.0/3.0)
|   |   TRAINING = High: No_Gap (6.0/1.0)
|   ORG_TYPE = Electronics: Gap (16.0/3.0)
|   ORG_TYPE = Pharmacy
|   |   EXTERNAL_TOOLS = Low: Gap (8.0)
|   |   EXTERNAL_TOOLS = High: No_Gap (4.0/1.0)
|   ORG_TYPE = Other: No_Gap (7.0/2.0)
MONITOR        = High
|   EXTERNAL_TOOLS = Low
|   |   USG_PERIOD = <6m: Gap (4.0)
|   |   USG_PERIOD = 6m_1y
|   |   |   ERP_SYS = SAP: Gap (2.0)
|   |   |   ERP_SYS = Oracle: Gap (3.0)
|   |   |   ERP_SYS = Microsoft_Dyn: No_Gap (6.0/1.0)
|   |   |   ERP_SYS = Odoo: No_Gap (0.0)
|   |   |   ERP_SYS = Other: No_Gap (1.0)
|   |   USG_PERIOD = 1_2y: No_Gap (5.0/1.0)
|   |   USG_PERIOD = 2_5y
|   |   |   TRAINING = Low: Gap (11.0/4.0)
|   |   |   TRAINING = High: No_Gap (5.0/1.0)
|   |   USG_PERIOD = >5y: No_Gap (22.0/4.0)
|   EXTERNAL_TOOLS = High: No_Gap (17.0/1.0)

Number of Leaves  :     18

Size of the tree :      26
```

Figure 25: pruned tree classification rules and tree dimensions (Survey)

To ensure that the model's findings are generalizable, a 10-fold cross-validation method was performed. The analysis yielded a correctly classified instance rate of 71.62%, indicating that the selected attributes provide a solid foundation for prediction. In machine learning research involving behavioural and survey-based datasets, accuracy levels between 70% and 80% are commonly considered acceptable and practically meaningful due to the presence of subjective human responses, overlapping behavioural patterns, and limited sample sizes. Unlike highly structured technical datasets, social and organisational datasets naturally contain variability and uncertainty, which makes achieving extremely high accuracy more difficult.

The J48 decision tree model in this study achieved an accuracy of 71.62% with a Kappa statistic of 0.4346 using 10-fold cross-validation. This result indicates that the model performs substantially better than random classification while maintaining balanced prediction capability across both classes ("Gap" and "No_Gap"). The obtained Kappa value reflects moderate agreement between predicted and actual outcomes, which is generally considered acceptable in exploratory organisational and behavioural studies.

Furthermore, the achieved accuracy is supported by relatively balanced class performance, where the model correctly identified 66.2% of the "No_Gap" cases and 77.5% of the "Gap" cases. This

balance is particularly important because the objective of the study is not only maximizing prediction accuracy, but also identifying interpretable organisational patterns influencing ERP underutilisation.

Previous studies in ERP adoption and behavioural prediction frequently report moderate predictive performance when using survey-driven datasets because employee behaviour is influenced by multiple human, organisational, and contextual factors that cannot always be perfectly separated by machine learning models. Therefore, the achieved accuracy level is considered academically acceptable for supporting decision-making and identifying the major predictors of ERP usage gaps within the Bahraini retail sector.

Although the Random Forest model achieved lower performance on the retail dataset (60.81%), the J48 model maintained stronger interpretability and more stable classification behaviour, while also outperforming Naïve Bayes (68.24%). This further supports the suitability of the selected J48 model for explaining ERP usage gap patterns within the context of this study. Additionally, the Kappa statistics were calculated at 0.4346, reflecting a moderate and reliable consistency between the predicted and actual outcomes. This level of accuracy, supported by a balanced True Positive (TP) rate across both classes, confirms that the model is effective at identifying the specific organizational and behavioral patterns that lead to an ERP usage gap.

Table 10: Confusion matrix of J48 algorithm (Survey)

	PREDICTED: LOW (A)	**PREDICTED: HIGH (B)**
ACTUAL: LOW (A)	51	26
ACTUAL: HIGH (B)	16	55

Table 10 demonstrates that the model achieved a solid level of correct classification across both categories, reflecting the strong predictive capability of the decision tree. However, there is a moderate error rate present, including a Mean Absolute Error of 0.3593. The minor misclassifications observed, particularly within the "No_Gap" category, where 26 instances were incorrectly labelled, may indicate that certain behavioral nuances or outlier examples were not fully captured by the current attribute set. Despite these variances, the model's ability to correctly identify 106 out of 148 instances confirms its overall effectiveness in mapping the ERP usage gap

within the retail sector.

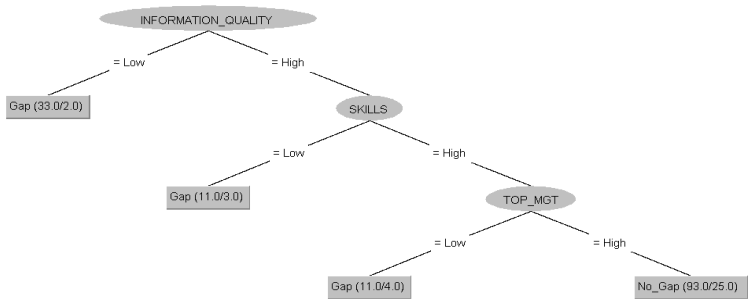

Figure 26: J48 Scale model decision tree of the survey generated by Weka (Survey)

Figure 26 offers a data-driven look at the ERP usage gap in Bahrain's retail industry, achieving a strong 75.68% accuracy. It reveals that Information Quality is the most important factor the foundation upon which everything else is built. From there, the model shows that success depends on a balance of User Skills and Management Support. Essentially, the tree provides a clear roadmap for organizations (if you want to eliminate usage gaps, you must first ensure your data is reliable, then empower your staff with the right skills and the backing of their leaders).

Finally, the decision to split the analysis into two distinct trees was a deliberate effort to avoid the 'one-size-fits-all' trap common in ERP research. While the initial model highlighted the systemic importance of active monitoring, the subsequent model offered a more intimate look at the user's experience, specifically how tenure and training either bridge or widen the performance gap. This dual-layered interpretation proved essential, as it revealed that ERP success isn't dictated by a single factor, but by a complex interplay of organizational pressure and individual competence.

Consequently, the findings offer a far more practical roadmap for retail managers, shifting the focus from 'whether' the system is used to 'how' and 'why' users engage with it.

Table 11: The Scale confusion matrix of the J48 algorithm (Survey)

	PREDICTED: LOW (A)	PREDICTED: HIGH (B)
ACTUAL: LOW (A)	68	9
ACTUAL: HIGH (B)	27	44

Table 11 demonstrates that the scale-based J48 model achieved a strong classification performance across both classes, correctly identifying 68 instances of the “No_Gap” category and 44 instances of the “Gap” category. The confusion matrix indicates that the model was particularly effective in recognizing organisations without ERP usage gaps, reflected by the high recall rate of 88.3% for the “No_Gap” class. However, 27 “Gap” instances were incorrectly classified as “No_Gap,” suggesting that some gap-related patterns may overlap with organisations showing positive ERP usage characteristics. Despite these misclassifications, the model achieved an overall accuracy of 75.68% with a Kappa statistic of 0.5079, indicating a moderate to strong level of agreement between the predicted and actual classifications. From the above results, it can be deduced that factors like Information Quality, Skills, and Top Management Support are very vital in classifying the ERP gaps in the retail industry.

4.4.4.2 Random Forest

Random Forest is an ensemble learning technique that constructs a multitude of individual decision trees during training and merges their results. It was employed to enhance predictive accuracy and ensure the robustness of the findings. By aggregating the votes of many different trees, the Random Forest model reduces the risk of errors associated with a single tree and confirms the stability of the patterns identified within the retail context.

The Random Forest model achieved a classification performance with 90 correctly classified instances, resulting in an overall accuracy of 60.81%. A total of 58 instances were misclassified, corresponding to an error rate of 39.19%. The detailed classification results are presented in the

table below, which illustrates the confusion matrix of the Random Forest model for the retail dataset.

Table 12: Confusion matrix of the Random Forest algorithm (Survey)

	PREDICTED: LOW (A)	PREDICTED: HIGH (B)
ACTUAL: LOW (A)	47	30
ACTUAL: HIGH (B)	28	43

Table 12: The Random Forest model achieved a weighted average F-measure of 0.608 and a Kappa statistic of 0.2158, reflecting a fair predictive consistency. The model also produced an ROC Area of 0.647, indicating its ability to distinguish between classes. Furthermore, the analysis showed a Mean Absolute Error of 0.4402 and a Root Mean Squared Error of 0.4936, representing the model's overall prediction error and precision.

4.4.4.3 *Naive Bayes*

Naive Bayes is a probabilistic classifier based on Bayes' Theorem, which assumes independence between all predictors in the dataset. It served as a high-speed baseline model to verify the consistency of the classification results. By calculating the statistical probability of a "Usage Gap" occurring based on the presence of certain features, it provided a mathematical validation of the findings that complemented the logic used by the tree-based algorithms.

In the survey study, the same dataset and 10-fold cross validation method were used to follow the same experimental design applied to the previous algorithms. The Naive Bayes model also achieved a solid classification performance, with 101 instances correctly classified, resulting in an overall accuracy of 68.24%. However, a total of 47 instances were misclassified, corresponding to an error rate of 31.76%. The detailed classification results are presented in the table below, which illustrates the confusion matrix of the Naive Bayes model.

Table 13: Confusion matrix of the Naïve Bayes algorithm (Survey)

	PREDICTED: LOW (A)	PREDICTED: HIGH (B)
ACTUAL: LOW (A)	54	23
ACTUAL: HIGH (B)	24	47

4.4.4.4 Feature Importance Analysis

The analysis of the decision tree models revealed that organisational variables played the most significant role in predicting ERP usage gaps within the Bahraini retail sector. Monitoring practices emerged as the strongest predictor in the general survey model, followed by training availability and the use of external tools. These variables consistently appeared near the top levels of the J48 decision tree, indicating their strong influence on ERP utilisation behaviour.

Within the scaled model, Information Quality, ICT Skills, and Top Management Support demonstrated the highest predictive importance. The results suggest that employees are more likely to underutilize ERP systems when information generated by the system is perceived as unreliable or when sufficient organisational support is absent.

The findings further indicate that organisational and human-related variables contributed more strongly to ERP gap prediction than purely technical variables. This supports the TOE framework, where organisational conditions significantly influence technology utilisation behaviour.

4.4.5 Phase 5: Evaluation

The evaluation phase focuses on assessing the developed machine learning model to determine its suitability for supporting the defined study objectives, specifically the prediction of ERP usage gaps. The purpose of this phase was to evaluate the models' ability to classify ERP users into 'Gap' and 'No_Gap' categories using the prepared dataset.

Due to limited size of the available dataset, the 10-fold cross-validation method was applied in Weka to ensure the reliability of the assessment. This approach allowed all the 148 records in the dataset to be used for both training and testing in different iterations, providing a more robust estimate of generalization performance and reducing the bias associated with a single train-test split.

Initially, three classification algorithms were evaluated in order to identify the most suitable algorithm for the research problem which are: J48 Decision Tree, Random Forest, and Naive Bayes. The evaluation revealed distinct performance variations across the three algorithms during both the case study-based model and the survey-based model. Notably, the models trained on the locally collected dataset exhibited lower predictive performance compared to the reference dataset, indicating differences in data characteristics and underlying patterns between the two contexts.

However, the accuracy numbers effectively narrowed the choice to be only between J48 Decision Tree and Naïve Bayes, as Random Forest experienced a large performance drop compared to the other algorithms when it was tested on a local dataset. This indicates a greater sensitivity to dataset size and data quality collected by much more than other algorithms.

Between these two finalists, the J48 Decision Tree was ultimately selected as the most suitable model for this study. While Naïve Bayes followed closely in terms of raw accuracy (68.24%), it operates as a probabilistic model that does not explicitly map the relationships between variables. In contrast, J48 achieved the highest accuracy of 71.62% and, more importantly, offered superior interpretability. It provided a clear visual representation of the decision-making process.

The white-box nature of the J48 Decision Tree algorithm allows retail managers to see the exact logic path that leads a user to a "Gap" or "No_Gap" classification. It started with MONITOR (organizational monitoring) as a central factor in determining classification and using the other factors, such as ORG_TYPE (organization type), TRAINING, EXTERNAL_TOOLS, and USG_PERIOD (usage period) to classify. In business-oriented problems such as ERP usage analysis, this plays a critical role, as decision-makers require clear and explainable insights rather than purely predictive outputs.

Furthermore, the reported Kappa statistic of 0.434 has confirmed the statistical validity of this choice as it indicates a moderate and reliable level of agreement that far exceeds random chance. This can also be proved if we looked at the ROC Area of 0.697 for the J48 model, it indicates a moderate ability to distinguish between the two classes. This level of performance suggests that the model is reasonably effective in differentiating between Gap and No_Gap instances, although there remains room for improvement.

Table 14: Evaluation models (Survey)

Model	Accuracy	Precision	Recall	F-measure	Kappa	ROC Area
J48	71.62%	0.722	0.716	0.716	0.434	0.697
Naive Bayes	68.24%	0.682	0.682	0.682	0.363	0.706
Random Forest	60.81%	0.609	0.608	0.608	0.215	0.647

The results from the confusion matrices provided further information on the performance of the classifier. The J48 model had an even spread across the two classes, having correctly classified 51 "No_Gap" cases as True Negatives and 55 "Gap" cases as True Positives. Although the Naive Bayes model showed some bias towards classifying people without a gap (54 cases), the J48 model had the most consistent predictive accuracy. With Kappa and ROC values being moderate, it is evident that this model was appropriate for the data in question.

4.4.6 Phase 6: Deployment

The deployment phase is the last phase of the CRISP-DM methodology. It is meant to link between the analytical findings and the actionable strategies in the real-world context. Since this research

uses a primary survey-based dataset, the deployment focus was not on integrating a new system, but on the creation of a Diagnostic Framework. This framework was used as a Decision Support System (DSS) that allows management to identify and mitigate ERP underutilization based on user-reported behavior, observations and perceptions.

As mentioned in the previous phase, what makes the J48 model particularly useful in a real-world context is its ability to support actionable decision-making. When survey data is entered, the model helps identify whether a user is experiencing underutilization. More importantly, the J48 tree explicitly reveals the reasoning behind this classification by highlighting the contributing factors. This allows the deployment to move from simple "prediction" to "prescription," which is an advanced type of analysis that could tell managers exactly which organizational lever to pull to close the gap.

4.5 Strategic Management Interventions

Based on the diagnostic findings of the J48 model, three specific deployment strategies are proposed for Bahraini retail organizations:

- **Targeted Monitoring Protocols:** Since monitoring was identified as the key factor, managers can use simple regular, lightweight follow-ups to address ERP usage gaps. If a department is flagged to have a high risk of underutilization, introducing regular reporting can help spread the culture of using ERP systems as part of the daily routine.
- **Resource Optimization for Training:** Rather than adopting company-wide training, which is an expensive option, the model uses the survey results to identify specific clusters, like certain organization types or usage periods that cause high usage gaps and allows for targeted training.
- **Policy-Driven Transition from External Tools:** For departments where the model classifies them as external tools users, deployment could involve administrative policies that gradually reduce the use of these tools in order to increase the utilization of ERP modules.

4.5.1 Proof of Concept

Although this study is academic in nature, it demonstrates how a continuous feedback approach could work in practice. In a real-world setting, retail companies could run short quarterly ERP health surveys and use the results to monitor system usage. The data could then be analyzed using the J48 model to create a simple management dashboard, highlighting areas of underutilization and potentially notify them when the usage status of the system significantly changes. This allows decision-makers to address issues early before they lead to operational problems or financial losses.

4.6 Chapter Summary

This chapter presented the systematic application of the CRISP-DM methodology to analyze the ERP usage gap within the Bahraini retail sector. The study progressed from a validation case study, which achieved an accuracy of 89.03%, to the main analysis of a locally collected survey dataset.

During the data preparation and modeling phases, tree-based machine learning algorithms were applied to identify the behavioral and organizational factors influencing ERP underutilization. The J48 Decision Tree emerged as the most suitable model for the Bahraini context, achieving an accuracy of 71.62% while also offering strong interpretability. The model identified Organizational Monitoring as the primary factor driving the usage gap, supported by some other secondary variables.

By successfully classifying users into “Gap” and “No_Gap” categories, this chapter provided empirical evidence that helps bridge the gap between ERP system capability and actual organizational use. These findings form the foundation for the discussion and recommendations presented in the following chapters.

Chapter 5: Results analysis and Discussion

5.1 Introduction

This chapter builds on the results presented in the previous chapter by moving beyond the description of findings toward their interpretation and meaning. While Chapter 4 focused on analysing the data and identifying patterns in ERP usage, this chapter explains why these patterns occur and what they imply within the context of retail organisations in Bahrain. The discussion is structured to provide a deeper understanding of the ERP usage gap identified in this study. It begins by justifying the use of both the case study and survey datasets, followed by a comparison between expected usage (intention) and actual usage in practice. The chapter then interprets the key findings by linking them to established theoretical frameworks, including the TOE framework and the DeLone and McLean IS Success Model. Additionally, the issue concerning the ERP utilization gap was investigated with an intensive gap analysis, as well as its impact on organizational performance. Through combining these issues, the chapter offers a detailed insight into the reasons and aspects contributing to the ERP systems' underutilization problem. Thus, in general, the chapter defined the problem and explained the underlying reasons and implications derived from the analysis.

Furthermore, the chapter includes an evaluation of the research hypotheses based on the empirical findings and machine learning results to validate the proposed relationships within the study model.

5.2 Overview of finding

This section provides a brief synthesis of the key findings from Chapter 5, focusing on how the results contribute to understanding ERP usage behaviour rather than simply restating model performance.

In the case study analysis, the models demonstrated impressive predictive power. The Random Forest algorithm took the lead with an accuracy of 89.03%, closely followed by J48 at 87.34% and Naïve Bayes at 86.50%. These strong results indicate that ERP usage behavior in this particular case study is quite structured and can be effectively captured using ensemble learning techniques. Whereas in the latter situation, the predictive performance was comparatively lower but still more complicated than in the former situation. In this context, J48 was observed to be the most efficient predictor model with an accuracy rate of 71.62%, followed by Naïve Bayes with 68.24%. Interestingly enough, the Random Forest was the least accurate among the three, with 60.81%. Nevertheless, despite this, decision tree algorithms were found very useful because of their interpretative value. They highlighted clear behavioral and organizational patterns that help explain why these gaps in ERP usage exist in the first place.

Additionally, we found that scaling the J48 model improved its performance to 75.68%. This tells us that grouping variables into broader categories like Information Quality, Skills, and Top Management Support doesn't just make the model clearer; it actually makes it better at generalizing these patterns to the real world. These differences in model performance highlight the importance of understanding the nature of the data sources used in this study, which is discussed in the following section.

5.3 Results Summary

The deep dive into ERP usage gaps in the Bahraini retail sector uncovered some key insights from both the initial survey data and the machine learning case study analysis. Here's a quick rundown of the main findings:

Organizational Factors Take Center Stage: Both sets of data pointed out that the "usage gap" in ERP systems is mainly influenced by organizational and human factors, rather than just technical glitches. This suggests that how well a system is used after it's implemented largely depends on management's involvement.

Predictive Model Performance: The machine learning analysis demonstrated that ERP usage

patterns could be predicted quite accurately using the J48 and Random Forest algorithms. The J48 Decision Tree stood out as the best model for this dataset, hitting an accuracy rate of 71.62%, with Naive Bayes following at 68.24%. These results back up the idea that the survey attributes chosen are significant indicators of how users behave.

The Role of Monitoring: During the case study analysis, researchers found that Organizational Monitoring is the most crucial factor influencing ERP usage. The results suggest that without proper monitoring, employees are likely to overlook the capabilities of the ERP systems.

Complexity and Training Connection: A recurring theme in both datasets was the negative correlation between system complexity and user skill. The findings show that if a system is too complex, it can widen the usage gap unless there's a strong emphasis on structured training and ongoing support. The findings demonstrated that employees' ICT skills were not the primary barrier to bridging the usage gap; instead, Perceived Usefulness (PUS) and System Quality (SYQ) were found to be the most significant predictors of full utilization.

Reliance on External Tools: The results highlighted that many employees often turn to manual solutions and third-party tools like Excel to bypass the complexities of ERP functionalities. This trend is particularly noticeable among lower-level employees, especially when they feel unsupported by upper management.

Consistency in Classification: The agreement among the three chosen factors remained steady throughout the analysis, highlighting the significance of these insights. The evaluated algorithms (J48, Naive Bayes, and Random Forest) confirm the dataset's stability. This shows that the identified predictors, like Management Support, Position, and Usage Period, hold strong across various analytical frameworks.

5.4 Conceptual Justification of Data Sources

The study utilises a case study from South Korea alongside a survey conducted in the Bahraini retail sector to support the research objective.

The case study focuses on users' intention to continue using ERP systems. This represents expected behaviour, meaning how users are likely to act when they are satisfied with the system and perceive it as useful. In this context, ERP usage is influenced mainly by system-related and perceptual factors, which makes user behaviour more stable and predictable.

In contrast, the survey dataset reflects actual ERP usage within retail organisations in Bahrain. It captures what is occurring in practice, where system usage is affected by organisational conditions such as training, monitoring, and management support. These factors introduce challenges that may prevent users from fully utilising the system.

These two datasets serve the main function of distinguishing between user intention and actual user behavior. The case study indicates that most users actually have a positive intention toward the continuation of ERP system use, but the results from the survey indicate that this intention does not lead to effective usage in real organizational practice.

This difference helps to explain the ERP usage gap. The gap does not arise because users reject the system, but because organisational and operational factors limit proper usage. Therefore, combining the two datasets provides a clearer understanding of the mismatch between expected ERP usage and actual practice, particularly in the context of the Bahraini retail sector.

5.5 Comparison for Insight (Interpretation)

A more meaningful comparison of the results can be achieved by aligning the findings with the core research objective, which is to identify the gap between expected and actual ERP usage in the Bahraini retail sector and to explain the reasons behind this gap.

An important distinction in this study lies in the difference between the target variables used in each dataset. The case study, which was conducted in South Korea, focuses on predicting users' intention to continue using the ERP system, representing expected or future-oriented behavior. In contrast, the survey dataset focuses on identifying the actual ERP usage gap within Bahraini retail organizations, reflecting real-world system utilization in practice. This distinction allowed the study to capture both what users intended to do and what occurred in organizational settings. This comparison is not intended to evaluate model performance across identical conditions, but rather to provide complementary insights by contrasting expected user behaviour with actual organizational practices.

At the algorithm level, the case study dataset achieved high predictive performance, with Random Forest producing the best results. This suggests that users' intention to continue using ERP systems follows relatively stable and predictable patterns. These patterns are largely influenced by technological and perceptual factors such as perceived usefulness, system quality, and satisfaction, which are consistent with the DeLone and McLean IS Success Model. Under such conditions, users demonstrate a generally positive orientation toward ERP systems, reflecting strong expected usage.

However, the survey dataset reveals a different and more complex reality. The J48 model provided the most meaningful results, highlighting clear decision paths that lead to the presence of an ERP usage gap. The structure of the decision tree shows that actual ERP usage is highly dependent on organisational and contextual conditions such as monitoring practices, training availability, user experience, and the use of external tools. While the concept of behavioural intention seems quite stable, usage is more influenced by operational considerations as well as workplace politics. The observations made here clearly fit into the TOE framework, since organisational and environmental considerations are essential in explaining technology usage.

At the dataset level, the comparison between intention (case study) and actual usage (survey) provides direct evidence of the ERP gap. While the case study indicates that users are generally inclined to continue using ERP systems, the survey results show that this intention does not always translate into effective system usage. This mismatch confirms the existence of a gap between

expected behaviour (intention) and actual practice (usage), particularly within the context of Bahraini retail organisations.

To further clarify this distinction, Table 15 presents a structured comparison between the case study and survey models.

Table 15: Comparison Between Case Study and Survey

Aspect	**Case Study (South Korea)**	**Survey (General Model)**	**Survey (Scaled Model)**
Target Variable	Continuance Intention	Actual ERP Usage Gap	ERP Gap (Construct-based)
Best Model	Random Forest (~89%)	J48 (~71.6%)	J48 (~75.7%)
Nature of Data	Behavioural/perceptual	Organisational + behavioural	Theoretical constructs
Dominant Factors	Usefulness, system quality, satisfaction	Monitoring, training, and external tools	Information quality, skills, and top management
Framework Alignment	DeLone & McLean	TOE (operational level)	TOE + IS Success (integrated view)
Key Insight	Users are willing to use ERP	ERP is not always used effectively	Root causes of the usage gap

Following Table 15, the comparison clearly highlights a critical insight that directly supports the research objective. While the case study (South Korea) demonstrates that users generally exhibit a strong intention to continue using ERP systems, the survey results from the Bahraini retail sector reveal that this intention is not consistently translated into actual system usage.

This distinction reveals that the problem of ERP use discrepancy stems not from users' attitude towards the system, but from the organizational context in which the system operates. As for the case study, the most influential factors are predominantly perceptual and system-specific, and this increases predictability and hence accuracy of model predictions. On the contrary, when using

survey data, the most important factors become organizational (monitoring, training, management support), and they introduce greater variability.

More importantly, the scaled model also makes things more complex since it demonstrates that regardless of the strength of the variables related to the system, such as the quality of information, lack of user skill or senior management support may result in the non-use of the system. Therefore, one can state that it is not only a matter of having a good system but being able to use it.

Therefore, the comparison confirms that the ERP usage gap in the Bahraini retail sector is primarily driven by organisational and human factors rather than technological limitations. It also explains why organisations may invest in advanced ERP systems and still fail to achieve the expected benefits. The gap exists because the conditions required to translate intention into effective usage are not fully met in practice.

While the general survey model identifies where the ERP usage gap occurs, the scaled model provides a deeper explanation by highlighting the underlying constructs, such as information quality, user skills, and top management support, that contribute to this gap.

An interesting observation is the decline in Random Forest performance within the Bahraini survey dataset compared to the case study dataset. In the case study, Random Forest achieved the highest accuracy because the dataset reflected more stable behavioural and perceptual patterns related to continuance intention. However, in the Bahraini retail survey, the data contained more organisational variability and contextual diversity, including differences in monitoring practices, training quality, management support, and employee behaviour across organisations.

Since Random Forest relies on combining multiple decision trees and detecting stable repeated patterns, the increased variability and relatively smaller dataset size reduced its ability to generalize effectively. In contrast, the J48 decision tree performed better because it was more capable of capturing direct and interpretable organisational decision paths related to ERP usage gaps.

The superior performance of J48 in the survey dataset is largely attributed to its interpretability and ability to model direct organisational decision paths. Unlike Random Forest, which performs better with highly stable and repetitive patterns, J48 was more suitable for capturing the diverse

behavioural and organisational conditions present within the Bahraini retail sector. This made the resulting model easier to interpret and more aligned with the research objective of explaining ERP usage gaps rather than only maximizing prediction accuracy.

5.6 Hypotheses Evaluation

This section evaluates the proposed research hypotheses based on the machine learning classification results, descriptive findings, and behavioural patterns identified from the ERP survey data collected from retail organizations in Bahrain.

H1: Information quality significantly contributes to reducing ERP system usage gaps.

Supported, the scaled J48 decision tree identified INFORMATION_QUALITY as the root node and the most influential predictor of ERP usage gaps. The findings showed that low information quality was strongly associated with ERP underutilization and the existence of usage gaps.

H2: Perceived usefulness (PUS) of the ERP system positively influences ERP utilization and reduces underutilization among retail employees.

Partially Supported, although perceived usefulness appeared within the scaled decision tree, the results indicated that usefulness alone was insufficient to eliminate ERP usage gaps when user skills were low. This suggests that perceived usefulness contributes to ERP utilization but must be supported by adequate ICT skills and organizational support.

H3: Higher levels of top management support (TMS) increase ERP system adoption and reduce reliance on manual or external tools.

Supported, Top management support appeared as an important decision factor within the scaled classification model. The findings demonstrated that strong managerial encouragement and organizational support contributed to reducing ERP usage gaps, particularly when employees already possessed sufficient ICT skills.

H4: Active organizational monitoring of ERP activities reduces the likelihood of ERP usage gaps.

Strongly Supported, the original J48 decision tree identified MONITOR as the root node, indicating that organizational monitoring was the most influential factor in predicting ERP usage gaps. Organizations with weak monitoring practices demonstrated significantly higher levels of ERP underutilization.

H5: Specialized ERP training positively influences employees' ability to utilize ERP modules effectively.

Supported, training repeatedly appeared as an influential factor within the decision tree structure. The results showed that employees with higher levels of ERP training demonstrated lower ERP usage gaps, even under conditions of weak monitoring or longer ERP usage periods.

H6: Organizational factors demonstrate higher predictive significance in identifying ERP usage gaps compared to technical factors when analysed using machine learning techniques.

Supported, the classification models consistently highlighted organizational and behavioural variables such as monitoring, management support, skills, and training as major predictors of ERP usage gaps. These factors appeared more influential than purely technical variables within the machine learning models.

H7: ERP usage gaps negatively influence users' continuance intention toward ERP systems.

Partially Supported, although the results identified ERP usage gaps and underutilization within retail organizations, most respondents still expressed positive continuance intention toward ERP systems, including their willingness to continue, increase, and regularly rely on ERP usage in the future. This indicates that ERP usage gaps may affect the effectiveness of ERP utilization; however, they did not completely reduce users' overall intention to continue using ERP systems.

Overall, the findings generally support the proposed hypotheses and demonstrate that ERP underutilization within Bahrain's retail sector is influenced primarily by organizational, behavioural, and user-related factors. The machine learning models further confirmed the effectiveness of classification techniques in identifying ERP usage gap patterns and interpreting ERP behavioural dynamics within retail organizations.

5.7 Discussion

The results of this research demonstrate that ERP gap utilization is not determined by one factor, but rather by a combination of technological, organizational, and human factors. It correlates perfectly with both the TOE framework and the DeLone and McLean IS success model.

Regarding technological issues, information quality and perceived usefulness are critical factors. The scaled model clearly demonstrates that information quality is the most dominant predictor of ERP usage gaps. This reluctance signifies a digital trust crisis within the operational workflow; users perceive data inaccuracies as a direct threat to their professional efficiency, leading them to adopt manual workarounds as a rational risk-mitigation strategy rather than a mere rejection of technology. This directly supports the IS success model, which identifies information quality as a key determinant of system usage and success.

However, the results indicate that technological factors alone are insufficient to explain ERP gaps. Organizational factors appear to play a more significant role. The decision tree identifies monitoring as the most critical variable in the survey dataset. Organisations that actively monitor ERP usage are significantly less likely to experience gaps. Monitoring acts as a control mechanism that enforces system usage and reduces reliance on informal practices.

In practical terms, organisational monitoring refers to activities such as ERP usage audits, login tracking, manager follow-ups, performance reviews, and monthly usage reporting. These practices help organisations ensure that employees consistently utilise ERP functionalities instead of relying on external or manual tools. The results suggest that organisations with stronger monitoring mechanisms are more likely to achieve effective ERP integration and reduce usage gaps.

Top management support also emerges as a key determinant. When even the highly skilled end-users lack the backing from management, then their under-utilization is bound to occur. This shows the role of having the backing from management in ensuring that the ERP implementation succeeds. From the view of the TOE model, this refers to the organizational aspect.

Human factors further explain the presence of ERP gaps. Training and user skills consistently appear as critical variables across both models. The findings show that the evidence proves that

general professional experience cannot compensate for a lack of specialized system knowledge. The persistent gaps in complex modules like Procurement and HR suggest that these functions require precise technical competencies that experience alone does not provide, framing continuous training as a strategic capital investment rather than a secondary support activity. This suggests that ERP adoption is not a one-time process but requires continuous learning and support.

An important insight from this study is the difference between intention and actual behaviour. The case study shows that users may have a high intention to continue using ERP systems. However, this discrepancy reveals a state of 'cognitive dissonance' among the workforce; while employees acknowledge the system's strategic value (intention), they are deterred by high-pressure environments or process rigidity (actual behavior). This underscores the need for training to evolve from 'technical operation' to 'seamless integration' within fast-paced retail tasks.

Another interesting finding relates to the role of external tools. Rather than indicating system failure, external tools appear to complement ERP systems. When combined with strong monitoring, they enhance performance. When monitoring is weak, they act as a workaround to compensate for system limitations. This demonstrates the environmental perspective of the TOE theory, which shows how organizations function within a wider environment of technologies.

Moreover, user experience is another key factor. Novices will be more prone to ERP gap occurrences as a result of their unfamiliarity with systems, while experts show effective use of systems. Nevertheless, experience may not suffice without training.

On the whole, the analysis reveals that ERP utilization gaps are mainly caused by organizational activities and human factors rather than just technology constraints. The reason for that is explained by the fact that the survey dataset is more insightful regarding the causes of ERP underutilization.

5.7.1 Comparison with GCC Studies

The findings obtained in this study correlate generally with previous ERP-focused studies carried out in the GCC region. Several studies carried out in Saudi Arabia and the UAE show that organizational support, training, and monitoring remain some of the key factors influencing ERP systems' performance and usage (Alkraijia, et al., 2020); (Salih, et al., 2022).

Similar to the results obtained in this research, previous GCC studies also identified that many organisations continue to experience ERP underutilisation even after successful system implementation. Employees often rely on external tools or partially use ERP modules due to insufficient training, lack of management follow-up, or operational pressures within the workplace.

An empirical investigation done in Saudi Arabia found employee training and continuous monitoring by managers as some of the strongest predictors of ERP success, particularly in retail and manufacturing firms. Similarly, research carried out in firms based in the UAE found employees in these firms avoiding the use of sophisticated features of the ERP systems due to their complexity. The above observations are consistent with those obtained from the current research, where it is found that monitoring and training are key in bridging ERP implementation gaps.

Despite these similarities, the Bahraini retail context demonstrates several unique characteristics. Retail organisations in Bahrain operate within highly dynamic and fast-paced business environments where employees frequently prioritize speed and operational convenience over strict system compliance. This may explain the continued dependence on external tools despite positive perceptions toward ERP systems. Therefore, the findings suggest that Bahraini organisations may require more structured monitoring mechanisms and continuous practical support compared to some other GCC sectors where ERP processes are more standardised and less operationally intensive.

However, this study differs from many regional studies by applying Artificial Intelligence and Machine Learning techniques to analyse ERP usage behaviour and classify ERP usage gaps. While most GCC studies focus mainly on adoption or implementation challenges, this research provides

a more analytical and predictive perspective by examining actual ERP utilisation patterns within the Bahraini retail sector.

5.7.2 Unexpected Findings and Observations

One unexpected finding observed in this study was the partial influence of perceived usefulness on ERP utilisation behaviour. While previous studies consider perceived usefulness as the key driver for system adoption, it can be inferred from the current results that even a positive attitude towards the ERP system does not guarantee its effective utilization. Employees might find the system useful, but they would still resort to using external tools because of operational pressures, lack of monitoring, or limited training.

The other interesting point relates to the impact of external tools. Initially, reliance on external applications such as spreadsheets was expected to indicate complete rejection of ERP systems. However, the findings revealed that external tools often functioned as supplementary workarounds rather than replacements for ERP systems, particularly in organisations with weak monitoring practices.

5.7.3 Future Analytical Implications

The findings also open opportunities for integrating Artificial Intelligence techniques with ERP behavioural analysis. By combining predictive modelling with ERP usage monitoring, organisations may be able to identify early warning signs of underutilization and provide proactive support for employees. Additionally, integrating the TOE framework with the DeLone and McLean IS Success Model offers a more comprehensive perspective for analysing ERP behaviour within the Bahraini retail environment.

Furthermore, ERP analytics in the future can incorporate real-time methods for detecting anomalies that would analyze the performance of employees at all times instead of depending only on their survey answers periodically. For example, machine learning algorithms might detect unusual trends like sharp drops in ERP usage, excessive use of spreadsheet applications, or persistent skipping of specific ERP modules. This would allow organizations to act quickly to provide necessary assistance, refreshment courses, or management intervention prior to

developing significant gaps in ERP usage. This would improve the efficiency of the company and ensure sustainable ERP operations in the retail industry.

The next important aspect is the possibility of integrating Natural Language Processing and smart support systems into ERP frameworks. Possible research directions may involve the analysis of complaints or feedback from employees regarding common issues with using ERP systems. Moreover, reinforcement learning might be explored in an effort to make recommendations for employee training depending on their personal usage and performance experience. The above-mentioned approaches can lead to designing adaptive ERP environments that will keep improving user engagement and preventing underutilization.

5.8 Gap Analysis

The findings of this study indicate that ERP usage gaps exist across multiple areas within retail organisations in Bahrain. These gaps are not random; rather, they can be clearly linked to specific organisational, technical, and user-related factors identified through both the decision tree models and the survey responses.

The analysis shows that one of the most critical points where the gap exists is in the inconsistency of system usage among employees. While ERP systems are implemented to standardise processes, actual usage varies significantly depending on the conditions within the organisation. This creates a clear gap between expected system use and what is observed in practice.

From an organisational perspective, a lack of monitoring the use of ERP systems and poor management support become significant factors that contribute to the issue. According to the decision tree, one of the most prevalent factors is monitoring, which means that in case a company fails to monitor the process, it would be more common for people to employee to rely on alternative methods.

Training is another major source of the gap. Both the model results and survey responses confirm that employees who receive limited or inadequate training are unable to fully utilise ERP functionalities. This results in partial adoption of the system despite the fact that the system itself

has the potential for effective support of the business processes. Furthermore, there were indications in some of the responses that indicated a lack of user awareness, suggesting that employees may not fully understand the purpose or benefits of the ERP system.

Technical factors also contribute to the gap. Some of the survey responses mentioned problems of complexity and poor performance of the ERP system, which can discourage users from relying on the ERP system. When systems are perceived as difficult to use or inefficient, users tend to avoid them and seek alternative solutions.

The usage gap for ERP systems can be seen from the underutilization of system modules. From the survey results obtained, some major modules such as Human Resources (HR), analytics reporting, and Customer Relationship Management (CRM) were noted to be underutilized. This shows that the organizations do not maximize on the available ERP capabilities within their organizations. Underutilization of these modules shows that there is a lack of sufficient training on them or a lack of awareness of their significance.

Another strong indicator of the gap is the reliance on external or manual tools. The use of spreadsheets or any other non-integrated tool is a deviation from the appropriate use of ERP systems. Although the tool may facilitate specific activities, it reduces system integration and results in inconsistency in data and processes.

Overall, the ERP usage gap can be attributed to three key factors. Some of these include inconsistency in system utilization, underutilization of modules in the system, and the utilization of external tools. Such gaps arise as a result of poor organizational practices, low user skills, and system issues.

5.9 Impact on organizational Performance

The ERP adoption gaps discovered in this research have considerable effects on organizational effectiveness among retail organizations in Bahrain. The gap does not just affect the usage of ERP but also influences other aspects such as organizational efficiency, reliability of data, and the decision-making process.

One of the first effects that occurs is in terms of operational effectiveness. The aim of ERP software is to create standardization, automate tasks and reduce manual effort, but since not everyone will use the system equally, it fails to achieve this goal. As a result, it becomes more difficult for employees to collaborate effectively, causing duplication of efforts, inefficiencies in accomplishing their tasks, and making them rely on manual processes again.

This effect is also observed in terms of data quality and consistency. With the use of external tools like spreadsheet applications in conjunction with the ERP system, data will become fragmented and scattered among different sources. It will then become challenging for an organization to ensure consistency in data. Over time, leading to inconsistent reporting, departmental misunderstanding and confusion between them, and a lack of trust in the system. Consequently, the ERP system will lose its status as the main source of truth within the organisation.

This problem directly impacts the decision-making process. ERP systems are supposed to provide timely, relevant, and integrated information for supporting decision-making at both operational and strategic levels. But if the use of the ERP system is not complete or consistent, then the decision-maker must depend on incomplete or old information that may have been manually compiled.

There is another significant effect that comes out of the underuse of the important system modules. such as the HR, analytics/reporting and CRM modules, among others, have an important role to play when it comes to helping organizations conduct their operations. However, if this does not happen, then there will be a failure to leverage benefits such as enhanced performance, improved understanding, and stronger relationships with customers. This limits the organisation's ability to achieve a return on its investment and that means less value will be generated by the entire ERP system.

Moreover, the ERP usage gap creates a discrepancy between what is expected by the organisation and what actually happens in reality. Even though management expects ERP software to contribute positively to enhancing performance and operational efficiency, failure to use the ERP systems properly results in a failure to meet these expectations, and that may lead to dissatisfaction with the system and may influence future technology-related decisions within the organisation.

Organisational factors like training and management support only increase the significance of these effects. Lack of proper training leads to lack of user confidence and a higher probability of mistakes; inadequate management support results in less emphasis on system usage. Lack of organizational commitment makes ERP systems go unutilized, irrespective of their potential capabilities.

In the long term, these combined may affect the competitiveness of the organisation. The inefficient process, unreliable data, and poor decision-making will adversely affect the ability to adapt, innovate, and make the organisation compete effectively. This is an indication that ERP adoption gaps go beyond just being operational issues. They are strategic problems that affect overall business performance.

Overall, the results highlight that ERP use deficiencies have diverse impacts on efficiency, data accuracy, decision-making quality, and overall value derived from ERP systems. Addressing these gaps is therefore essential for organisations seeking to fully benefit from their ERP investments.

5.10 Chapter Summary

This chapter interpreted the study findings in relation to the research objective of identifying ERP usage gaps in the retail sector in Bahrain. The comparison between the case study and the survey clarified the difference between expected usage, reflected in users' intention to continue using ERP systems, and actual usage in practice. The results showed that although users had a generally positive attitude towards ERP systems, this did not always translate into consistent or effective use within organisations.

Moreover, it was found that the reason behind these gaps is basically associated with issues concerning organisations and users in terms of insufficient training, inadequate monitoring, and a lack of management support. These consequences can be felt through daily work, when inefficiencies, inconsistencies, and poor decision-making start becoming apparent. In essence, what the chapter has shown is that these gaps not only affect the usage of ERP systems but their effectiveness as well. Essentially, what has been learned in this chapter is that maximizing benefits from the use of ERP systems involves much more than the systems themselves.

Finally, the evaluation of hypotheses further strengthened the study's findings by confirming the significant role of organisational and human factors in explaining ERP usage gaps within the Bahraini retail sector.

Chapter 6: Conclusion and Recommendations

6.1 Conclusion

This research focused on identifying and analyzing the ERP Usage Gap within the retail sector of Bahrain, specifically the disconnect between the technical potential of Enterprise Resource Planning systems and their actual daily application by employees. By integrating the TOE framework with the DeLone and McLean IS Success Model, the study established a theoretical foundation for understanding how system quality and organizational support influence usage outcomes.

To bridge the theory and practice gap, the study utilized a quantitative survey of 148 retail professionals and a validation case study of 237 responses, structured around the CRISP-DM methodology:

- **Business & Data Understanding:** Identifying the specific usage gap in Bahrain's retail sector through survey distribution and initial data exploration.
- **Data Preparation:** Cleaning and pre-processing the survey results to ensure compatibility with the WEKA machine learning environment.
- **Modelling:** Application of machine learning techniques such as random forest (with an accuracy of 89.03%) and J48 to find the key factors behind underutilization.
- **Evaluation & Deployment:** validation of the model's performance and subsequent deployment of the knowledge gained from the data into managerial actions.

The conclusions from this study show that the implementation of ERP alone does not guarantee or ensure successful system utilisation. However, while many retail companies invest heavily in ERP technology, the expected operational benefits may remain limited when employees do not consistently engage with the system during daily activities. The results highlight the importance of combining the technological competencies with organization's preparedness, continuous monitoring, and user support systems. Moreover, the results indicate the applicability of using Artificial Intelligence and Machine Learning to conduct analysis on ERP usage behaviour. The

models developed provided valuable insights into the causes of ERP usage gaps and proved the ability of data analytics to make informed decisions in the retail industry.

6.2 Summary of the study

This study examined the factors contributing to the underutilization of ERP systems within retail organizations in Bahrain. Despite the advanced capabilities of ERP systems, a significant usage gap exists, where employees rely primarily on basic functionalities while underutilizing advanced features.

The research was grounded in the integration of two well-known academic frameworks: TOE and the DeLone & McLean Success Model, allowing for a comprehensive analysis of both technical and organizational factors influencing ERP usage.
Data was collected using a two-stage approach:

- A primary survey of 148 retail professionals in Bahrain
- A validation case study consisting of 237 responses to ensure model reliability

The analysis followed the CRISP-DM methodology, where the data was prepared and analyzed using the WEKA platform.

The Random Forest model achieved the highest predictive performance (89.03%) in the case study, while J48 decision trees provided interpretable insights into the factors influencing ERP usage. The findings indicate that while core modules such as Sales and Inventory are widely used, advanced modules such as Reporting and Human Resources remain significantly underutilized. Overall, the results demonstrate that the ERP usage gap is not purely a technical issue, but is strongly influenced by human and organizational factors, particularly perceived usefulness, training, and top management support.

6.3 Recommendations

Based on the results obtained in the previous chapter, this section offers recommendations that would be useful in addressing the ERP implementation gaps and improving system effectiveness in retail companies operating in Bahrain.

The area that needs to be considered is monitoring the use of the system. From the findings, the lack of monitoring in the use of ERP systems leads to increased dependency on other techniques, such as the use of Excel sheets for carrying out certain tasks. The system consistency and efficiency may become an issue without monitoring. There are various ways that monitoring can be done. It might be through internal or external audits. These activities do not have to be stringent, just enough to keep everyone accountable. For instance, companies could produce monthly ERP utilization reports per department or have their team leaders monitor the systems' activities periodically to ensure compliance.

The other recommendation concerns the use of training techniques. It is evident from the results that limited or one-time training does not contribute much towards achieving effective use of ERP. Although employees may be familiar with the basic features of the software, without continuous support, they are less likely to explore advanced features or fully utilise the system. Organizations need to take into consideration regular training depending on the various roles of individuals using the software. Such refresher sessions, practical classes, workshops, and internal manuals will definitely help in increasing users' confidence and improving the acceptance rate of the system, and organisations can arrange such short refresher courses at an interval of one month. Video tutorials for frequently performed activities by employees can also be produced or provide quick-reference guides that employees can access during their daily work to achieve effective use of ERP.

Support from management is another important consideration in ensuring that the usage of the ERP system is effective. In cases where there is no support from management, there could be a tendency for the usage of the ERP system to take a back seat in day-to-day activities at work. Employees usually concentrate on what is expected and monitored by their supervisors. As a result, managers need to get involved in encouraging the use of the ERP system, either by making expectations clear, following up, or simply by serving as good examples. This would help in creating a work environment where ERP is viewed as a necessary component.

Additionally, organisations can implement motivational techniques like rewarding those employees who regularly use the ERP system efficiently, by providing rewards every month or small bonuses.

Moreover, the report also indicates the need for enhancing awareness regarding the capabilities of the system. The fact that the use of modules such as HR management, analytics/reporting, and CRM is below par indicates the low awareness levels regarding the capabilities of the system among employees. Organizations can overcome this challenge by conducting awareness campaigns within the organization or by providing case studies and examples that would demonstrate the usefulness of these modules in completing various tasks. Employees' awareness regarding their potential to gain from employing the system will be greatly improved. For instance, department managers can conduct small workshops or short sessions indicating the utility of each module in completing routine tasks efficiently, such as reporting tools, which can simplify tasks like performance tracking or decision-making.

Finally, organisations should not neglect the issue of dependency on external tools. The problem itself should not be considered merely negative, but rather, there is a need to figure out why this occurs. Sometimes, external tools are actually more comfortable and easier for employees to utilise when performing a specific task. This means that there is something wrong with how the ERP system works. Therefore, once the problem area is determined, it is possible to fix it, reduce the need for workarounds and improve overall system integration. For example, when employees use Excel to create reports, it indicates that the reporting module of the ERP system should be simplified or that employees need more training.

In summary, enhancing the effectiveness of ERP use in retail enterprises involves the gradual improvement of multiple associated aspects. For instance, raising awareness of the system and encouraging employees to take full advantage of all available modules could assist organizations in leveraging the resources of their existing systems. On the other hand, involvement from the management, as well as simplistic methods like rewarding usage of the system, make the use of ERP a part of the routine activities for employees, as opposed to being a secondary activity. In addition, exploring the reasons behind the dependence of employees on external tools could lead to finding possible problems with the ERP system that require modifications. At the same time, monitoring and training will ensure more stable usage of the system in the future. When these

efforts come together, organisations are more likely to reduce usage gaps and achieve more reliable and effective use of their ERP systems in practice.

6.3.1 Recommendation Prioritization Matrix

To support practical implementation, the proposed recommendations were prioritized according to their expected organisational impact and implementation difficulty. This helps organisations focus first on the most influential improvements for reducing ERP usage gaps.

Table 16: Recommendation Prioritization Matrix for ERP Usage Gap Reduction

Recommendation	Expected Impact	Implementation Difficulty	Priority
Improve ERP monitoring practices	High	Low	High
Enhance employee ERP training	High	Medium	High
Strengthen top management support	Medium	Medium	Medium
Improve ERP information quality	High	High	Medium

The prioritization results indicate that monitoring practices and employee training should be addressed first due to their high organisational impact and relatively manageable implementation requirements. These areas were also repeatedly identified by the machine learning models as major contributors to ERP usage effectiveness.

6.3.2 Implementation KPIs and Success Metrics

To evaluate the effectiveness of the proposed recommendations, several Key Performance Indicators (KPIs) were identified based on the main factors highlighted by the machine learning models. These indicators provide organizations with measurable methods to monitor ERP

improvement and assess whether the implemented strategies successfully reduce ERP usage gaps over time.

Table 17: ERP Implementation KPIs and Success Metrics

Recommendation	KPI	Target Outcome
Improve ERP training	Employee ERP training completion rate	Above 85% completion
Strengthen ERP monitoring	Frequency of ERP usage monitoring reports	Regular monthly monitoring
Improve information quality	User satisfaction with ERP information	Satisfaction above 80%
Enhance top management support	Employee perception of management support	Improved ERP adoption levels

These KPIs provide organisations with a practical framework for evaluating ERP improvement initiatives over time. By continuously monitoring these indicators, organisations can identify areas requiring further support and ensure that ERP adoption remains stable and effective.

6.4 Limitations

While this research provides a validated framework for understanding and predicting ERP usage gaps, several limitations should be acknowledged to better contextualize the findings.

6.4.1 Sample Size and Sectoral Scope

The research was conducted using 148 valid responses collected from the Bahraini retail industry. Although this sample size was sufficient to develop and evaluate the machine learning models used in this study, it represents only a single sector within the broader economy. Retail

organizations typically have specific operational characteristics, such as shorter transaction cycles, higher customer interaction, and different system usage patterns compared to other industries.

In contrast, sectors such as manufacturing, oil and gas and finance often involve more complex processes, stricter regulatory requirements and different levels of ERP system integration and usage. These differences can significantly influence how ERP systems are adopted, monitored and utilized. Therefore, the patterns and relationships identified in this study may not fully reflect the realities of other industries. As a result, the findings should be interpreted with caution when considering their applicability beyond the retail sector.

6.4.2 Geographic and Cultural Specificity

In addition to the sectoral limitations discussed in Section 6.4.1, this study is also context-specific to the Bahraini environment. The research was designed around the Bahraini retail sector, where local management styles, levels of digital maturity and organizational culture play an important role in shaping ERP usage. These contextual factors directly influenced key variables in the model, particularly Top Management Support and Monitoring.

Since organizational behaviors and culture may differ greatly from one region to another, the links revealed in this study might be different in other geographical locations. Consequently, the rules generated by the model would require some adjustment or recalibration for application to retail businesses in other countries.

6.4.3 Subjectivity of Perception-Based Data

The study relied on primary survey data, which captures user perceptions, attitudes and self-reported behaviors. While perceptions are the primary drivers of technology adoption, they are inherently subjective. Which means that the collected data is not necessarily reflect reality as the study did not have access to objective automated system event logs from the organizations

involved as users become more familiar with the system, as new features are implemented, or as management introduces new policies or training initiatives.

6.4.4 Cross-Sectional Constraint

ERP utilization is a dynamic process that evolves as employees become more experienced or as the software is updated. However, the data collected during this research only represents a snapshot that captures a single point in time. It cannot capture how usage patterns develop over time.

6.4.5 Technical Deployment Boundaries

In accordance with the academic scope of this project, the Deployment Phase of the CRISP-DM methodology remained conceptual. The models were developed, evaluated, and validated within the WEKA environment rather than being integrated into a live, real-time ERP system. While this study demonstrates a proof of concept, it does not account for the practical challenges associated with real-world implementation.

In reality, implementing a model will have to include system integration via API, data gathering continuously, as well as creating user-friendly dashboards for decision-makers. These aspects could affect the results significantly, since they play a significant role in the success of any implementation.

Thus, although this model shows a lot of promise as a decision-making aid, further research is required to test its performance and feasibility within a fully integrated, real-time ERP setting.

6.5 Future Work

Based on the insights gained and the limitations identified in this study, the following areas are suggested for future research to further bridge the ERP usage gap:

6.5.1 Integration of Objective Process Mining

Future research should build on the process mining concepts introduced in Section 2.5.1 by incorporating objective system-generated data alongside perception-based survey data. By analyzing event logs extracted from ERP systems, researchers can understand how processes are actually performed in practice based on real user interactions.

This would enable a more precise comparison between intended system workflows and actual system usage. It also allows for the identification of inefficiencies, deviations and technical bottlenecks with greater accuracy. Such an approach would enhance the reliability of ERP usage gap analysis and provide a more comprehensive understanding of system utilization in practice.

6.5.2 Long-term Analysis of Usage Trends

To address the gaps left by a one-time snapshot, this study suggests a long-term monitoring period of one to two years. Following the same organizational groups over this duration allows for a clearer look at the 'usage gap' as it develops. More importantly, it provides a realistic way to evaluate how management interventions and system updates influence user behavior over time, rather than just seeing a momentary reaction.

6.5.3 Cross-Sectoral Comparative Studies

The framework would also benefit from being applied to other major sectors in Bahrain, such as healthcare, finance, or manufacturing. Since these industries operate under very different regulatory pressures and technical complexities, the ERP usage gap likely looks different in each one. Comparing these sectors is the only way to see if the drivers identified in this study are universal truths or if they change depending on the industry. These findings would eventually allow for more practical, sector-specific strategies rather than a 'one-size-fits-all' approach to ERP adoption. In addition to supporting organizations in designing interventions that are better aligned with their operational environment.

6.5.4 Development of Real-Time AI Dashboards

Translating the predictive models developed in this study into practical, real-time systems represents an important avenue for future research. Investigations into the use of APIs would facilitate the development of a system that is able to detect in real time when employees' engagement levels start falling and send out corresponding alarms to the management team.

6.5.5 Exploring Qualitative Depth

Whereas this current study employs a quantitative approach to identify patterns in ERP usage, future research could adopt qualitative methods such as interviews or focus group discussions with participants. Engaging directly with employees and ERP project managers would allow researchers to gain deeper “insider knowledge” of the factors influencing system use. Particularly, these issues, such as resistance to change, lack of motivation or informal work practices that may not be fully captured through survey data.

This approach would provide richer contextual insights into user behavior and help explain the reasons behind the patterns identified in the quantitative results. Combining qualitative data with predictive models would help future studies understand both the patterns and the reasons behind ERP usage gaps.

Appendix A: List of Measurement Items

This appendix presents the measurement items used in the survey instrument. The items are adapted from established models, including the DeLone and McLean IS Success Model and the TOE framework. All items were measured using a five-point Likert scale ranging from 1 = Strongly Disagree to 5 = Strongly Agree.

Perceived Usefulness

PUS1: Using ERP helps me complete tasks more quickly

PUS2: Using ERP improves my productivity

Ease of Use

PEOU1: The ERP system is easy to use

PEOU2: Learning to use ERP is easy for me

System Quality

SYQ1: The ERP system is reliable and stable

SYQ2: ERP functions meet my work requirements

Information Quality

INQ1: ERP provides accurate information

INQ2: ERP provides useful and relevant information

Top Management Support

TMS1: Top management supports the use of ERP

TMS2: Top management encourages employees to use ERP

ICT Skills

ISK1: I have sufficient skills to use ERP effectively

ISK2: I have received adequate training to use ERP

Continuance Intention

COI1: I intend to continue using ERP in the future

COI2: I plan to increase my use of ERP

COI3: I will rely on ERP regularly in my work

Appendix B: Questionnaire

This appendix presents the full questionnaire used to collect data for this study.

Title: ERP System Usage and Underutilization in Retail Organizations in Bahrain
This questionnaire is part of an academic research project that aims to examine ERP system usage and identify potential usage gaps within retail organizations in Bahrain. ERP systems are integrated information systems designed to support and manage various organizational processes, enhance efficiency, and improve decision-making.
The purpose of this study is to understand how ERP systems are used in practice and to identify the factors that may limit their effective utilization. The findings of this research will contribute to providing insights that may support organizations in improving ERP usage and achieving better performance outcomes.
Participation in this survey is voluntary. All responses will be kept strictly confidential and will be used for academic purposes only. No personal or identifiable information will be collected or disclosed.
The estimated time to complete this questionnaire is approximately 3–7 minutes.
Thank you for your participation.

Please answer the following questions.

Q1. What type of retail organization do you operate?
- Supermarket / Hypermarket
- Clothing / Fashion
- Electronics
- Pharmacy
- Other

Q2. What is your current position in the organization?
- Clerk / Staff

- Supervisor
- Manager
- Senior Management

Q3. How long have you been using the ERP system in your organization?

- Less than 6 months
- 6 months – 1 year
- 1 – 2 years
- 2 – 5 years
- More than 5 years

Q4. Which ERP system is currently used in your organization?

- SAP
- Oracle
- Microsoft Dynamics
- Odoo
- Other

Q5. Which ERP modules do you use regularly? (Select all that apply)

- Inventory
- Sales
- Procurement
- Finance
- Human Resources
- Reporting
- Other

Q6. How frequently do you use the ERP system during your workday?

- Rarely
- Sometimes
- Often
- Very frequently

Q7. How often do you complete tasks entirely within the ERP system (without using external tools)?

- Rarely

- Sometimes
- Often
- Always

Q8. Are there ERP features or modules that you do not use?

- Yes
- No

Q9. If yes, which ERP modules or functions are underutilized? (Short answer)

Q10. What are the main reasons for not fully utilizing the ERP system? (Select all that apply)

- Lack of training
- System complexity
- Poor system performance
- Lack of management support
- Preference for manual/external tools
- Other

Q11. Does your organization monitor ERP usage?

- Yes, regularly
- Sometimes
- Rarely
- No

Please indicate your level of agreement with the following statements: Likert scale ranging from 1 = Strongly Disagree to 5 = Strongly Agree.

Perceptions (Likert Scale)

Table 18: Survey: Likert Scale Questions

No	Questions	1	2	3	4	5
1	Using ERP helps me complete tasks more quickly					
2	Using ERP improves my productivity					
3	The ERP system is easy to use					
4	Learning to use ERP is easy for me					
5	The ERP system is reliable and stable					
6	ERP functions meet my work requirements					
7	ERP provides accurate information					
8	ERP provides useful and relevant information					
9	Top management supports the use of ERP					
10	Top management encourages employees to use ERP					
11	I have sufficient skills to use ERP effectively					
12	I have received adequate training to use ERP					
13	I intend to continue using ERP in the future					
14	I plan to increase my use of ERP					
15	I will rely on ERP regularly in my work					
Thank you for your time						

Appendix C: WEKA Modelling Environment

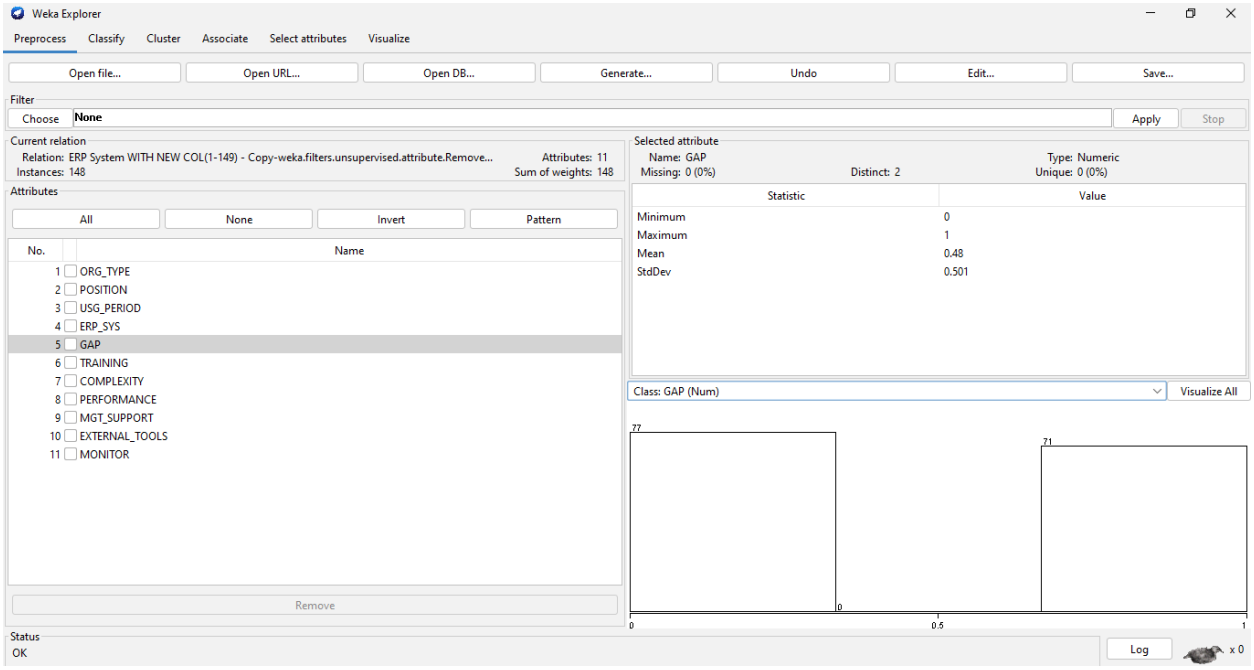

Figure 27: Final Prepared Dataset attributes selection in Weka (survey)

The figure shows the final prepared dataset attributes selection before completing the preprocessing for machine learning modelling.

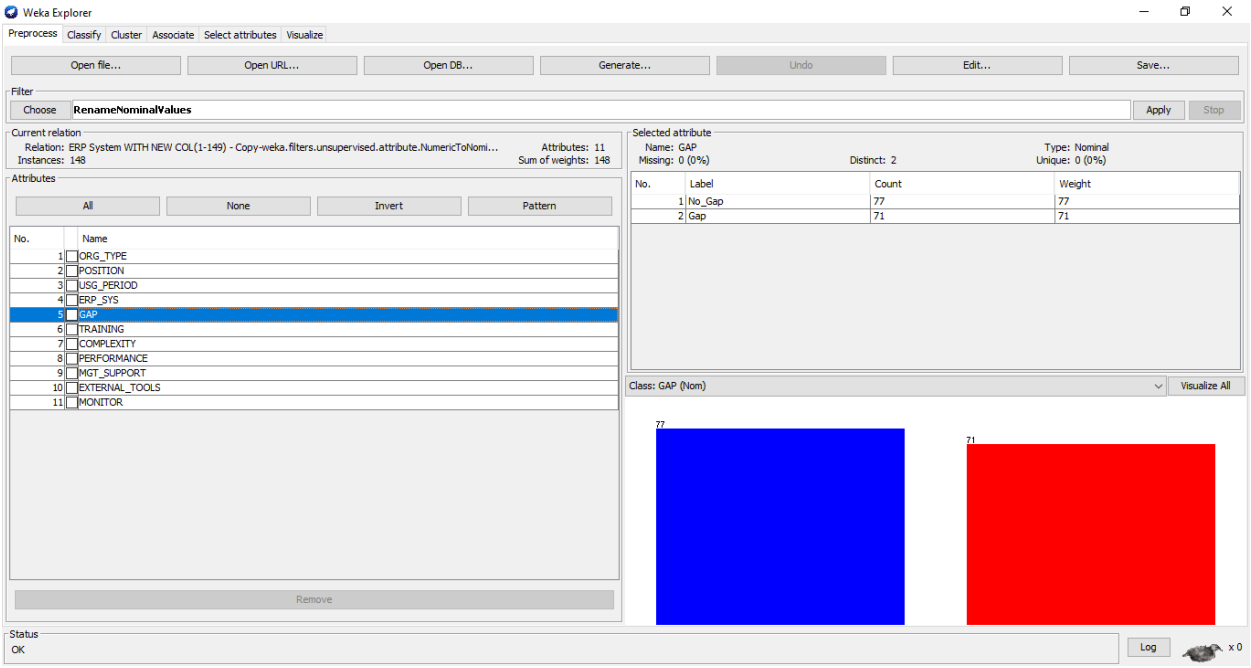

Figure 28: RenameNominalValues Filter Configuration (survey)

This filter was used to rename nominal class values and improve the readability and consistency of the dataset within WEKA.

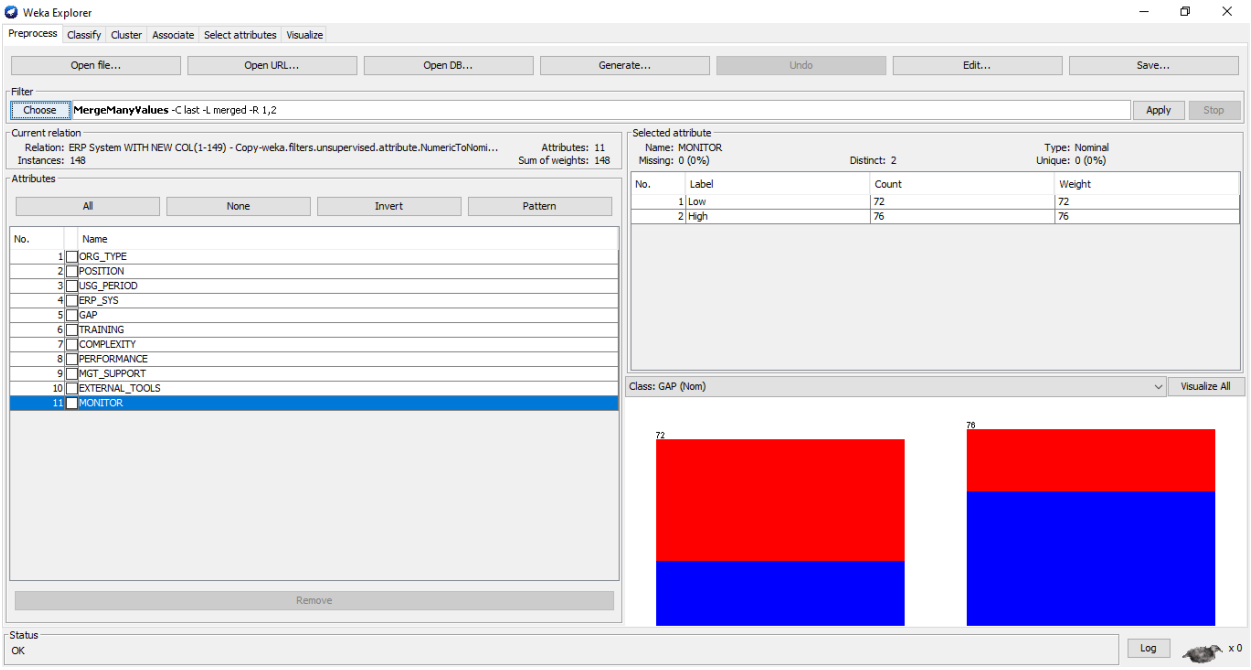

Figure 29: MergeManyValues Filter Used for Monitoring Categories (survey)

The MergeManyValues filter was applied to combine similar monitoring categories into broader classes to simplify the classification process.

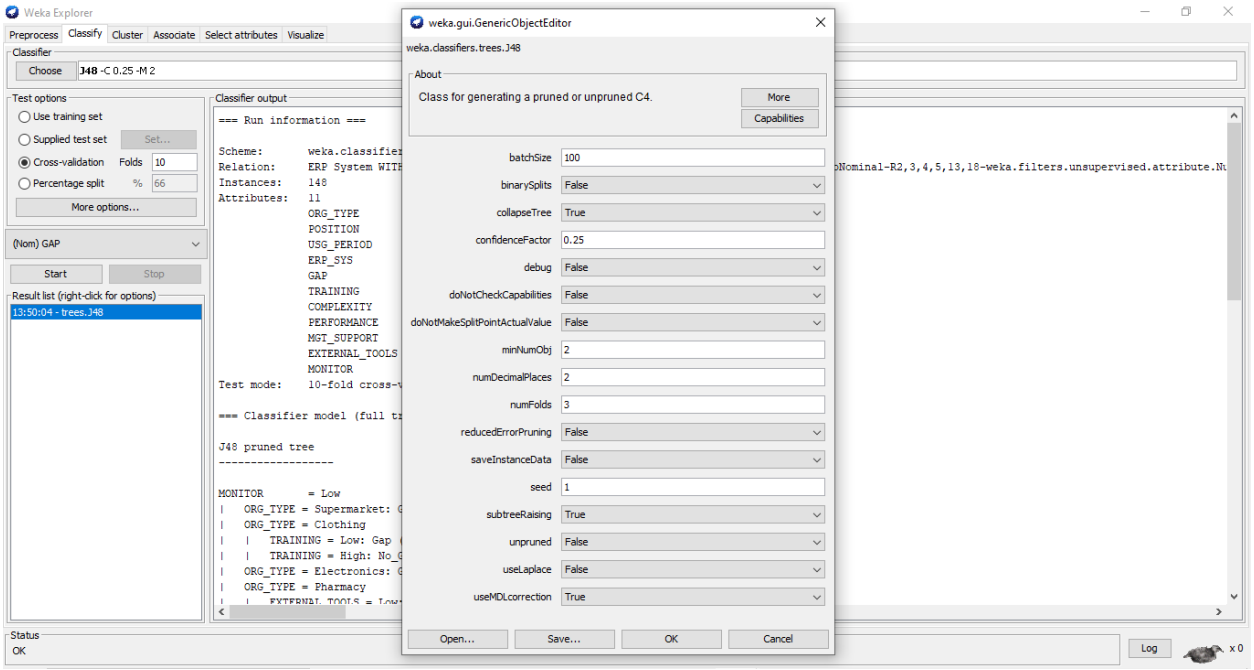

Figure 30: J48 Classification Configuration (survey)

This figure presents the configuration settings used for the J48 Decision Tree classifier during the modelling stage.

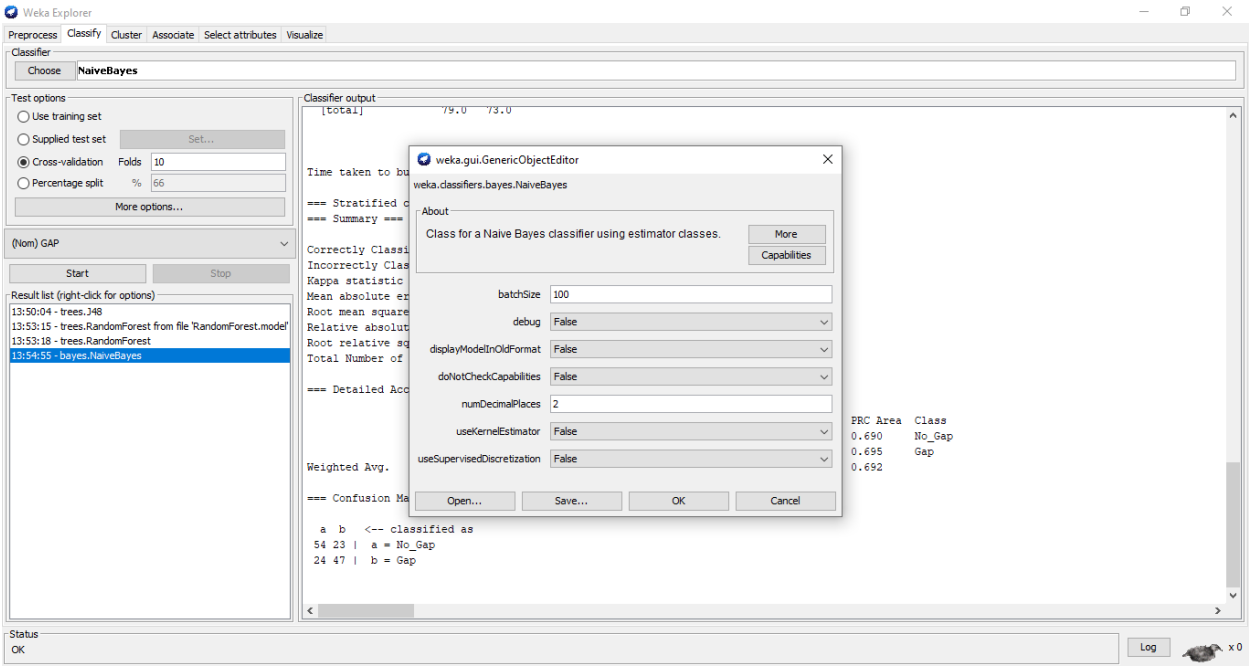

Figure 31: Random Forest Configuration (survey)

The figure illustrates the parameter settings used for the Random Forest classifier in WEKA for predictive analysis.

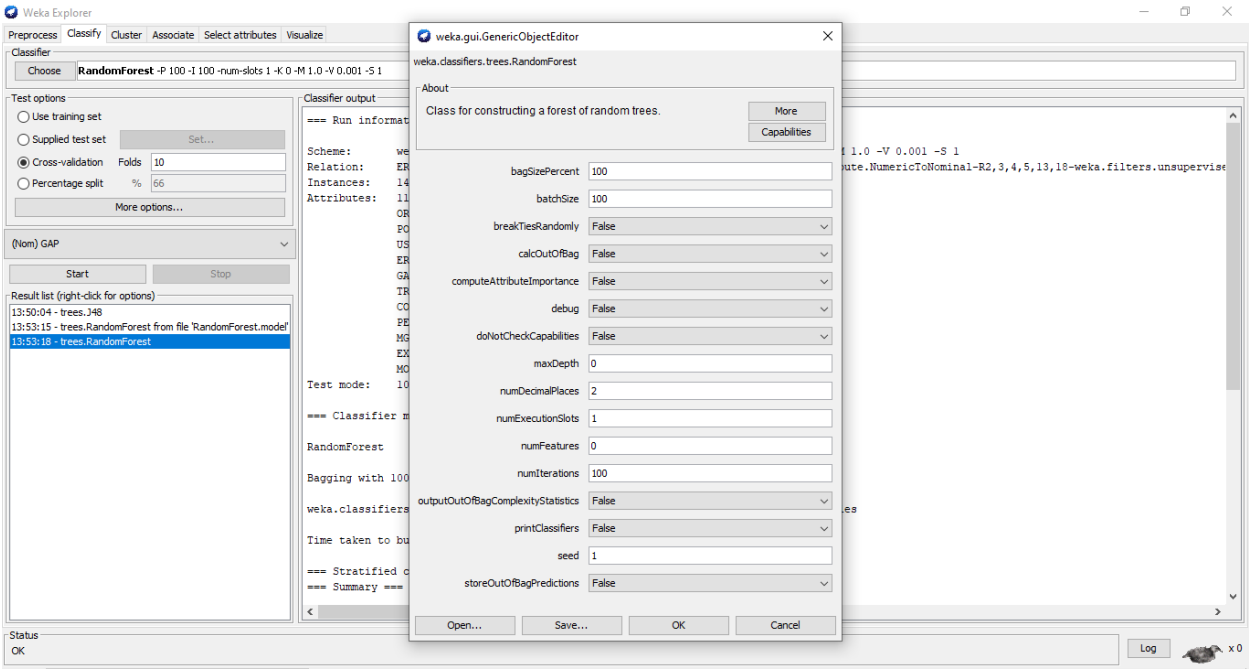

Figure 32: Naïve Bayes Configuration (survey)

This figure displays the WEKA configuration settings used for implementing the Naïve Bayes classification algorithm.

Appendix D: SYSTEM IMPLEMENTATION

This appendix presents the complete implementation of the ERP Usage Gap prediction system developed using the J48 machine learning model trained with WEKA. The Java application acts as an interface between the user and the trained model, receiving input data from the web interface, processing it, and passing it to the model for classification. The resulting prediction is then returned and displayed to the user through the application interface.

```
package com.research.erp;


import weka.classifiers.Classifier;
import weka.core.*;
import java.util.*;
import static spark.Spark.*;

public class App {

    static Classifier model;
    static Instances data;

    public static void main(String[] rags) {

        port(8080);
        staticFiles.location("/public");

        try {
            model = (Classifier)
SerializationHelper.read("models/usage_gap.model");

            List<String> hl = List.of(" High ", " Low ");

            List<Attribute> attrs = new ArrayList<>(List.of(

                    new Attribute("ORG_TYPE", List.of(
                            " Supermarket / Hypermarket ",
                            " Clothing / Fashion ",
                            " Electronics ",
                            " Pharmacy ",
                            " Other "
                    )),

                    new Attribute("POSITION", List.of(
                            " Clerk / Staff ",
                            " Manager ",
                            " Supervisor ",
                            " Senior Management "
```

```
                    )),

                    new Attribute("USG_PERIOD", List.of(
                            " <6m ", " 6m_1y ", " >5y ", " 1_2y ", " 2_5y "
                    )),

                    new Attribute("ERP_SYS", List.of(
                            " Oracle ",
                            " Microsoft Dynamics ",
                            " SAP ",
                            " Odoo "
                    )),

                    new Attribute("GAP", List.of(
                            " No_Gap ",
                            " Gap "
                    ))
            ));

            for (String f : List.of(
                    "TRAINING", "COMPLEXITY", "PERFORMANCE",
                    "MGT_SUPPORT", "EXTERNAL_TOOLS", "MONITOR"
            )) {
                attrs.add(new Attribute(f, hl));
            }

            data = new Instances("UsageGapPrediction", new
ArrayList<>(attrs), 0);
            data.setClassIndex(4); // GAP is now index 4

        } catch (Exception e) {
            e.printStackTrace();
            return;
        }

        get("/", (q, r) -> {
            r.redirect("/index.html");
            return null;
        });

        post("/predict", (req, res) -> {
            try {
                Instance i = new DenseInstance(data.numAttributes());
                sedatest(data);

                set(i, "ORG_TYPE", req.queryParams("ORG_TYPE"));
                set(i, "POSITION", req.queryParams("POSITION"));

                String period = switch (req.queryParams("USG_PERIOD")) {
```

```
                    case "lt6m" -> "<6m";
                    case "gt5y" -> ">5y";
                    case "6m_1y" -> "6m_1y";
                    case "1_2y" -> "1_2y";
                    case "2_5y" -> "2_5y";
                    default -> req.queryParams("USG_PERIOD");
                };

                set(i, "USG_PERIOD", period);
                set(i, "ERP_SYS", req.queryParams("ERP_SYS"));

                for (String f : List.of(
                        "TRAINING", "COMPLEXITY", "PERFORMANCE",
                        "MGT_SUPPORT", "EXTERNAL_TOOLS", "MONITOR"
                )) {
                    set(i, f, req.queryParams(f));
                }

                String pred = data.classAttribute()
                        .value((int) model.classifyInstance(i))
                        .trim();

                return page(pred);

            } catch (Exception e) {
                return "<h3>Error:</h3>" + e.getMessage();
            }
        });
    }

    static void set(Instance i, String attr, String val) {
        i.setValue(data. Attribute(attr), " " + val + " ");
    }

    static String page(String p) {
        String c = p.equals("Gap") ? "#e74c3c" : "#2ecc71";

        return String.format("""
                <!DOCTYPE html>
                <html>
                <head>
                    <meta charset='UTF-8'>
                    <meta name='viewport' content='width=device-width,
initial-scale=1.0'>
                    <title>Prediction Result</title>
                    <style>
                        body{
                            margin:0;
                            font-family:Arial;
```

```
                            display: flex;
                            justify-content: center;
                            align-items:center;
                            height:100vh;
                            background:linear-
gradient(135deg,#74ebd5,#acb6e5);
                        }
                        .card{
                            background:#off;
                            padding:35px;
                            border-radius:16px;
                            text-align:center;
                            box-shadow:0 10px 30px grab(0,0,0,.15)
                        }
                        .badge{
                            padding:10px 20px;
                            border-radius:30px;
                            color:#fff;
                            background:%s;
                            font-weight:bold;
                        }
                        a{
                            display:inline-block;
                            margin-top:20px;
                            padding:10px 18px;
                            background:#007BFF;
                            color:#fff;
                            text-decoration:none;
                            border-radius:8px;
                        }
                    </style>
                </head>
                <body>
                    <div class='card'>
                        <h2>Prediction Result</h2>
                        <div class='badge'>%s</div><br>
                        <a href='/index.html'>Back</a>
                    </div>
                </body>
                </html>
                """, c, p);
    }
}
```

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